

comsol-freidd  
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# Глава 1

## Алфавитный указатель пространств имен

### 1.1 Пространства имен

Полный список пространств имен.

|                            |    |
|----------------------------|----|
| add_command                | 9  |
| bezier_dialog              | 9  |
| boolean_item               | 9  |
| boolean_operations         | 9  |
| boundary_conditions_dialog | 9  |
| command                    | 10 |
| command_service            | 10 |
| comsol                     | 10 |
| data_parser                | 10 |
| delete_command             | 11 |
| dialog                     | 11 |
| dialog_factory             | 11 |
| doxytest                   | 11 |
| drawing_service            | 11 |
| edge_item                  | 11 |
| editable_bezier            | 12 |
| ellipse_dialog             | 13 |
| ellipse_item               | 13 |
| event_handler              | 13 |
| gmsh_mesh_builder          | 13 |
| graphics_view              | 13 |
| grid_scene                 | 14 |
| initial_conditions_dialog  | 14 |
| line_dialog                | 14 |
| line_item                  | 14 |
| main                       | 14 |
| main_window                | 15 |
| material_dialog            | 16 |
| mesh_dialog                | 16 |
| move_command               | 16 |
| mpl_canvas                 | 16 |
| parametric_curve_item      | 16 |
| parametric_dialog          | 16 |
| plotter                    | 17 |
| rect_dialog                | 17 |

|                                           |                    |
|-------------------------------------------|--------------------|
| <a href="#">rectangle_item</a>            | <a href="#">17</a> |
| <a href="#">rotate_command</a>            | <a href="#">17</a> |
| <a href="#">selection_service</a>         | <a href="#">17</a> |
| <a href="#">template</a>                  | <a href="#">17</a> |
| <a href="#">transformation_operations</a> | <a href="#">17</a> |
| <a href="#">turbulence_dialog</a>         | <a href="#">18</a> |
| <a href="#">turbulence_models</a>         | <a href="#">18</a> |

## Глава 2

# Иерархический список классов

### 2.1 Иерархия классов

Иерархия классов.

#### ABC

|                                                     |     |
|-----------------------------------------------------|-----|
| command.Command . . . . .                           | 39  |
| add_command.AddCommand . . . . .                    | 19  |
| delete_command.DeleteCommand . . . . .              | 46  |
| move_command.MoveCommand . . . . .                  | 125 |
| boolean_operations.BooleanOperations . . . . .      | 26  |
| turbulence_models.BoundaryConditions . . . . .      | 31  |
| turbulence_models.InletBoundaryConditions . . . . . | 97  |
| turbulence_models.OpenBoundaryConditions . . . . .  | 129 |
| turbulence_models.WallBoundaryConditions . . . . .  | 170 |

#### Command

|                                          |     |
|------------------------------------------|-----|
| rotate_command.RotateCommand . . . . .   | 146 |
| command_service.CommandService . . . . . | 41  |
| data_parser.DataParser . . . . .         | 44  |

#### Dialog

|                                                               |     |
|---------------------------------------------------------------|-----|
| bezier_dialog.BezierDialog . . . . .                          | 22  |
| boundary_conditions_dialog.BoundaryConditionsDialog . . . . . | 33  |
| ellipse_dialog.EllipseDialog . . . . .                        | 66  |
| initial_conditions_dialog.InitialConditionsDialog . . . . .   | 93  |
| line_dialog.LineDialog . . . . .                              | 99  |
| material_dialog.MaterialDialog . . . . .                      | 119 |
| mesh_dialog.MeshDialog . . . . .                              | 122 |
| parametric_dialog.ParametricDialog . . . . .                  | 134 |
| rect_dialog.RectDialog . . . . .                              | 143 |
| turbulence_dialog.TurbulenceDialog . . . . .                  | 156 |
| dialog_factory.DialogFactory . . . . .                        | 50  |
| doxytest.DoxyTest . . . . .                                   | 51  |
| drawing_service.DrawingService . . . . .                      | 56  |

#### Enum

|                                                   |     |
|---------------------------------------------------|-----|
| turbulence_models.BoundaryConditionType . . . . . | 38  |
| turbulence_models.TurbulenceModel . . . . .       | 159 |
| turbulence_models.WallType . . . . .              | 172 |
| event_handler.EventHandler . . . . .              | 73  |
| FigureCanvasQTAgg                                 |     |
| mpl_canvas.MplCanvas . . . . .                    | 127 |

|                                                              |     |
|--------------------------------------------------------------|-----|
| gmsh_mesh_builder.GmshMeshBuilder . . . . .                  | 78  |
| turbulence_models.InitialConditions . . . . .                | 92  |
| turbulence_models.Material . . . . .                         | 118 |
| object                                                       |     |
| comsol.Ui_MainWindow . . . . .                               | 162 |
| template.Ui_MainWindow . . . . .                             | 166 |
| plotter.Plotter . . . . .                                    | 137 |
| QDialog                                                      |     |
| dialog.Dialog . . . . .                                      | 49  |
| QGraphicsEllipseItem                                         |     |
| editable_bezier.DraggablePoint . . . . .                     | 52  |
| ellipse_item.EllipseItem . . . . .                           | 69  |
| QGraphicsLineItem                                            |     |
| line_item.LineItem . . . . .                                 | 102 |
| QGraphicsPathItem                                            |     |
| boolean_item.BooleanShapeItem . . . . .                      | 29  |
| edge_item.EdgeItem . . . . .                                 | 59  |
| editable_bezier.EditableBezierCurveItem . . . . .            | 62  |
| parametric_curve_item.ParametricCurveItem . . . . .          | 131 |
| QGraphicsRectItem                                            |     |
| rectangle_item.RectangleItem . . . . .                       | 139 |
| QGraphicsScene                                               |     |
| grid_scene.GridScene . . . . .                               | 87  |
| QGraphicsView                                                |     |
| graphics_view.GraphicsView . . . . .                         | 81  |
| QMainWindow                                                  |     |
| main_window.MainWindow . . . . .                             | 106 |
| main_window.MainWindow . . . . .                             | 106 |
| selection_service.SelectionService . . . . .                 | 149 |
| transformation_operations.TransformationOperations . . . . . | 154 |
| turbulence_models.TurbulenceParams . . . . .                 | 161 |

## Глава 3

# Алфавитный указатель классов

### 3.1 Классы

Классы с их кратким описанием.

|                                                                     |     |
|---------------------------------------------------------------------|-----|
| <a href="#">add_command.AddCommand</a>                              | 19  |
| <a href="#">bezier_dialog.BezierDialog</a>                          | 22  |
| <a href="#">boolean_operations.BooleanOperations</a>                | 26  |
| <a href="#">boolean_item.BooleanShapeItem</a>                       | 29  |
| <a href="#">turbulence_models.BoundaryConditions</a>                | 31  |
| <a href="#">boundary_conditions_dialog.BoundaryConditionsDialog</a> | 33  |
| <a href="#">turbulence_models.BoundaryConditionType</a>             | 38  |
| <a href="#">command.Command</a>                                     | 39  |
| <a href="#">command_service.CommandService</a>                      | 41  |
| <a href="#">data_parser.DataParser</a>                              | 44  |
| <a href="#">delete_command.DeleteCommand</a>                        | 46  |
| <a href="#">dialog.Dialog</a>                                       | 49  |
| <a href="#">dialog_factory.DialogFactory</a>                        | 50  |
| <a href="#">doxytest.DoxyTest</a>                                   | 51  |
| <a href="#">editable_bezier.DraggablePoint</a>                      | 52  |
| <a href="#">drawing_service.DrawingService</a>                      | 56  |
| <a href="#">edge_item.EdgeItem</a>                                  | 59  |
| <a href="#">editable_bezier.EditableBezierCurveItem</a>             | 62  |
| <a href="#">ellipse_dialog.EllipseDialog</a>                        | 66  |
| <a href="#">ellipse_item.EllipseItem</a>                            | 69  |
| <a href="#">event_handler.EventHandler</a>                          | 73  |
| <a href="#">gmsh_mesh_builder.GmshMeshBuilder</a>                   | 78  |
| <a href="#">graphics_view.GraphicsView</a>                          | 81  |
| <a href="#">grid_scene.GridScene</a>                                | 87  |
| <a href="#">turbulence_models.InitialConditions</a>                 | 92  |
| <a href="#">initial_conditions_dialog.InitialConditionsDialog</a>   | 93  |
| <a href="#">turbulence_models.InletBoundaryConditions</a>           | 97  |
| <a href="#">line_dialog.LineDialog</a>                              | 99  |
| <a href="#">line_item.LineItem</a>                                  | 102 |
| <a href="#">main_window.MainWindow</a>                              | 106 |
| <a href="#">turbulence_models.Material</a>                          | 118 |
| <a href="#">material_dialog.MaterialDialog</a>                      | 119 |
| <a href="#">mesh_dialog.MeshDialog</a>                              | 122 |
| <a href="#">move_command.MoveCommand</a>                            | 125 |
| <a href="#">mpl_canvas.MplCanvas</a>                                | 127 |

|                                                    |     |
|----------------------------------------------------|-----|
| turbulence_models.OpenBoundaryConditions           | 129 |
| parametric_curve_item.ParametricCurveItem          | 131 |
| parametric_dialog.ParametricDialog                 | 134 |
| plotter.Plotter                                    | 137 |
| rectangle_item.RectangleItem                       | 139 |
| rect_dialog.RectDialog                             | 143 |
| rotate_command.RotateCommand                       | 146 |
| selection_service.SelectionService                 | 149 |
| transformation_operations.TransformationOperations | 154 |
| turbulence_dialog.TurbulenceDialog                 | 156 |
| turbulence_models.TurbulenceModel                  | 159 |
| turbulence_models.TurbulenceParams                 | 161 |
| comsol.Ui_MainWindow                               | 162 |
| template.Ui_MainWindow                             | 166 |
| turbulence_models.WallBoundaryConditions           | 170 |
| turbulence_models.WallType                         | 172 |



## Глава 4

# Список файлов

### 4.1 Файлы

Полный список файлов.

|                                                          |     |
|----------------------------------------------------------|-----|
| modules/data/main.py                                     | 175 |
| modules/data/src/event_handler.py                        | 179 |
| modules/data/src/main_window.py                          | 179 |
| modules/data/src/commands/add_command.py                 | 175 |
| modules/data/src/commands/command.py                     | 176 |
| modules/data/src/commands/delete_command.py              | 176 |
| modules/data/src/commands/move_command.py                | 176 |
| modules/data/src/commands/rotate_command.py              | 176 |
| modules/data/src/dialogs/bezier_dialog.py                | 177 |
| modules/data/src/dialogs/boundary_conditions_dialog.py   | 177 |
| modules/data/src/dialogs/dialog.py                       | 177 |
| modules/data/src/dialogs/dialog_factory.py               | 177 |
| modules/data/src/dialogs/ellipse_dialog.py               | 177 |
| modules/data/src/dialogs/initial_conditions_dialog.py    | 178 |
| modules/data/src/dialogs/line_dialog.py                  | 178 |
| modules/data/src/dialogs/material_dialog.py              | 178 |
| modules/data/src/dialogs/mesh_dialog.py                  | 178 |
| modules/data/src/dialogs/parametric_dialog.py            | 179 |
| modules/data/src/dialogs/rect_dialog.py                  | 179 |
| modules/data/src/dialogs/turbulence_dialog.py            | 179 |
| modules/data/src/operations/boolean_operations.py        | 180 |
| modules/data/src/operations/transformation_operations.py | 180 |
| modules/data/src/physics/turbulence_models.py            | 181 |
| modules/data/src/services/command_service.py             | 181 |
| modules/data/src/services/drawing_service.py             | 181 |
| modules/data/src/services/gmsh_mesh_builder.py           | 181 |
| modules/data/src/services/selection_service.py           | 182 |
| modules/data/src/shapes/boolean_item.py                  | 182 |
| modules/data/src/shapes/ellipse_item.py                  | 182 |
| modules/data/src/shapes/line_item.py                     | 182 |
| modules/data/src/shapes/parametric_curve_item.py         | 183 |
| modules/data/src/shapes/rectangle_item.py                | 183 |
| modules/data/src/ui/template.py                          | 183 |
| modules/data/src/widgets/edge_item.py                    | 183 |
| modules/data/src/widgets/editable_bezier.py              | 184 |

|                                                             |     |
|-------------------------------------------------------------|-----|
| modules/data/src/widgets/graphics_view.py . . . . .         | 184 |
| modules/data/src/widgets/grid_scene.py . . . . .            | 184 |
| modules/doxytest/doxytest.py . . . . .                      | 185 |
| modules/visualization/main.py . . . . .                     | 175 |
| modules/visualization/src/main_window.py . . . . .          | 180 |
| modules/visualization/src/mpl_canvas.py . . . . .           | 185 |
| modules/visualization/src/services/data_parser.py . . . . . | 185 |
| modules/visualization/src/services/plotter.py . . . . .     | 185 |
| modules/visualization/src/ui/comsol.py . . . . .            | 185 |

## Глава 5

# Пространства имен

### 5.1 Пространство имен `add_command`

#### Классы

- class [AddCommand](#)

### 5.2 Пространство имен `bezier_dialog`

#### Классы

- class [BezierDialog](#)

### 5.3 Пространство имен `boolean_item`

#### Классы

- class [BooleanShapeItem](#)

### 5.4 Пространство имен `boolean_operations`

#### Классы

- class [BooleanOperations](#)

### 5.5 Пространство имен `boundary_conditions_dialog`

#### Классы

- class [BoundaryConditionsDialog](#)

## 5.6 Пространство имен command

### Классы

- class [Command](#)

## 5.7 Пространство имен command\_service

### Классы

- class [CommandService](#)

## 5.8 Пространство имен comsol

### Классы

- class [Ui\\_MainWindow](#)

## 5.9 Пространство имен data\_parser

### Классы

- class [DataParser](#)

### Переменные

- [parser](#) = [DataParser](#)(' ../data/semicolon.txt')
- [data](#) = parser.parse()
- int [c](#) = 1

### 5.9.1 Переменные

#### 5.9.1.1 c

```
int data_parser.c = 1
```

#### 5.9.1.2 data

```
data_parser.data = parser.parse()
```

#### 5.9.1.3 parser

```
data_parser.parser = DataParser(' ../data/semicolon.txt')
```

## 5.10 Пространство имен delete\_command

### Классы

- class [DeleteCommand](#)

## 5.11 Пространство имен dialog

### Классы

- class [Dialog](#)

## 5.12 Пространство имен dialog\_factory

### Классы

- class [DialogFactory](#)

## 5.13 Пространство имен doxytest

### Классы

- class [DoxyTest](#)

## 5.14 Пространство имен drawing\_service

### Классы

- class [DrawingService](#)

## 5.15 Пространство имен edge\_item

### Классы

- class [EdgeItem](#)

### Функции

- int [\\_next\\_edge\\_id](#) ()

## Переменные

- `_edge_id_counter` = `itertools.count(1)`
- `p` = `QPainterPath()`
- `e` = `EdgeItem(p)`
- `p1` = `QPointF(0.0, 0.0)`
- `p2` = `QPointF(0.0, 0.0)`

### 5.15.1 Функции

#### 5.15.1.1 `_next_edge_id()`

```
int edge_item._next_edge_id () [protected]
```

### 5.15.2 Переменные

#### 5.15.2.1 `_edge_id_counter`

```
edge_item._edge_id_counter = itertools.count(1) [protected]
```

#### 5.15.2.2 `e`

```
edge_item.e = EdgeItem(p)
```

#### 5.15.2.3 `p`

```
edge_item.p = QPainterPath()
```

#### 5.15.2.4 `p1`

```
edge_item.p1 = QPointF(0.0, 0.0)
```

#### 5.15.2.5 `p2`

```
edge_item.p2 = QPointF(0.0, 0.0)
```

## 5.16 Пространство имен `editable_bezier`

### Классы

- class `DraggablePoint`
- class `EditableBezierCurveItem`

## 5.17 Пространство имен ellipse\_dialog

### Классы

- class [EllipseDialog](#)

## 5.18 Пространство имен ellipse\_item

### Классы

- class [EllipseItem](#)

### Переменные

- `el = EllipseItem(QRectF(0, 0, 100, 100))`

### 5.18.1 Переменные

#### 5.18.1.1 el

`ellipse_item.el = EllipseItem(QRectF(0, 0, 100, 100))`

## 5.19 Пространство имен event\_handler

### Классы

- class [EventHandler](#)

## 5.20 Пространство имен gmsh\_mesh\_builder

### Классы

- class [GmshMeshBuilder](#)

## 5.21 Пространство имен graphics\_view

### Классы

- class [GraphicsView](#)

## 5.22 Пространство имен grid\_scene

### Классы

- class [GridScene](#)

## 5.23 Пространство имен initial\_conditions\_dialog

### Классы

- class [InitialConditionsDialog](#)

## 5.24 Пространство имен line\_dialog

### Классы

- class [LineDialog](#)

## 5.25 Пространство имен line\_item

### Классы

- class [LineItem](#)

## 5.26 Пространство имен main

### Переменные

- [level](#)
- [DEBUG](#)
- [format](#)
- [stream](#)
- [app](#) = QApplication(sys.argv)
- [window](#) = MainWindow()
- dict [font](#)

### 5.26.1 Переменные

#### 5.26.1.1 app

```
main.app = QApplication(sys.argv)
```



#### 5.26.1.2 DEBUG

main.DEBUG

#### 5.26.1.3 font

dict main.font

Инициализатор

```
00001 = {  
00002     'family': 'normal',  
00003     'size': 20  
00004 }
```

#### 5.26.1.4 format

main.format

#### 5.26.1.5 level

main.level

#### 5.26.1.6 stream

main.stream

#### 5.26.1.7 window

main.window = MainWindow()

### 5.27 Пространство имен main\_window

Классы

- class [MainWindow](#)

Переменные

- [app](#) = QApplication([])
- [window](#) = [MainWindow](#)()

#### 5.27.1 Переменные

##### 5.27.1.1 app

main\_window.app = QApplication([])

#### 5.27.1.2 window

```
main_window.window = MainWindow()
```

### 5.28 Пространство имен material\_dialog

#### Классы

- class [MaterialDialog](#)

### 5.29 Пространство имен mesh\_dialog

#### Классы

- class [MeshDialog](#)

### 5.30 Пространство имен move\_command

#### Классы

- class [MoveCommand](#)

### 5.31 Пространство имен mpl\_canvas

#### Классы

- class [MplCanvas](#)

### 5.32 Пространство имен parametric\_curve\_item

#### Классы

- class [ParametricCurveItem](#)

### 5.33 Пространство имен parametric\_dialog

#### Классы

- class [ParametricDialog](#)

## 5.34 Пространство имен `plotter`

### Классы

- class [Plotter](#)

## 5.35 Пространство имен `rect_dialog`

### Классы

- class [RectDialog](#)

## 5.36 Пространство имен `rectangle_item`

### Классы

- class [RectangleItem](#)

## 5.37 Пространство имен `rotate_command`

### Классы

- class [RotateCommand](#)

## 5.38 Пространство имен `selection_service`

### Классы

- class [SelectionService](#)

## 5.39 Пространство имен `template`

### Классы

- class [Ui\\_MainWindow](#)

## 5.40 Пространство имен `transformation_operations`

### Классы

- class [TransformationOperations](#)

## 5.41 Пространство имен turbulence\_dialog

### Классы

- class [TurbulenceDialog](#)

## 5.42 Пространство имен turbulence\_models

### Классы

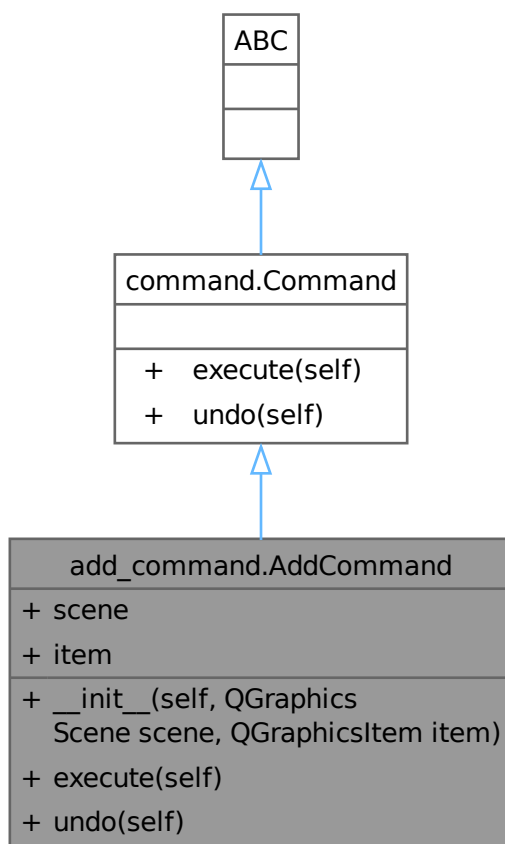
- class [BoundaryConditions](#)
- class [BoundaryConditionType](#)
- class [InitialConditions](#)
- class [InletBoundaryConditions](#)
- class [Material](#)
- class [OpenBoundaryConditions](#)
- class [TurbulenceModel](#)
- class [TurbulenceParams](#)
- class [WallBoundaryConditions](#)
- class [WallType](#)

## Глава 6

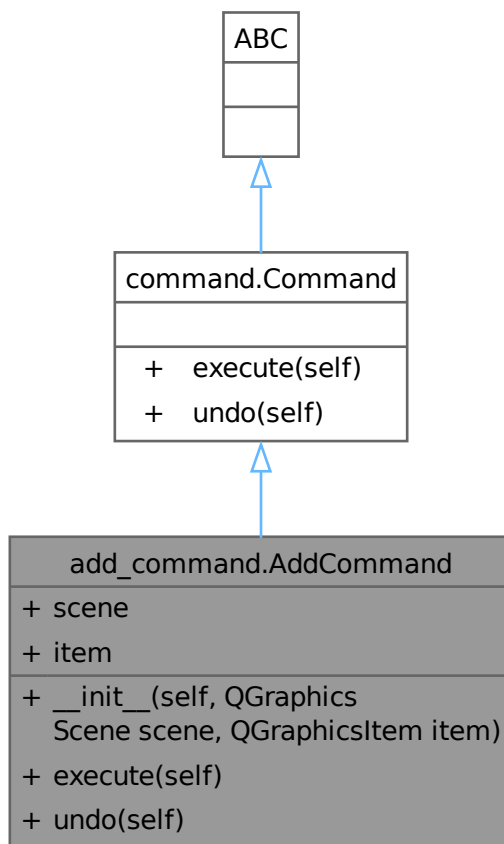
# Классы

### 6.1 Класс `add_command.AddCommand`

Граф наследования:`add_command.AddCommand`:



Граф связей класса `add_command.AddCommand`:



Открытые члены

- `__init__` (self, QGraphicsScene `scene`, QGraphicsItem `item`)
- `execute` (self)
- `undo` (self)

Открытые атрибуты

- `scene` = scene
- `item` = item

## 6.1.1 Конструктор(ы)

### 6.1.1.1 `__init__`()

```

add_command.AddCommand.__init__ (
    self,
    QGraphicsScene scene,
    QGraphicsItem item)
  
```

## 6.1.2 Методы

### 6.1.2.1 `execute()`

```
add_command.AddCommand.execute (  
    self)
```

Переопределяет метод предка [command.Command](#).

### 6.1.2.2 `undo()`

```
add_command.AddCommand.undo (  
    self)
```

Переопределяет метод предка [command.Command](#).

## 6.1.3 Данные класса

### 6.1.3.1 `item`

```
add_command.AddCommand.item = item
```

### 6.1.3.2 `scene`

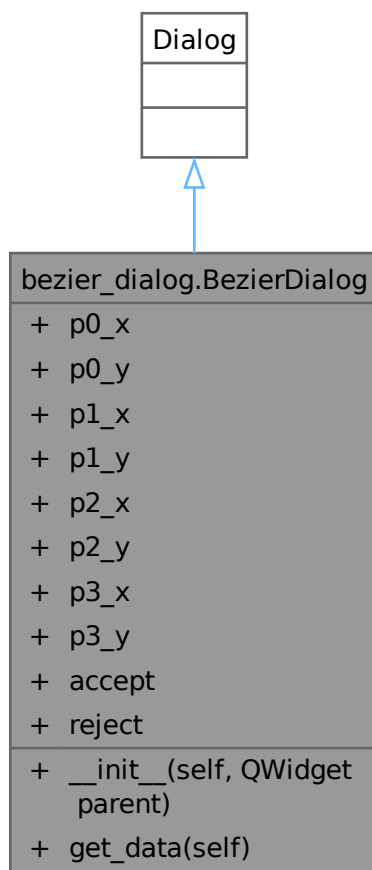
```
add_command.AddCommand.scene = scene
```

Объявления и описания членов класса находятся в файле:

- `modules/data/src/commands/add\_command.py`

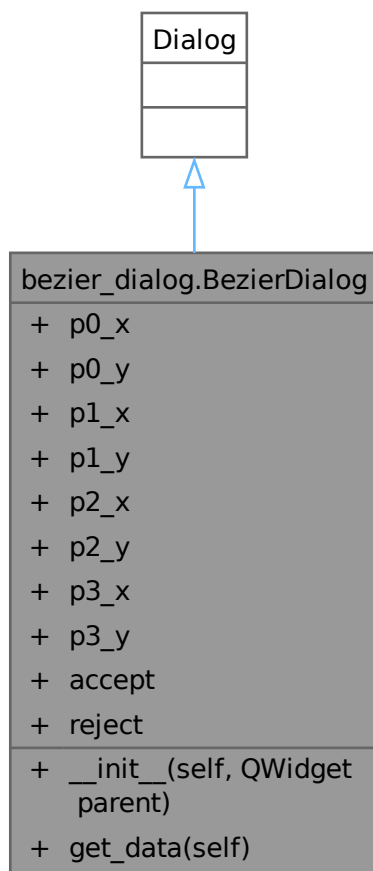
## 6.2 Класс bezier\_dialog.BezierDialog

Граф наследования:bezier\_dialog.BezierDialog:





Граф связей класса bezier\_dialog.BezierDialog:



Открытые члены

- `__init__` (self, QWidget parent)
- `get_data` (self)

Открытые атрибуты

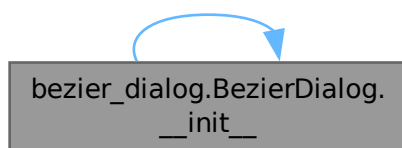
- `p0_x` = QLineEdit('0')
- `p0_y` = QLineEdit('0')
- `p1_x` = QLineEdit('50')
- `p1_y` = QLineEdit('100')
- `p2_x` = QLineEdit('150')
- `p2_y` = QLineEdit('100')
- `p3_x` = QLineEdit('200')
- `p3_y` = QLineEdit('0')
- `accept`
- `reject`

## 6.2.1 Конструктор(ы)

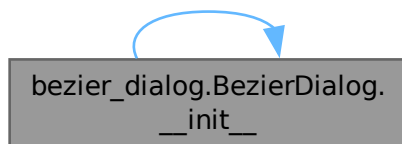
### 6.2.1.1 `__init__()`

```
bezier_dialog.BezierDialog.__init__ (  
    self,  
    QWidget parent)
```

Граф вызовов:



Граф вызова функции:



## 6.2.2 Методы

### 6.2.2.1 `get_data()`

```
bezier_dialog.BezierDialog.get_data (  
    self)
```

## 6.2.3 Данные класса

### 6.2.3.1 `accept`

```
bezier_dialog.BezierDialog.accept
```

## 6.2.3.2 p0\_x

```
bezier_dialog.BezierDialog.p0_x = QLineEdit('0')
```

## 6.2.3.3 p0\_y

```
bezier_dialog.BezierDialog.p0_y = QLineEdit('0')
```

## 6.2.3.4 p1\_x

```
bezier_dialog.BezierDialog.p1_x = QLineEdit('50')
```

## 6.2.3.5 p1\_y

```
bezier_dialog.BezierDialog.p1_y = QLineEdit('100')
```

## 6.2.3.6 p2\_x

```
bezier_dialog.BezierDialog.p2_x = QLineEdit('150')
```

## 6.2.3.7 p2\_y

```
bezier_dialog.BezierDialog.p2_y = QLineEdit('100')
```

## 6.2.3.8 p3\_x

```
bezier_dialog.BezierDialog.p3_x = QLineEdit('200')
```

## 6.2.3.9 p3\_y

```
bezier_dialog.BezierDialog.p3_y = QLineEdit('0')
```

## 6.2.3.10 reject

```
bezier_dialog.BezierDialog.reject
```

Объявления и описания членов класса находятся в файле:

- `modules/data/src/dialogs/bezier_dialog.py`

### 6.3 Класс `boolean_operations.BooleanOperations`

Граф связей класса `boolean_operations.BooleanOperations`:

| <code>boolean_operations.BooleanOperations</code>                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>+ <code>parent</code></li> <li>+ <code>scene</code></li> <li>+ <code>command_service</code></li> <li>+ <code>drawing_service</code></li> <li>+ <code>selection_service</code></li> </ul>                                                                                                                                      |
| <ul style="list-style-type: none"> <li>+ <code>__init__(self, QWidget parent, GridScene scene, CommandService command_service, DrawingService drawing_service, SelectionService selection_service)</code></li> <li>+ <code>perform_union(self)</code></li> <li>+ <code>perform_difference(self)</code></li> <li>+ <code>perform_intersection(self)</code></li> </ul> |

Открытые члены

- `__init__` (self, QWidget `parent`, GridScene `scene`, CommandService `command_service`, DrawingService `drawing_service`, SelectionService `selection_service`)
- `perform_union` (self)
- `perform_difference` (self)
- `perform_intersection` (self)

Открытые атрибуты

- `parent` = parent
- `scene` = scene
- `command_service` = command\_service
- `drawing_service` = drawing\_service
- `selection_service` = selection\_service

### 6.3.1 Конструктор(ы)

#### 6.3.1.1 `__init__()`

```
boolean_operations.BooleanOperations.__init__ (
    self,
    QWidget parent,
    GridScene scene,
    CommandService command_service,
    DrawingService drawing_service,
    SelectionService selection_service)
```

### 6.3.2 Методы

#### 6.3.2.1 `perform_difference()`

```
boolean_operations.BooleanOperations.perform_difference (
    self)
```

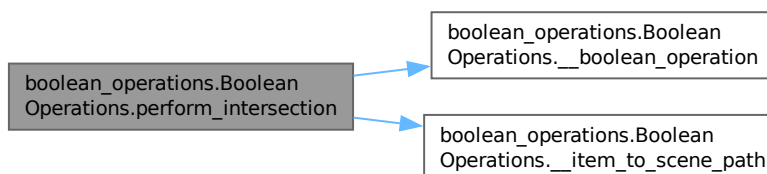
Граф вызовов:



#### 6.3.2.2 `perform_intersection()`

```
boolean_operations.BooleanOperations.perform_intersection (
    self)
```

Граф вызовов:



### 6.3.2.3 perform\_union()

```
boolean_operations.BooleanOperations.perform_union (  
    self)
```

Граф вызовов:



## 6.3.3 Данные класса

### 6.3.3.1 command\_service

```
boolean_operations.BooleanOperations.command_service = command_service
```

### 6.3.3.2 drawing\_service

```
boolean_operations.BooleanOperations.drawing_service = drawing_service
```

### 6.3.3.3 parent

```
boolean_operations.BooleanOperations.parent = parent
```

### 6.3.3.4 scene

```
boolean_operations.BooleanOperations.scene = scene
```

### 6.3.3.5 selection\_service

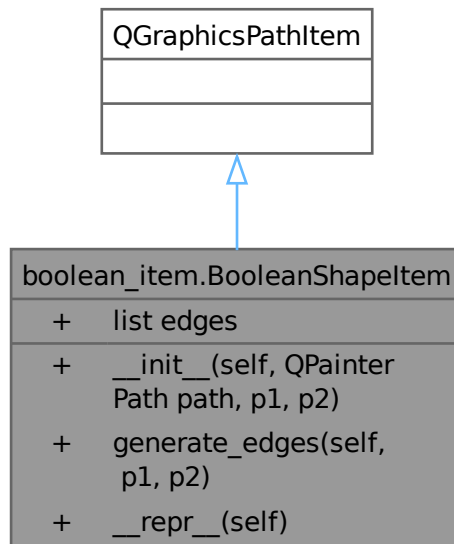
```
boolean_operations.BooleanOperations.selection_service = selection_service
```

Объявления и описания членов класса находятся в файле:

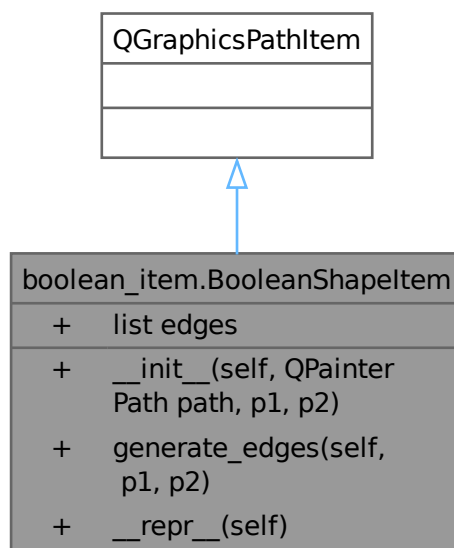
- `modules/data/src/operations/boolean\_operations.py`

## 6.4 Класс `boolean_item.BooleanShapeItem`

Граф наследования: `boolean_item.BooleanShapeItem`:



Граф связей класса `boolean_item.BooleanShapeItem`:



## Открытые члены

- `__init__` (self, QPainterPath path, p1, p2)
- `generate_edges` (self, p1, p2)
- `__repr__` (self)

## Открытые атрибуты

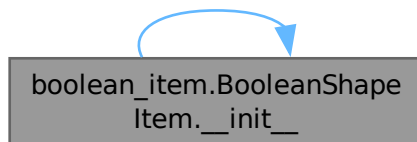
- list `edges` = []

## 6.4.1 Конструктор(ы)

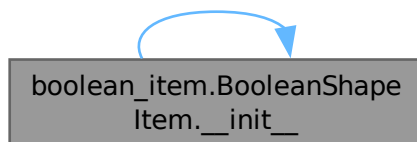
6.4.1.1 `__init__`()

```
boolean_item.BooleanShapeItem.__init__ (
    self,
    QPainterPath path,
    p1,
    p2)
```

Граф вызовов:



Граф вызова функции:





## 6.4.2 Методы

### 6.4.2.1 `__repr__()`

```
boolean_item.BooleanShapeItem.__repr__ (
    self)
```

### 6.4.2.2 `generate_edges()`

```
boolean_item.BooleanShapeItem.generate_edges (
    self,
    p1,
    p2)
```

## 6.4.3 Данные класса

### 6.4.3.1 `edges`

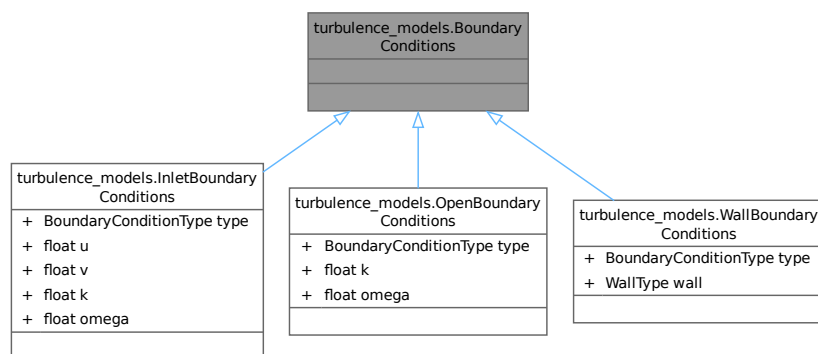
```
list boolean_item.BooleanShapeItem.edges = []
```

Объявления и описания членов класса находятся в файле:

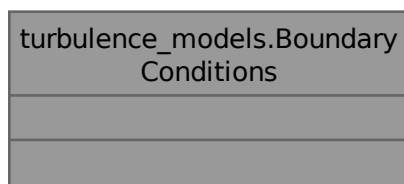
- `modules/data/src/shapes/boolean_item.py`

## 6.5 Класс `turbulence_models.BoundaryConditions`

Граф наследования:`turbulence_models.BoundaryConditions`:



Граф связей класса `turbulence_models.BoundaryConditions`:

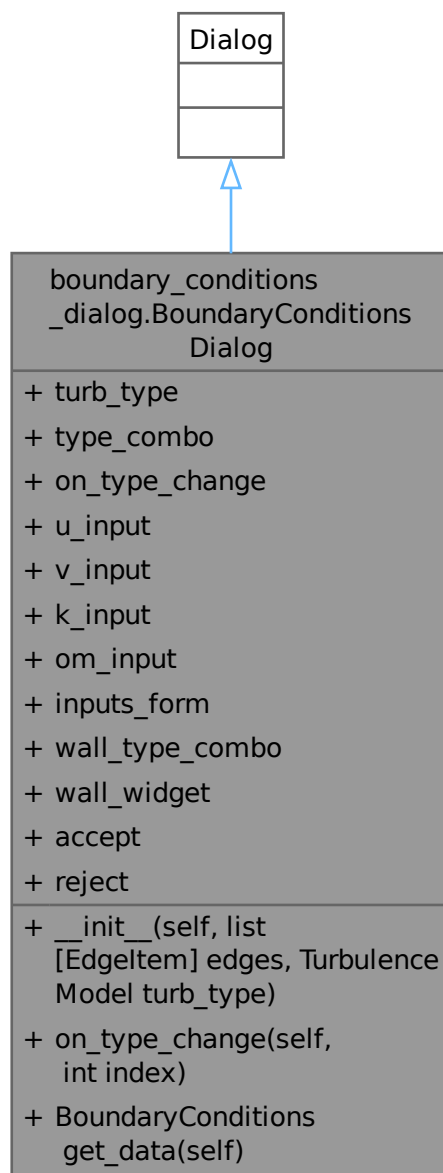


Объявления и описания членов класса находятся в файле:

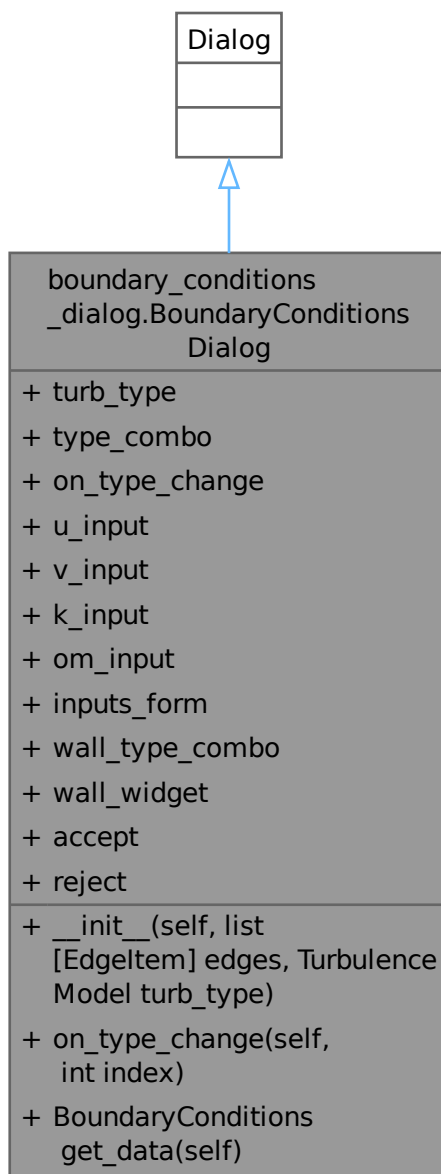
- `modules/data/src/physics/turbulence_models.py`

## 6.6 Класс boundary\_conditions\_dialog.BoundaryConditionsDialog

Граф наследования:boundary\_conditions\_dialog.BoundaryConditionsDialog:



Граф связей класса `boundary_conditions_dialog.BoundaryConditionsDialog`:



#### Открытые члены

- `__init__` (self, list[EdgeItem] edges, TurbulenceModel `turb_type`)
- `on_type_change` (self, int index)
- `BoundaryConditions.get_data` (self)

#### Открытые атрибуты

- `turb_type` = `turb_type`

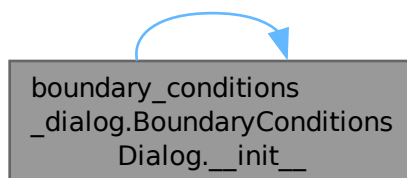
- `type_combo` = `QComboBox()`
- `on_type_change`
- `u_input` = `QLineEdit()`
- `v_input` = `QLineEdit()`
- `k_input` = `QDoubleSpinBox()`
- `om_input` = `QDoubleSpinBox()`
- `inputs_form` = `QFormLayout()`
- `wall_type_combo` = `QComboBox()`
- `wall_widget` = `QWidget()`
- `accept`
- `reject`

### 6.6.1 Конструктор(ы)

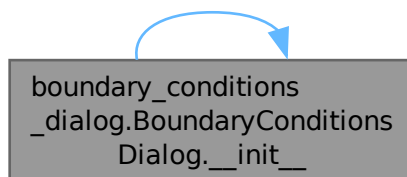
#### 6.6.1.1 `__init__()`

```
boundary_conditions_dialog.BoundaryConditionsDialog.__init__(
    self,
    list[EdgeItem] edges,
    TurbulenceModel turb_type)
```

Граф вызовов:



Граф вызова функции:



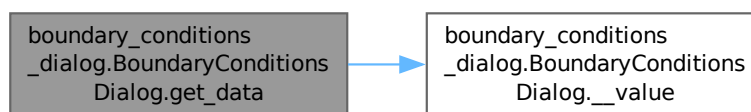
## 6.6.2 Методы

### 6.6.2.1 get\_data()

```
BoundaryConditions boundary_conditions_dialog.BoundaryConditionsDialog.get_data (
    self)
```

Возвращает конкретный объект условий.

Граф вызовов:



### 6.6.2.2 on\_type\_change()

```
boundary_conditions_dialog.BoundaryConditionsDialog.on_type_change (
    self,
    int index)
```

В зависимости от выбранного типа показываем нужные поля.

## 6.6.3 Данные класса

### 6.6.3.1 accept

```
boundary_conditions_dialog.BoundaryConditionsDialog.accept
```

### 6.6.3.2 inputs\_form

```
boundary_conditions_dialog.BoundaryConditionsDialog.inputs_form = QFormLayout()
```

### 6.6.3.3 k\_input

```
boundary_conditions_dialog.BoundaryConditionsDialog.k_input = QDoubleSpinBox()
```

#### 6.6.3.4 om\_input

```
boundary_conditions_dialog.BoundaryConditionsDialog.om_input = QDoubleSpinBox()
```

#### 6.6.3.5 on\_type\_change

```
boundary_conditions_dialog.BoundaryConditionsDialog.on_type_change
```

#### 6.6.3.6 reject

```
boundary_conditions_dialog.BoundaryConditionsDialog.reject
```

#### 6.6.3.7 turb\_type

```
boundary_conditions_dialog.BoundaryConditionsDialog.turb_type = turb_type
```

#### 6.6.3.8 type\_combo

```
boundary_conditions_dialog.BoundaryConditionsDialog.type_combo = QComboBox()
```

#### 6.6.3.9 u\_input

```
boundary_conditions_dialog.BoundaryConditionsDialog.u_input = QLineEdit()
```

#### 6.6.3.10 v\_input

```
boundary_conditions_dialog.BoundaryConditionsDialog.v_input = QLineEdit()
```

#### 6.6.3.11 wall\_type\_combo

```
boundary_conditions_dialog.BoundaryConditionsDialog.wall_type_combo = QComboBox()
```

#### 6.6.3.12 wall\_widget

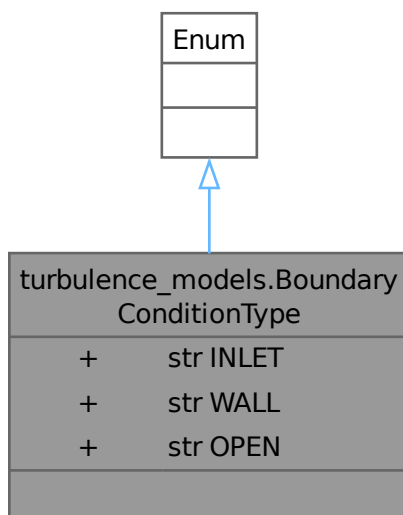
```
boundary_conditions_dialog.BoundaryConditionsDialog.wall_widget = QWidget()
```

Объявления и описания членов класса находятся в файле:

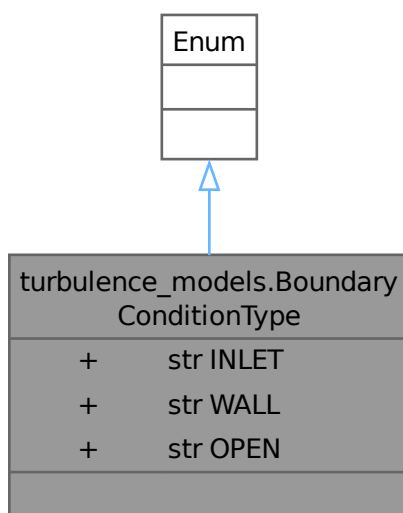
- `modules/data/src/dialogs/boundary_conditions_dialog.py`

## 6.7 Класс turbulence\_models.BoundaryConditionType

Граф наследования:turbulence\_models.BoundaryConditionType:



Граф связей класса turbulence\_models.BoundaryConditionType:





Статические открытые данные

- `str INLET = 'inlet'`
- `str WALL = 'wall'`
- `str OPEN = 'open boundary'`

### 6.7.1 Данные класса

#### 6.7.1.1 INLET

```
str turbulence_models.BoundaryConditionType.INLET = 'inlet' [static]
```

#### 6.7.1.2 OPEN

```
str turbulence_models.BoundaryConditionType.OPEN = 'open boundary' [static]
```

#### 6.7.1.3 WALL

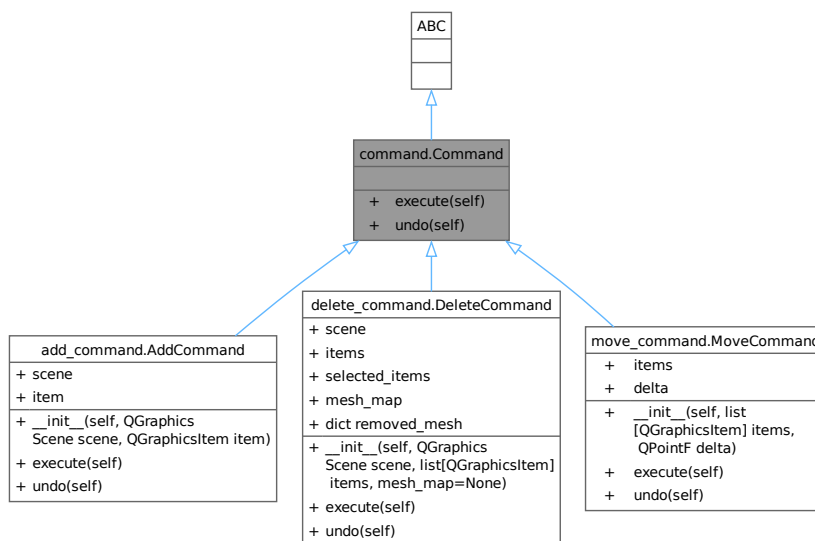
```
str turbulence_models.BoundaryConditionType.WALL = 'wall' [static]
```

Объявления и описания членов класса находятся в файле:

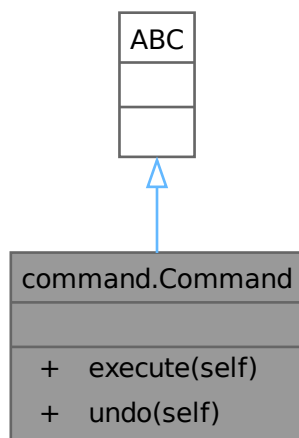
- `modules/data/src/physics/turbulence_models.py`

## 6.8 Класс `command.Command`

Граф наследования:`command.Command`:



Граф связей класса `command.Command`:



Открытые члены

- `execute` (self)
- `undo` (self)

## 6.8.1 Методы

### 6.8.1.1 `execute()`

```
command.Command.execute (  
    self)
```

Переопределяется в [add\\_command.AddCommand](#), [delete\\_command.DeleteCommand](#) и [move\\_command.MoveCommand](#)

### 6.8.1.2 `undo()`

```
command.Command.undo (  
    self)
```

Переопределяется в [add\\_command.AddCommand](#), [delete\\_command.DeleteCommand](#) и [move\\_command.MoveCommand](#)

Объявления и описания членов класса находятся в файле:

- `modules/data/src/commands/command.py`

## 6.9 Класс `command_service.CommandService`

Граф связей класса `command_service.CommandService`:

| <code>command_service.CommandService</code> |                                             |
|---------------------------------------------|---------------------------------------------|
| +                                           | list <code>undo_stack</code>                |
| +                                           | list <code>redo_stack</code>                |
| +                                           | <code>__init__(self)</code>                 |
| +                                           | <code>append(self, Command command)</code>  |
| +                                           | <code>execute(self, Command command)</code> |
| +                                           | <code>undo(self)</code>                     |
| +                                           | <code>redo(self)</code>                     |

Открытые члены

- `__init__` (self)
- `append` (self, Command command)
- `execute` (self, Command command)
- `undo` (self)
- `redo` (self)

Открытые атрибуты

- list `undo_stack` = []
- list `redo_stack` = []

### 6.9.1 Конструктор(ы)

#### 6.9.1.1 `__init__`()

```
command_service.CommandService.__init__ (
    self)
```

## 6.9.2 Методы

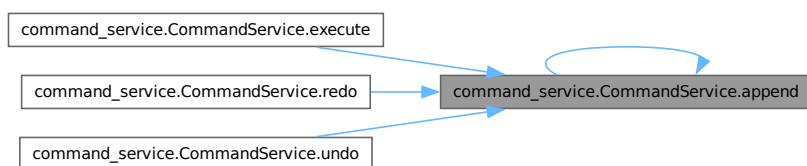
### 6.9.2.1 append()

```
command_service.CommandService.append (  
    self,  
    Command command)
```

Граф вызовов:



Граф вызова функции:



### 6.9.2.2 execute()

```
command_service.CommandService.execute (  
    self,  
    Command command)
```

Граф вызовов:



### 6.9.2.3 `redo()`

```
command_service.CommandService.redo (  
    self)
```

Граф вызовов:



### 6.9.2.4 `undo()`

```
command_service.CommandService.undo (  
    self)
```

Граф вызовов:



## 6.9.3 Данные класса

### 6.9.3.1 `redo_stack`

```
list command_service.CommandService.redo_stack = []
```

### 6.9.3.2 `undo_stack`

```
list command_service.CommandService.undo_stack = []
```

Объявления и описания членов класса находятся в файле:

- `modules/data/src/services/command_service.py`

## 6.10 Класс data\_parser.DataParser

Граф связей класса data\_parser.DataParser:

| data_parser.DataParser                             |
|----------------------------------------------------|
| + str file_path                                    |
| + str separator                                    |
| + __init__(self, str file_path, str separator=';') |
| + DataFrame parse(self)                            |

Открытые члены

- `__init__` (self, str `file_path`, str `separator`=';')
- DataFrame `parse` (self)

Открытые атрибуты

- str `file_path` = file\_path
- str `separator` = separator

### 6.10.1 Конструктор(ы)

#### 6.10.1.1 `__init__`()

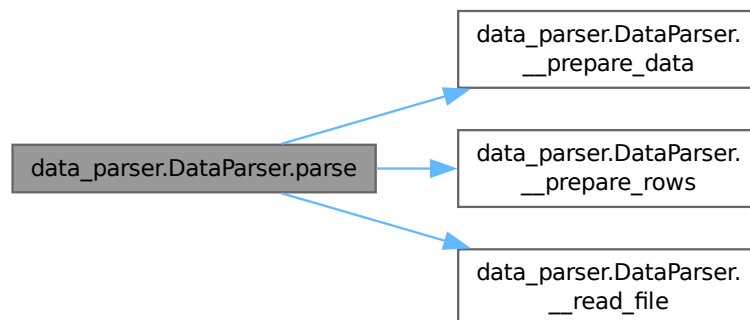
```
data_parser.DataParser.__init__ (
    self,
    str file_path,
    str separator = ';' )
```

### 6.10.2 Методы

#### 6.10.2.1 `parse`()

```
DataFrame data_parser.DataParser.parse (
    self)
```

Граф вызовов:



### 6.10.3 Данные класса

#### 6.10.3.1 file\_path

```
data_parser.DataParser.file_path = file_path
```

#### 6.10.3.2 separator

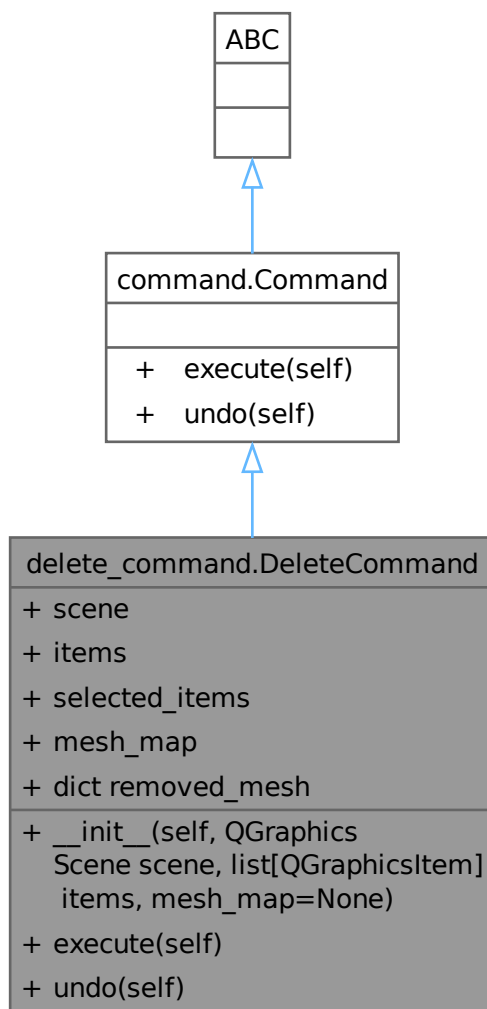
```
str data_parser.DataParser.separator = separator
```

Объявления и описания членов класса находятся в файле:

- `modules/visualization/src/services/data\_parser.py`

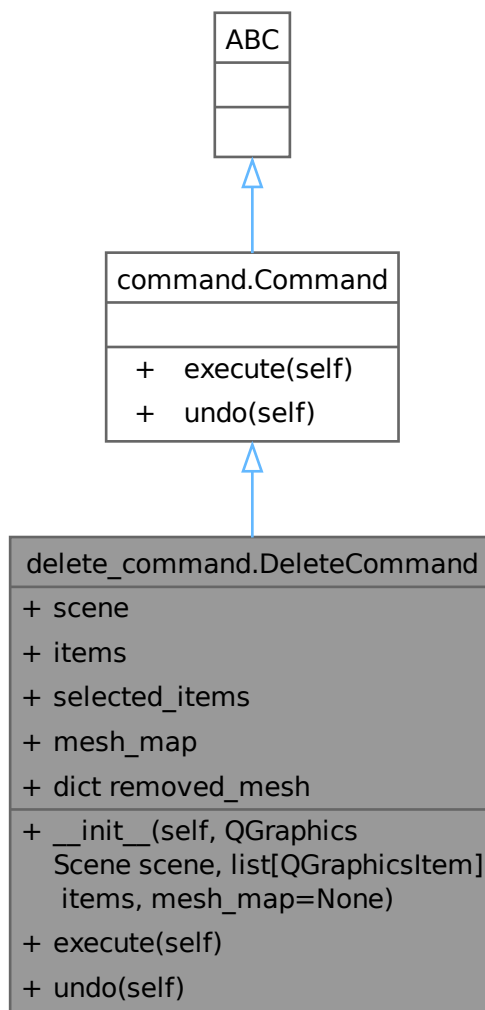
## 6.11 Класс delete\_command.DeleteCommand

Граф наследования:delete\_command.DeleteCommand:





Граф связей класса `delete_command.DeleteCommand`:



Открытые члены

- `__init__` (self, QGraphicsScene `scene`, list[QGraphicsItem] `items`, `mesh_map`=None)
- `execute` (self)
- `undo` (self)

Открытые атрибуты

- `scene` = scene
- `items` = items
- `selected_items` = set(`items`)
- `mesh_map` = mesh\_map or {}
- dict `removed_mesh` = {}

### 6.11.1 Конструктор(ы)

#### 6.11.1.1 `__init__()`

```
delete_command.DeleteCommand.__init__ (
    self,
    QGraphicsScene scene,
    list[QGraphicsItem] items,
    mesh_map = None)
```

### 6.11.2 Методы

#### 6.11.2.1 `execute()`

```
delete_command.DeleteCommand.execute (
    self)
```

Переопределяет метод предка [command.Command](#).

#### 6.11.2.2 `undo()`

```
delete_command.DeleteCommand.undo (
    self)
```

Переопределяет метод предка [command.Command](#).

### 6.11.3 Данные класса

#### 6.11.3.1 `items`

```
delete_command.DeleteCommand.items = items
```

#### 6.11.3.2 `mesh_map`

```
delete_command.DeleteCommand.mesh_map = mesh_map or {}
```

#### 6.11.3.3 `removed_mesh`

```
dict delete_command.DeleteCommand.removed_mesh = {}
```

#### 6.11.3.4 `scene`

```
delete_command.DeleteCommand.scene = scene
```

## 6.11.3.5 selected\_items

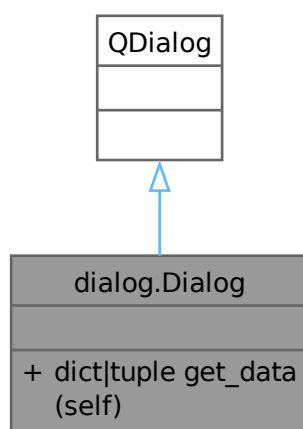
```
delete_command.DeleteCommand.selected_items = set(items)
```

Объявления и описания членов класса находятся в файле:

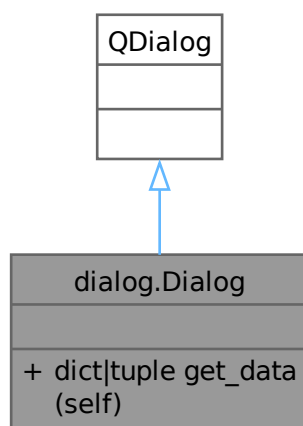
- `modules/data/src/commands/delete_command.py`

## 6.12 Класс dialog.Dialog

Граф наследования:dialog.Dialog:



Граф связей класса dialog.Dialog:



Открытые члены

- dict|tuple [get\\_data](#) (self)

### 6.12.1 Методы

#### 6.12.1.1 [get\\_data](#)()

dict | tuple `dialog.Dialog.get_data (self)`

Объявления и описания членов класса находятся в файле:

- `modules/data/src/dialogs/dialog.py`

## 6.13 Класс `dialog_factory.DialogFactory`

Граф связей класса `dialog_factory.DialogFactory`:

| <code>dialog_factory.DialogFactory</code>                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>+ Dialog None <code>create_dialog(str dialog_type, QWidget parent)</code></li> </ul> |

Открытые статические члены

- Dialog|None [create\\_dialog](#) (str dialog\_type, QWidget parent)

### 6.13.1 Методы

#### 6.13.1.1 `create_dialog`()

Dialog | None `dialog_factory.DialogFactory.create_dialog (str dialog_type, QWidget parent) [static]`

Объявления и описания членов класса находятся в файле:

- `modules/data/src/dialogs/dialog\_factory.py`

## 6.14 Класс doxytest.DoxyTest

Граф связей класса doxytest.DoxyTest:

| doxytest.DoxyTest                                                                                                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>+ None <code>__init__(self)</code></li> <li>+ str None <code>test(self, str timestamp, str priority, str priority_name, str message)</code></li> </ul> |

### Открытые члены

- None `__init__(self)`
- str|None `test(self, str timestamp, str priority, str priority_name, str message)`

### 6.14.1 Подробное описание

Класс для документации.

### 6.14.2 Конструктор(ы)

#### 6.14.2.1 `__init__()`

```
None doxytest.DoxyTest.__init__(
    self)
```

### 6.14.3 Методы

#### 6.14.3.1 `test()`

```
str | None doxytest.DoxyTest.test (
    self,
    str timestamp,
    str priority,
    str priority_name,
    str message)
```

Комментарий для документации.

Указаны параметры и тип возвр значения.

@param a Первое число

@param b Второе число

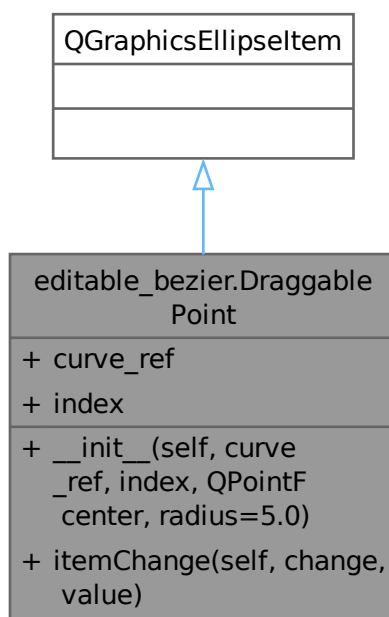
@return Сумма a и b

Объявления и описания членов класса находятся в файле:

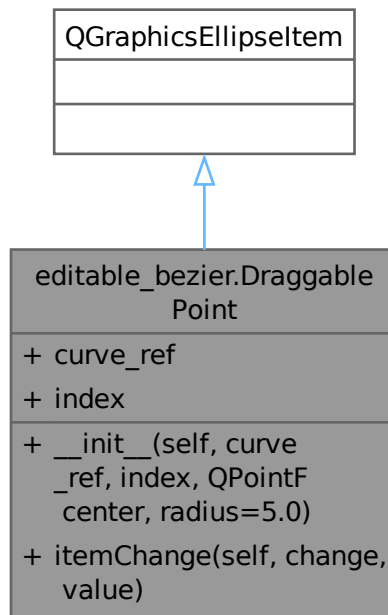
- `modules/doxytest/doxytest.py`

## 6.15 Класс `editable_bezier.DraggablePoint`

Граф наследования:`editable_bezier.DraggablePoint`:



Граф связей класса `editable_bezier.DraggablePoint`:



Открытые члены

- `__init__` (self, `curve_ref`, `index`, `QPointF center`, `radius=5.0`)
- `itemChange` (self, `change`, `value`)

Открытые атрибуты

- `curve_ref` = `curve_ref`
- `index` = `index`

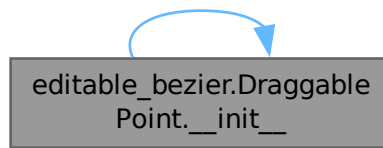
### 6.15.1 Конструктор(ы)

#### 6.15.1.1 `__init__()`

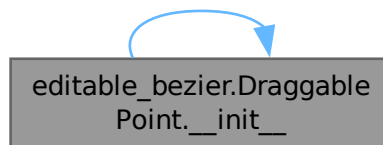
```

editable_bezier.DraggablePoint.__init__ (
    self,
    curve_ref,
    index,
    QPointF center,
    radius = 5.0)
  
```

Граф вызовов:



Граф вызова функции:

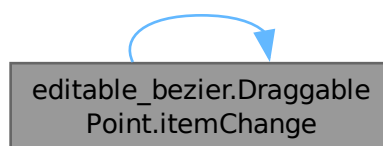


## 6.15.2 Методы

### 6.15.2.1 itemChange()

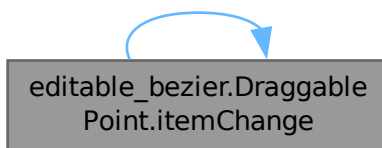
```
editable_bezier.DraggablePoint.itemChange (  
    self,  
    change,  
    value)
```

Граф вызовов:





Граф вызова функции:



### 6.15.3 Данные класса

#### 6.15.3.1 `curve_ref`

```
editable_bezier.DraggablePoint.curve_ref = curve_ref
```

#### 6.15.3.2 `index`

```
editable_bezier.DraggablePoint.index = index
```

Объявления и описания членов класса находятся в файле:

- `modules/data/src/widgets/editable_bezier.py`

## 6.16 Класс `drawing_service.DrawingService`

Граф связей класса `drawing_service.DrawingService`:

| drawing_service.DrawingService                                                                                                                                                                                                                                                   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| + scene<br>+ parent<br>+ command_service<br>+ selection_service<br>+ default_pen                                                                                                                                                                                                 |
| + __init__(self, QWidget parent, GridScene scene, CommandService command_service, SelectionService selection_service)<br>+ draw_line_by_params(self)<br>+ draw_rect_by_params(self)<br>+ draw_ellipse_by_params(self)<br>+ draw_curve_by_params(self)<br>+ draw_parametric(self) |

Открытые члены

- `__init__` (self, QWidget `parent`, GridScene `scene`, CommandService `command_service`, SelectionService `selection_service`)
- `draw_line_by_params` (self)
- `draw_rect_by_params` (self)
- `draw_ellipse_by_params` (self)
- `draw_curve_by_params` (self)
- `draw_parametric` (self)

Открытые атрибуты

- `scene` = scene
- `parent` = parent
- `command_service` = command\_service
- `selection_service` = selection\_service
- `default_pen` = QPen(Qt.black, 0)

### 6.16.1 Конструктор(ы)

#### 6.16.1.1 \_\_init\_\_()

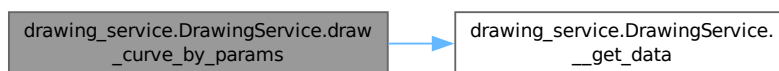
```
drawing_service.DrawingService.__init__ (
    self,
    QWidget parent,
    GridScene scene,
    CommandService command_service,
    SelectionService selection_service)
```

### 6.16.2 Методы

#### 6.16.2.1 draw\_curve\_by\_params()

```
drawing_service.DrawingService.draw_curve_by_params (
    self)
```

Граф вызовов:



#### 6.16.2.2 draw\_ellipse\_by\_params()

```
drawing_service.DrawingService.draw_ellipse_by_params (
    self)
```

Граф вызовов:



### 6.16.2.3 draw\_line\_by\_params()

```
drawing_service.DrawingService.draw_line_by_params (  
    self)
```

Граф вызовов:



### 6.16.2.4 draw\_parametric()

```
drawing_service.DrawingService.draw_parametric (  
    self)
```

Граф вызовов:



### 6.16.2.5 draw\_rect\_by\_params()

```
drawing_service.DrawingService.draw_rect_by_params (  
    self)
```

Граф вызовов:



## 6.16.3 Данные класса

### 6.16.3.1 command\_service

```
drawing_service.DrawingService.command_service = command_service
```

6.16.3.2 `default_pen`

```
drawing_service.DrawingService.default_pen = QPen(Qt.black, 0)
```

6.16.3.3 `parent`

```
drawing_service.DrawingService.parent = parent
```

6.16.3.4 `scene`

```
drawing_service.DrawingService.scene = scene
```

6.16.3.5 `selection_service`

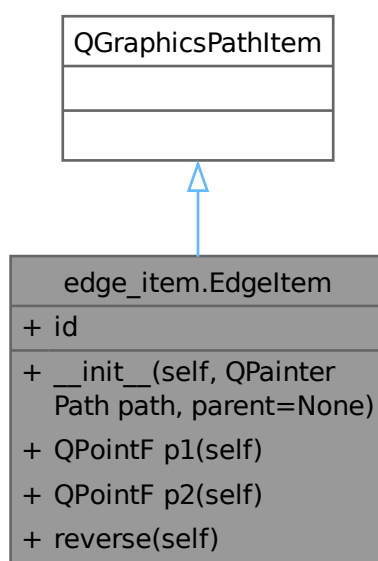
```
drawing_service.DrawingService.selection_service = selection_service
```

Объявления и описания членов класса находятся в файле:

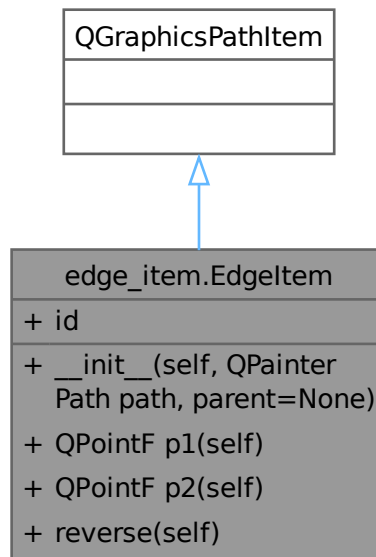
- `modules/data/src/services/drawing_service.py`

6.17 Класс `edge_item.EdgeItem`

Граф наследования: `edge_item.EdgeItem`:



Граф связей класса `edge_item.EdgeItem`:



Открытые члены

- `__init__` (self, QPainterPath path, parent=None)
- `QPointF p1` (self)
- `QPointF p2` (self)
- `reverse` (self)

Открытые атрибуты

- `id = _next_edge_id()`

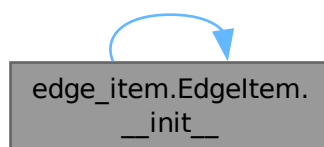
### 6.17.1 Конструктор(ы)

#### 6.17.1.1 `__init__()`

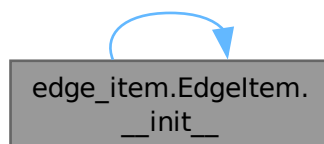
```

edge_item.EdgeItem.__init__(
    self,
    QPainterPath path,
    parent = None)
  
```

Граф вызовов:



Граф вызова функции:



## 6.17.2 Методы

### 6.17.2.1 p1()

```
QPointF edge_item.EdgeItem.p1 (  
    self)
```

### 6.17.2.2 p2()

```
QPointF edge_item.EdgeItem.p2 (  
    self)
```

### 6.17.2.3 reverse()

```
edge_item.EdgeItem.reverse (  
    self)
```

Поменять направление кривой на обратное.

### 6.17.3 Данные класса

#### 6.17.3.1 id

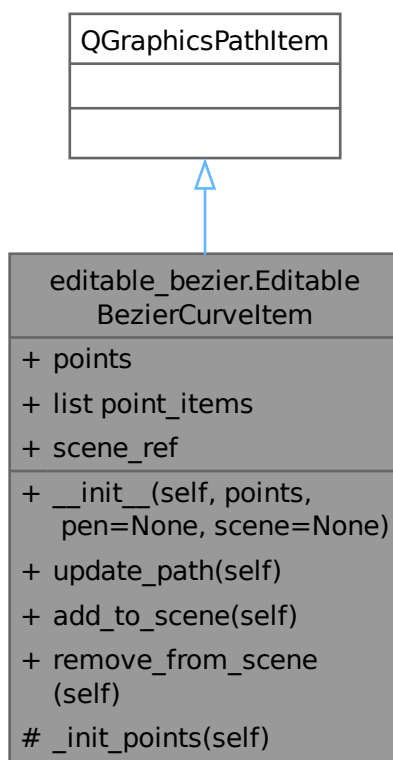
```
edge_item.EdgeItem.id = _next_edge_id()
```

Объявления и описания членов класса находятся в файле:

- `modules/data/src/widgets/edge_item.py`

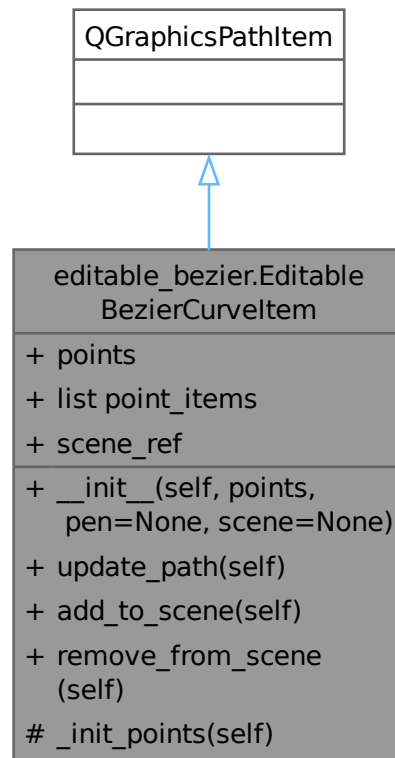
## 6.18 Класс `editable_bezier.EditableBezierCurveItem`

Граф наследования: `editable_bezier.EditableBezierCurveItem`:





Граф связей класса `editable_bezier.EditableBezierCurveItem`:



#### Открытые члены

- `__init__` (self, `points`, pen=None, scene=None)
- `update_path` (self)
- `add_to_scene` (self)
- `remove_from_scene` (self)

#### Открытые атрибуты

- `points` = `points[:]`
- list `point_items` = []
- `scene_ref` = `scene`

#### Защищенные члены

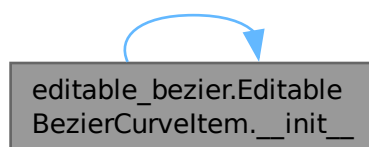
- `_init_points` (self)

## 6.18.1 Конструктор(ы)

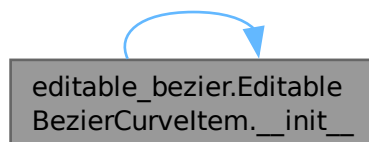
### 6.18.1.1 `__init__()`

```
editable_bezier.EditableBezierCurveItem.__init__ (
    self,
    points,
    pen = None,
    scene = None)
```

Граф вызовов:



Граф вызова функции:



## 6.18.2 Методы

### 6.18.2.1 `_init_points()`

```
editable_bezier.EditableBezierCurveItem._init_points (
    self) [protected]
```

### 6.18.2.2 `add_to_scene()`

```
editable_bezier.EditableBezierCurveItem.add_to_scene (
    self)
```

#### 6.18.2.3 `remove_from_scene()`

```
editable_bezier.EditableBezierCurveItem.remove_from_scene (  
    self)
```

#### 6.18.2.4 `update_path()`

```
editable_bezier.EditableBezierCurveItem.update_path (  
    self)
```

### 6.18.3 Данные класса

#### 6.18.3.1 `point_items`

```
editable_bezier.EditableBezierCurveItem.point_items = []
```

#### 6.18.3.2 `points`

```
editable_bezier.EditableBezierCurveItem.points = points[:]
```

#### 6.18.3.3 `scene_ref`

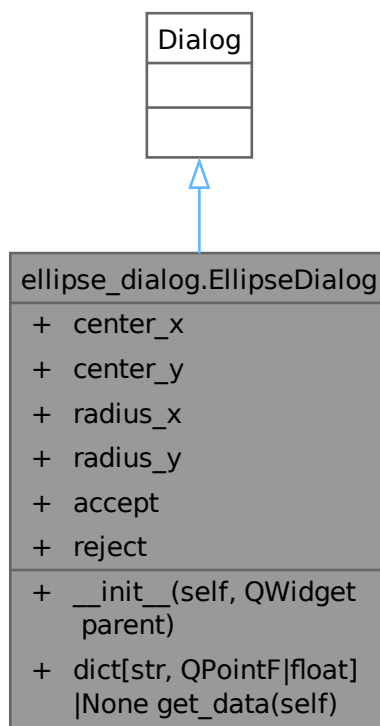
```
editable_bezier.EditableBezierCurveItem.scene_ref = scene
```

Объявления и описания членов класса находятся в файле:

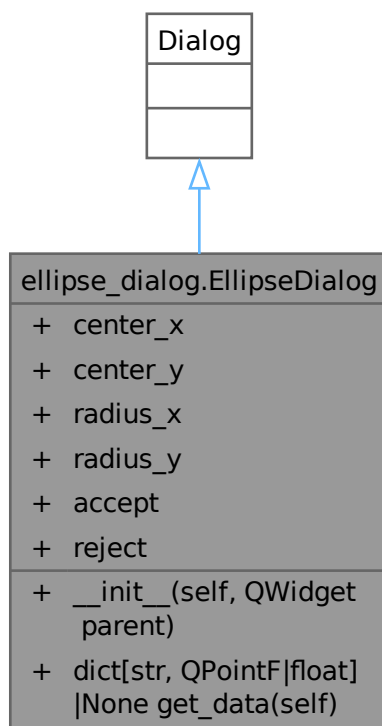
- `modules/data/src/widgets/`[editable\\_bezier.py](#)

## 6.19 Класс ellipse\_dialog.EllipseDialog

Граф наследования:ellipse\_dialog.EllipseDialog:



Граф связей класса ellipse\_dialog.EllipseDialog:



Открытые члены

- `__init__` (self, QWidget parent)
- dict[str, QPointF|float]|None `get_data` (self)

Открытые атрибуты

- `center_x` = QLineEdit('0')
- `center_y` = QLineEdit('0')
- `radius_x` = QLineEdit('180')
- `radius_y` = QLineEdit('180')
- `accept`
- `reject`

### 6.19.1 Конструктор(ы)

#### 6.19.1.1 `__init__`()

```

ellipse_dialog.EllipseDialog.__init__ (
    self,
    QWidget parent)
  
```

Граф вызовов:



Граф вызова функции:



## 6.19.2 Методы

### 6.19.2.1 get\_data()

```
dict[str, QPointF | float] | None ellipse_dialog.EllipseDialog.get_data (
    self)
```

## 6.19.3 Данные класса

### 6.19.3.1 accept

```
ellipse_dialog.EllipseDialog.accept
```

### 6.19.3.2 center\_x

```
ellipse_dialog.EllipseDialog.center_x = QLineEdit('0')
```

### 6.19.3.3 center\_y

```
ellipse_dialog.EllipseDialog.center_y = QLineEdit('0')
```

## 6.19.3.4 radius\_x

```
ellipse_dialog.EllipseDialog.radius_x = QLineEdit('180')
```

## 6.19.3.5 radius\_y

```
ellipse_dialog.EllipseDialog.radius_y = QLineEdit('180')
```

## 6.19.3.6 reject

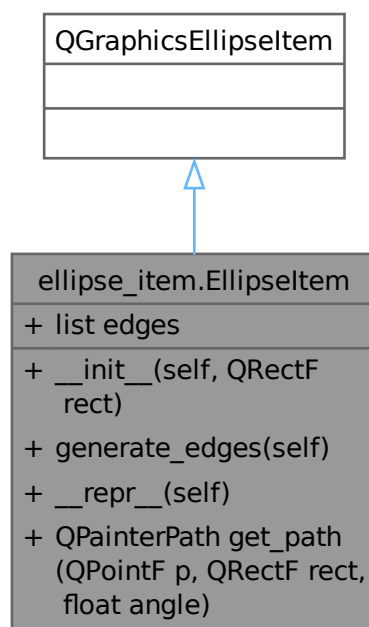
```
ellipse_dialog.EllipseDialog.reject
```

Объявления и описания членов класса находятся в файле:

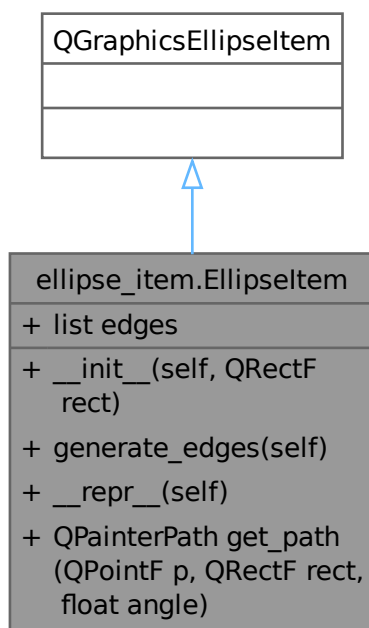
- `modules/data/src/dialogs/ellipse_dialog.py`

## 6.20 Класс ellipse\_item.EllipseItem

Граф наследования:ellipse\_item.EllipseItem:



Граф связей класса `ellipse_item.EllipseItem`:



Открытые члены

- `__init__` (self, `QRectF` rect)
- `generate_edges` (self)
- `__repr__` (self)

Открытые статические члены

- `QPainterPath` `get_path` (`QPointF` p, `QRectF` rect, float angle)

Открытые атрибуты

- list `edges` = []

## 6.20.1 Конструктор(ы)

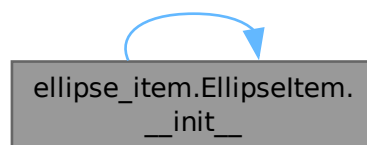
### 6.20.1.1 `__init__`()

```

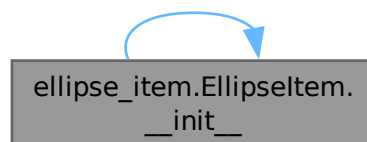
ellipse_item.EllipseItem.__init__ (
    self,
    QRectF rect)
  
```



Граф вызовов:



Граф вызова функции:



## 6.20.2 Методы

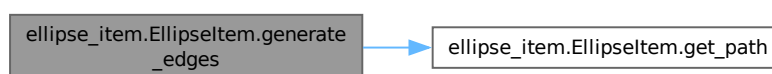
### 6.20.2.1 `__repr__()`

```
ellipse_item.EllipseItem.__repr__ (
    self)
```

### 6.20.2.2 `generate_edges()`

```
ellipse_item.EllipseItem.generate_edges (
    self)
```

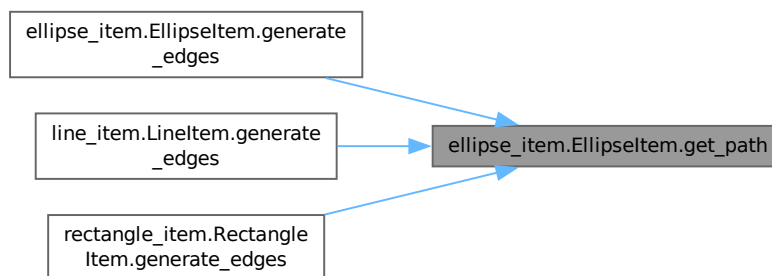
Граф вызовов:



### 6.20.2.3 get\_path()

```
QPainterPath ellipse_item.EllipseItem.get_path (  
    QPointF p,  
    QRectF rect,  
    float angle) [static]
```

Граф вызова функции:



## 6.20.3 Данные класса

### 6.20.3.1 edges

```
list ellipse_item.EllipseItem.edges = []
```

Объявления и описания членов класса находятся в файле:

- `modules/data/src/shapes/ellipse_item.py`

## 6.21 Класс event\_handler.EventHandler

Граф связей класса event\_handler.EventHandler:

| event_handler.EventHandler                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>+ main_window</div> <div>+ scene</div> <div>+ graphics_view</div> <div>+ properties_layout</div> <div>+ selection_service</div> <div>+ command_service</div> <div>+ bool moving_items</div> <div>+ move_start_pos</div> <div>+ move_initial_pos</div> <div>+ start_point</div> <div>+ temp_item</div> <div>+ list curve_points</div> <div>+ temp_curve_item</div>                                                                                                |
| <div>+ __init__(self, main_window, GridScene scene, GraphicsView graphics_view, QVBoxLayout properties_layout, SelectionService selection_service, CommandService command_service)</div> <div>+ bool mouse_press(self, QMouseEvent event)</div> <div>+ bool mouse_move(self, QMouseEvent event)</div> <div>+ bool mouse_release(self, QMouseEvent event)</div> <div>+ bool event_filter(self, QEvent event)</div> <div>+ key_press_event(self, QKeyEvent event)</div> |

Открытые члены

- `__init__` (self, [main\\_window](#), GridScene [scene](#), GraphicsView [graphics\\_view](#), QVBoxLayout [properties\\_layout](#), SelectionService [selection\\_service](#), CommandService [command\\_service](#))
- bool [mouse\\_press](#) (self, QMouseEvent event)

- `bool mouse_move (self, QMouseEvent event)`
- `bool mouse_release (self, QMouseEvent event)`
- `bool event_filter (self, QEvent event)`
- `key_press_event (self, QKeyEvent event)`

#### Открытые атрибуты

- `main_window = main_window`
- `scene = scene`
- `graphics_view = graphics_view`
- `properties_layout = properties_layout`
- `selection_service = selection_service`
- `command_service = command_service`
- `bool moving_items = False`
- `move_start_pos = QPointF()`
- `move_initial_pos = QPointF()`
- `start_point = None`
- `temp_item = None`
- `list curve_points = []`
- `temp_curve_item = None`

### 6.21.1 Конструктор(ы)

#### 6.21.1.1 \_\_init\_\_()

```
event_handler.EventHandler.__init__ (
    self,
    main_window,
    GridScene scene,
    GraphicsView graphics_view,
    QVBoxLayout properties_layout,
    SelectionService selection_service,
    CommandService command_service)
```

### 6.21.2 Методы

#### 6.21.2.1 event\_filter()

```
bool event_handler.EventHandler.event_filter (
    self,
    QEvent event)
```

Граф вызовов:



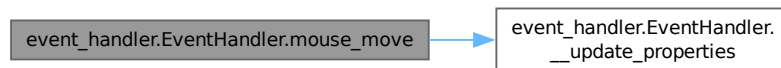
## 6.21.2.2 key\_press\_event()

```
event_handler.EventHandler.key_press_event (  
    self,  
    QKeyEvent event)
```

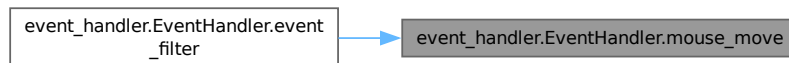
## 6.21.2.3 mouse\_move()

```
bool event_handler.EventHandler.mouse_move (  
    self,  
    QMouseEvent event)
```

Граф вызовов:



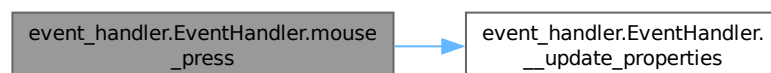
Граф вызова функции:



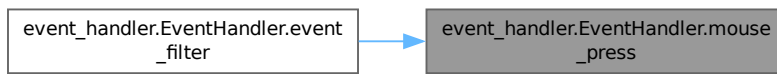
## 6.21.2.4 mouse\_press()

```
bool event_handler.EventHandler.mouse_press (  
    self,  
    QMouseEvent event)
```

Граф вызовов:



Граф вызова функции:

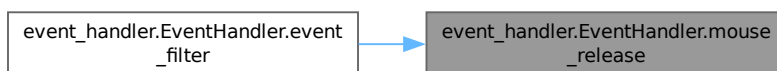


#### 6.21.2.5 mouse\_release()

```

bool event_handler.EventHandler.mouse_release (
    self,
    QMouseEvent event)
  
```

Граф вызова функции:



### 6.21.3 Данные класса

#### 6.21.3.1 command\_service

```
event_handler.EventHandler.command_service = command_service
```

#### 6.21.3.2 curve\_points

```
list event_handler.EventHandler.curve_points = []
```

#### 6.21.3.3 graphics\_view

```
event_handler.EventHandler.graphics_view = graphics_view
```

#### 6.21.3.4 main\_window

```
event_handler.EventHandler.main_window = main_window
```

## 6.21.3.5 move\_initial\_pos

```
event_handler.EventHandler.move_initial_pos = QPointF()
```

## 6.21.3.6 move\_start\_pos

```
event_handler.EventHandler.move_start_pos = QPointF()
```

## 6.21.3.7 moving\_items

```
bool event_handler.EventHandler.moving_items = False
```

## 6.21.3.8 properties\_layout

```
event_handler.EventHandler.properties_layout = properties_layout
```

## 6.21.3.9 scene

```
event_handler.EventHandler.scene = scene
```

## 6.21.3.10 selection\_service

```
event_handler.EventHandler.selection_service = selection_service
```

## 6.21.3.11 start\_point

```
event_handler.EventHandler.start_point = None
```

## 6.21.3.12 temp\_curve\_item

```
event_handler.EventHandler.temp_curve_item = None
```

## 6.21.3.13 temp\_item

```
event_handler.EventHandler.temp_item = None
```

Объявления и описания членов класса находятся в файле:

- `modules/data/src/event_handler.py`

## 6.22 Класс `gmsh_mesh_builder.GmshMeshBuilder`

Граф связей класса `gmsh_mesh_builder.GmshMeshBuilder`:

| <code>gmsh_mesh_builder.GmshMeshBuilder</code>                                      |
|-------------------------------------------------------------------------------------|
| + <code>grid_spacing</code>                                                         |
| + <code>filename</code>                                                             |
| + <code>dict boundary_conditions</code>                                             |
| + <code>__init__(self, int grid_spacing=1, str filename='mesh.msh')</code>          |
| + <code>add_loop(self, list [EdgeItem] loop, float max_element_size)</code>         |
| + <code>list[list[EdgeItem]] build_closed_loops (self, list[EdgeItem] edges)</code> |
| + <code>bool equal_points(self, QPointF a, QPointF b, tol=1e-3)</code>              |
| + <code>build_mesh(self, list [EdgeItem] edges, float max_element_size)</code>      |
| + <code>__del__(self)</code>                                                        |

Открытые члены

- `__init__` (self, int `grid_spacing`=1, str `filename`='mesh.msh')
- `add_loop` (self, list[EdgeItem] loop, float max\_element\_size)
- list[list[EdgeItem]] `build_closed_loops` (self, list[EdgeItem] edges)
- bool `equal_points` (self, QPointF a, QPointF b, tol=1e-3)
- `build_mesh` (self, list[EdgeItem] edges, float max\_element\_size)
- `__del__` (self)

Открытые атрибуты

- `grid_spacing` = grid\_spacing
- `filename` = filename
- dict `boundary_conditions` = {}



### 6.22.1 Конструктор(ы)

#### 6.22.1.1 \_\_init\_\_()

```
gmsh_mesh_builder.GmshMeshBuilder.__init__ (
    self,
    int grid_spacing = 1,
    str filename = 'mesh.msh')
```

#### 6.22.1.2 \_\_del\_\_()

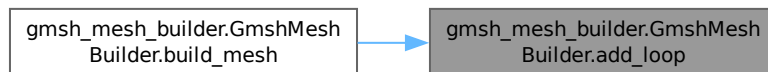
```
gmsh_mesh_builder.GmshMeshBuilder.__del__ (
    self)
```

### 6.22.2 Методы

#### 6.22.2.1 add\_loop()

```
gmsh_mesh_builder.GmshMeshBuilder.add_loop (
    self,
    list[EdgeItem] loop,
    float max_element_size)
```

Граф вызова функции:

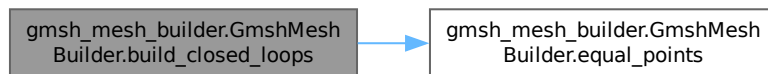


#### 6.22.2.2 build\_closed\_loops()

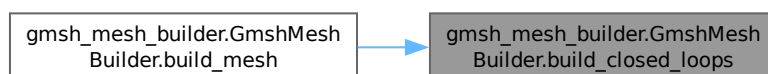
```
list[list[EdgeItem]] gmsh_mesh_builder.GmshMeshBuilder.build_closed_loops (
    self,
    list[EdgeItem] edges)
```

Вернёт список замкнутых циклов (каждый — список EdgeItem в порядке обхода). Если какая-то цепочка не закрылась (открытая), она в результат не попадёт.

Граф вызовов:



Граф вызова функции:

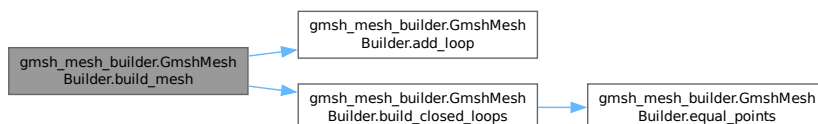


### 6.22.2.3 build\_mesh()

```

gmsh_mesh_builder.GmshMeshBuilder.build_mesh (
    self,
    list[EdgeItem] edges,
    float max_element_size)
  
```

Граф вызовов:

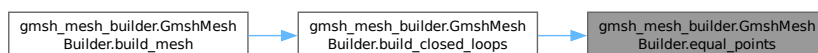


### 6.22.2.4 equal\_points()

```

bool gmsh_mesh_builder.GmshMeshBuilder.equal_points (
    self,
    QPointF a,
    QPointF b,
    tol = 1e-3)
  
```

Граф вызова функции:



### 6.22.3 Данные класса

#### 6.22.3.1 `boundary_conditions`

```
dict gmsh_mesh_builder.GmshMeshBuilder.boundary_conditions = {}
```

#### 6.22.3.2 `filename`

```
gmsh_mesh_builder.GmshMeshBuilder.filename = filename
```

#### 6.22.3.3 `grid_spacing`

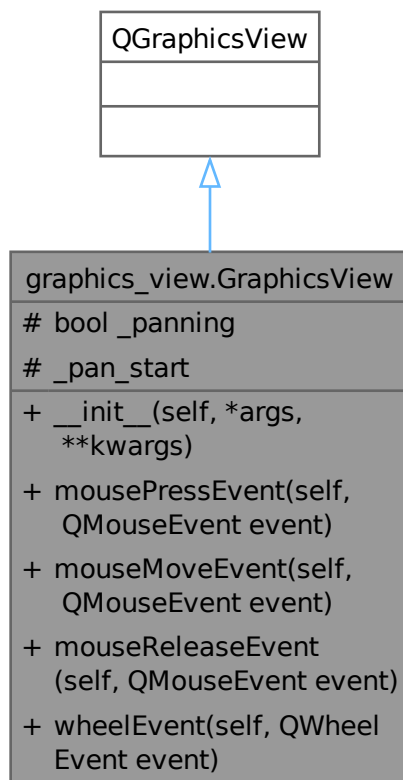
```
gmsh_mesh_builder.GmshMeshBuilder.grid_spacing = grid_spacing
```

Объявления и описания членов класса находятся в файле:

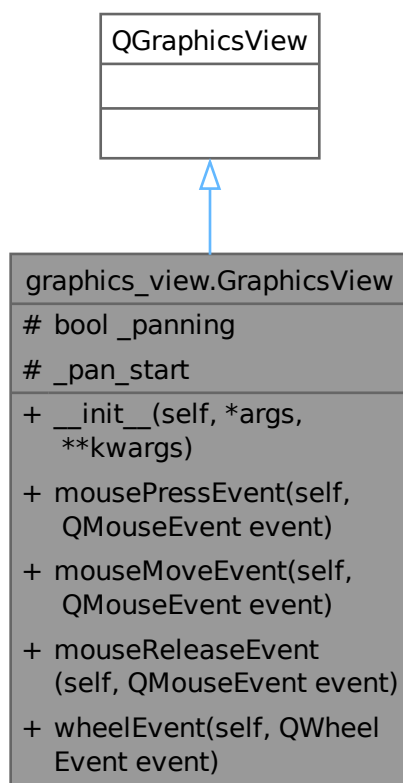
- `modules/data/src/services/gmsh_mesh_builder.py`

## 6.23 Класс `graphics_view.GraphicsView`

Граф наследования: `graphics_view.GraphicsView`:



Граф связей класса `graphics_view.GraphicsView`:



Открытые члены

- `__init__` (self, \*args, \*\*kwargs)
- `mousePressEvent` (self, QMouseEvent event)
- `mouseMoveEvent` (self, QMouseEvent event)
- `mouseReleaseEvent` (self, QMouseEvent event)
- `wheelEvent` (self, QWheelEvent event)

Защищенные данные

- `bool _panning` = False
- `_pan_start` = None

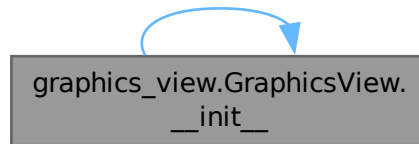
### 6.23.1 Конструктор(ы)

#### 6.23.1.1 `__init__`()

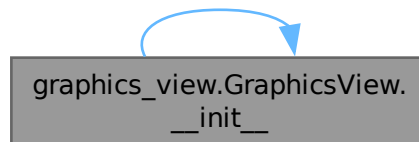
```
graphics_view.GraphicsView.__init__ (
    self,
```

```
* args,  
** kwargs)
```

Граф вызовов:



Граф вызова функции:

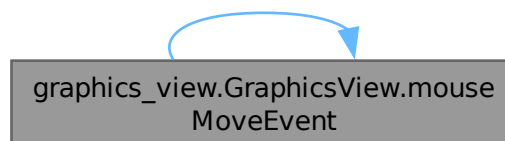


## 6.23.2 Методы

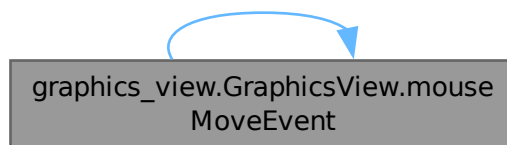
### 6.23.2.1 mouseMoveEvent()

```
graphics_view.GraphicsView.mouseMoveEvent (  
    self,  
    QMouseEvent event)
```

Граф вызовов:



Граф вызова функции:



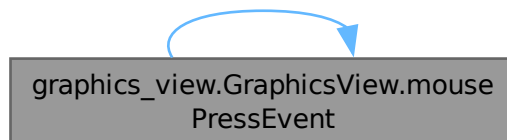
#### 6.23.2.2 mousePressEvent()

```
graphics_view.GraphicsView.mousePressEvent (  
    self,  
    QMouseEvent event)
```

Граф вызовов:



Граф вызова функции:



## 6.23.2.3 mouseReleaseEvent()

```
graphics_view.GraphicsView.mouseReleaseEvent (  
    self,  
    QMouseEvent event)
```

Граф вызовов:



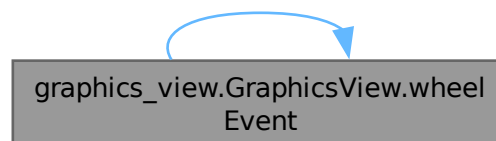
Граф вызова функции:



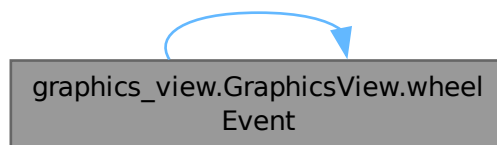
## 6.23.2.4 wheelEvent()

```
graphics_view.GraphicsView.wheelEvent (  
    self,  
    QWheelEvent event)
```

Граф вызовов:



Граф вызова функции:



### 6.23.3 Данные класса

#### 6.23.3.1 `_pan_start`

```
graphics_view.GraphicsView._pan_start = None [protected]
```

#### 6.23.3.2 `_panning`

```
bool graphics_view.GraphicsView._panning = False [protected]
```

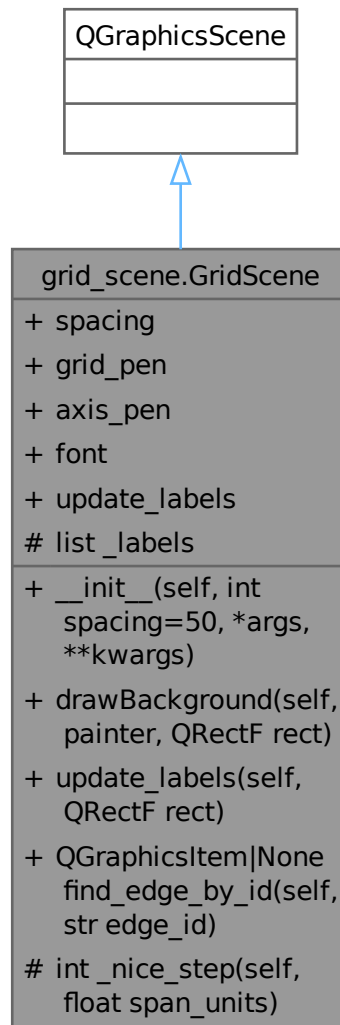
Объявления и описания членов класса находятся в файле:

- `modules/data/src/widgets/graphics_view.py`

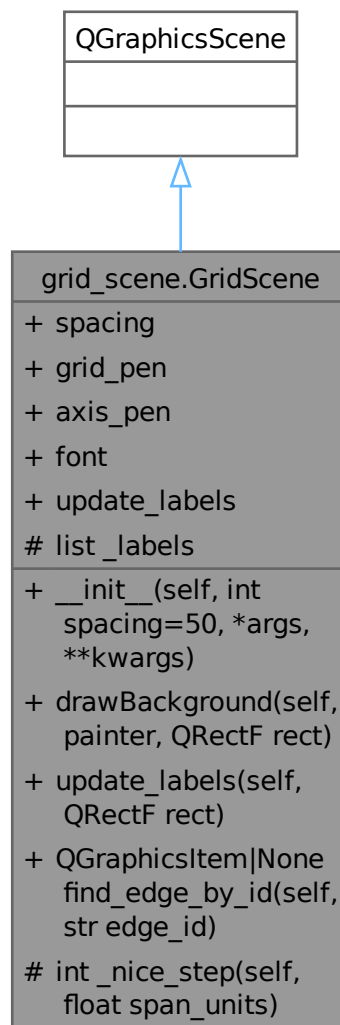


## 6.24 Класс `grid_scene.GridScene`

Граф наследования: `grid_scene.GridScene`:



Граф связей класса `grid_scene.GridScene`:



#### Открытые члены

- `__init__` (self, int `spacing`=50, \*args, \*\*kwargs)
- `drawBackground` (self, painter, QRectF rect)
- `update_labels` (self, QRectF rect)
- `QGraphicsItem|None find_edge_by_id` (self, str edge\_id)

#### Открытые атрибуты

- `spacing` = spacing
- `grid_pen` = QPen(Qt.lightGray, 0)
- `axis_pen` = QPen(Qt.black, 0)
- `font` = QFont("Arial", 20)
- `update_labels`

Защищенные члены

- `int _nice_step` (`self`, `float span_units`)

Защищенные данные

- `list _labels = []`

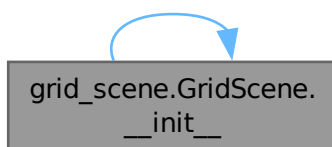
### 6.24.1 Конструктор(ы)

#### 6.24.1.1 `__init__()`

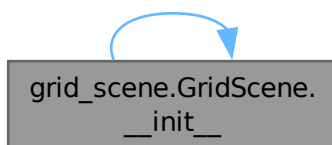
```
grid_scene.GridScene.__init__(  
    self,  
    int spacing = 50,  
    * args,  
    ** kwargs)
```

:param spacing: расстояние между линиями сетки в пикселях

Граф вызовов:



Граф вызова функции:



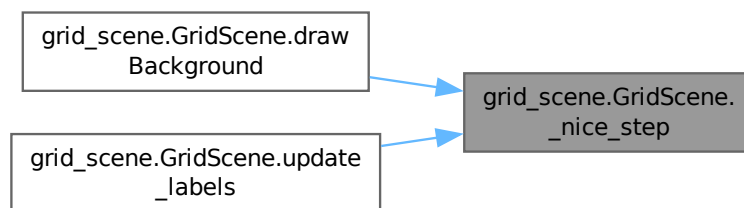
## 6.24.2 Методы

### 6.24.2.1 `_nice_step()`

```
int grid_scene.GridScene._nice_step (
    self,
    float span_units) [protected]
```

`span_units = (pixels_span / spacing)`  
хотим ~10 делений, но шаг выбираем из  $\{1,2,4,10\} \times 10^{\text{exp}}$

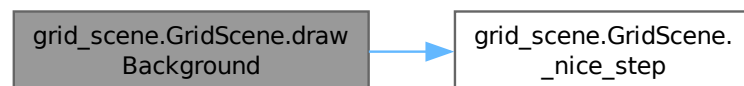
Граф вызова функции:



### 6.24.2.2 `drawBackground()`

```
grid_scene.GridScene.drawBackground (
    self,
    painter,
    QRectF rect)
```

Граф вызовов:



### 6.24.2.3 `find_edge_by_id()`

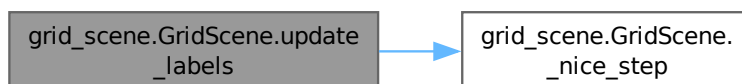
```
QGraphicsItem | None grid_scene.GridScene.find_edge_by_id (
    self,
    str edge_id)
```

Находит ребро на сцене по его ID.

6.24.2.4 `update_labels()`

```
grid_scene.GridScene.update_labels (  
    self,  
    QRectF rect)
```

Граф вызовов:



## 6.24.3 Данные класса

6.24.3.1 `_labels`

```
list grid_scene.GridScene._labels = [] [protected]
```

6.24.3.2 `axis_pen`

```
grid_scene.GridScene.axis_pen = QPen(Qt.black, 0)
```

6.24.3.3 `font`

```
grid_scene.GridScene.font = QFont("Arial", 20)
```

6.24.3.4 `grid_pen`

```
grid_scene.GridScene.grid_pen = QPen(Qt.lightGray, 0)
```

6.24.3.5 `spacing`

```
grid_scene.GridScene.spacing = spacing
```

6.24.3.6 `update_labels`

```
grid_scene.GridScene.update_labels
```

Объявления и описания членов класса находятся в файле:

- `modules/data/src/widgets/grid\_scene.py`

## 6.25 Класс `turbulence_models.InitialConditions`

Граф связей класса `turbulence_models.InitialConditions`:

| turbulence_models.InitialConditions |                          |
|-------------------------------------|--------------------------|
| +                                   | float <code>u</code>     |
| +                                   | float <code>v</code>     |
| +                                   | float <code>p</code>     |
| +                                   | float <code>k</code>     |
| +                                   | float <code>omega</code> |
|                                     |                          |

Статические открытые данные

- float `u` = 0
- float `v` = 0
- float `p` = 0
- float `k` = 4.184e-7
- float `omega` = 2.78

### 6.25.1 Данные класса

#### 6.25.1.1 `k`

```
float turbulence_models.InitialConditions.k = 4.184e-7 [static]
```

#### 6.25.1.2 `omega`

```
float turbulence_models.InitialConditions.omega = 2.78 [static]
```

#### 6.25.1.3 `p`

```
float turbulence_models.InitialConditions.p = 0 [static]
```

#### 6.25.1.4 `u`

```
float turbulence_models.InitialConditions.u = 0 [static]
```

6.25.1.5 `v`

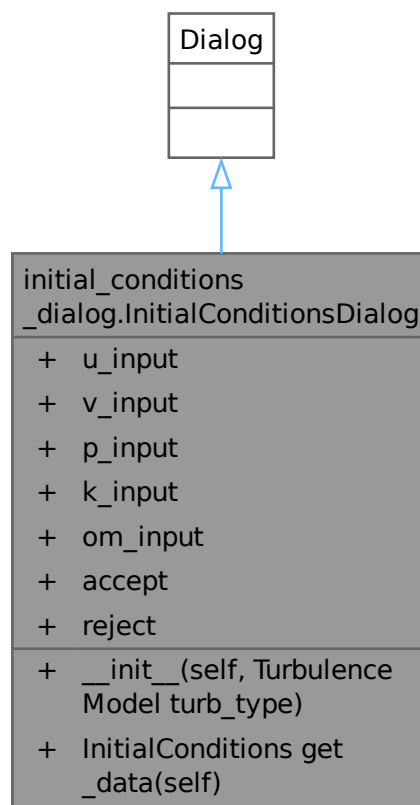
```
float turbulence_models.InitialConditions.v = 0 [static]
```

Объявления и описания членов класса находятся в файле:

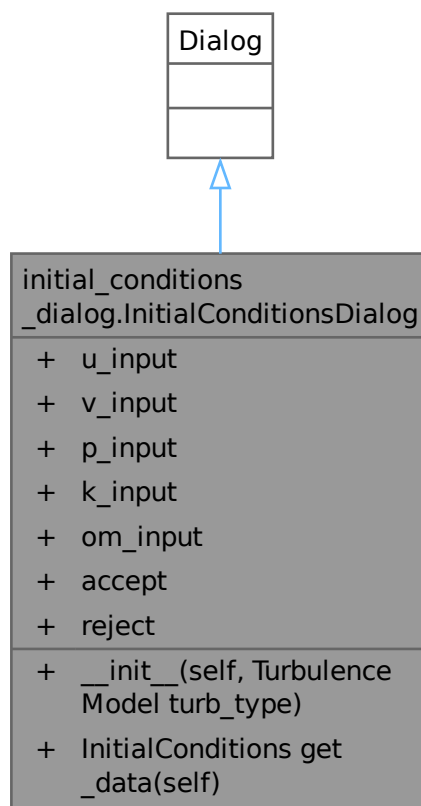
- `modules/data/src/physics/turbulence_models.py`

6.26 Класс `initial_conditions_dialog.InitialConditionsDialog`

Граф наследования:`initial_conditions_dialog.InitialConditionsDialog`:



Граф связей класса `initial_conditions_dialog.InitialConditionsDialog`:



#### Открытые члены

- `__init__` (self, TurbulenceModel turb\_type)
- `InitialConditions get_data` (self)

#### Открытые атрибуты

- `u_input` = `QDoubleSpinBox()`
- `v_input` = `QDoubleSpinBox()`
- `p_input` = `QDoubleSpinBox()`
- `k_input` = `QDoubleSpinBox()`
- `om_input` = `QDoubleSpinBox()`
- `accept`
- `reject`

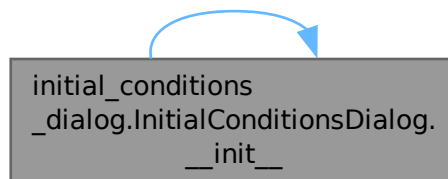


### 6.26.1 Конструктор(ы)

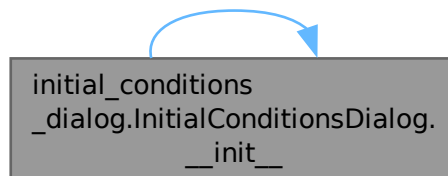
#### 6.26.1.1 \_\_init\_\_()

```
initial_conditions_dialog.InitialConditionsDialog.__init__ (  
    self,  
    TurbulenceModel turb_type)
```

Граф вызовов:



Граф вызова функции:



### 6.26.2 Методы

#### 6.26.2.1 get\_data()

```
InitialConditions initial_conditions_dialog.InitialConditionsDialog.get_data (  
    self)
```

### 6.26.3 Данные класса

#### 6.26.3.1 accept

```
initial_conditions_dialog.InitialConditionsDialog.accept
```

#### 6.26.3.2 k\_input

```
initial_conditions_dialog.InitialConditionsDialog.k_input = QDoubleSpinBox()
```

#### 6.26.3.3 om\_input

```
initial_conditions_dialog.InitialConditionsDialog.om_input = QDoubleSpinBox()
```

#### 6.26.3.4 p\_input

```
initial_conditions_dialog.InitialConditionsDialog.p_input = QDoubleSpinBox()
```

#### 6.26.3.5 reject

```
initial_conditions_dialog.InitialConditionsDialog.reject
```

#### 6.26.3.6 u\_input

```
initial_conditions_dialog.InitialConditionsDialog.u_input = QDoubleSpinBox()
```

#### 6.26.3.7 v\_input

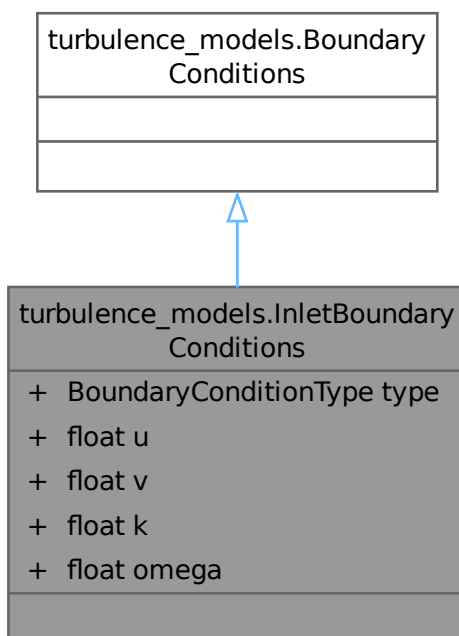
```
initial_conditions_dialog.InitialConditionsDialog.v_input = QDoubleSpinBox()
```

Объявления и описания членов класса находятся в файле:

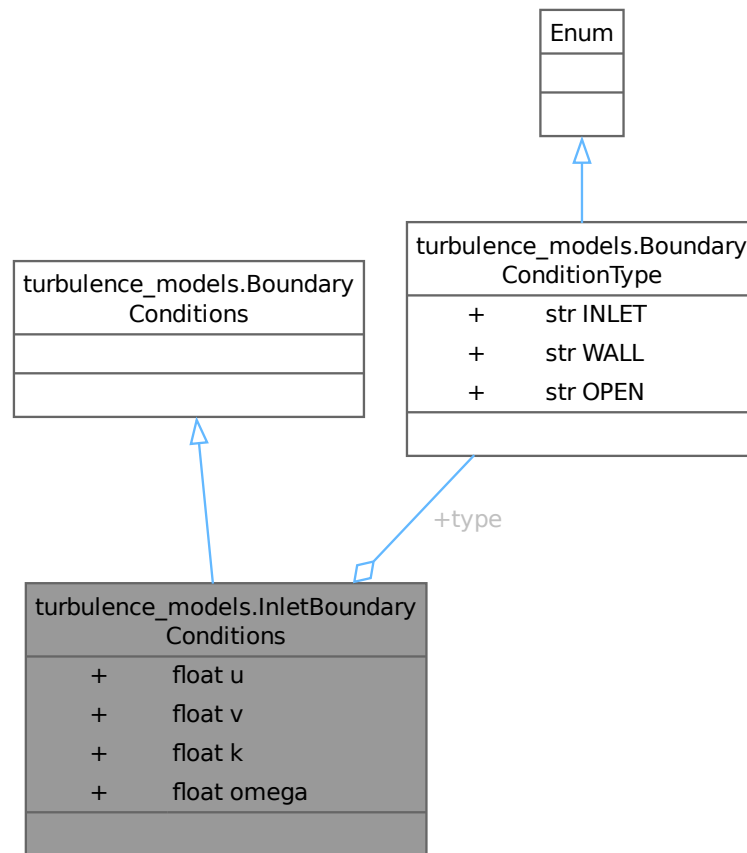
- `modules/data/src/dialogs/initial\_conditions\_dialog.py`

## 6.27 Класс turbulence\_models.InletBoundaryConditions

Граф наследования:turbulence\_models.InletBoundaryConditions:



Граф связей класса `turbulence_models.InletBoundaryConditions`:



Статические открытые данные

- `BoundaryConditionType type = BoundaryConditionType.INLET`
- `float u = 0`
- `float v = 0`
- `float k = 4.184e-7`
- `float omega = 2.78`

## 6.27.1 Данные класса

### 6.27.1.1 k

`float turbulence_models.InletBoundaryConditions.k = 4.184e-7 [static]`

### 6.27.1.2 omega

`float turbulence_models.InletBoundaryConditions.omega = 2.78 [static]`

## 6.27.1.3 type

`BoundaryConditionType` turbulence\_models.InletBoundaryConditions.type = `BoundaryConditionType.INLET` [static]

## 6.27.1.4 u

`float` turbulence\_models.InletBoundaryConditions.u = 0 [static]

## 6.27.1.5 v

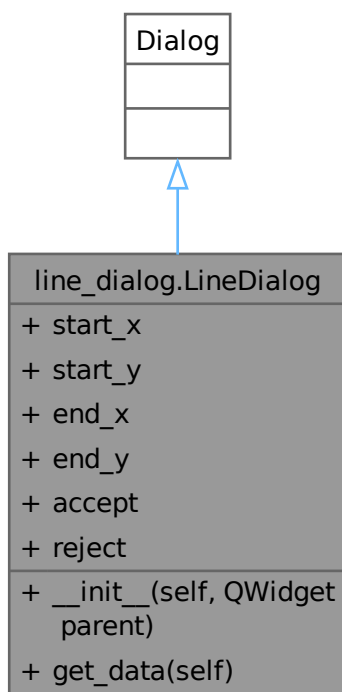
`float` turbulence\_models.InletBoundaryConditions.v = 0 [static]

Объявления и описания членов класса находятся в файле:

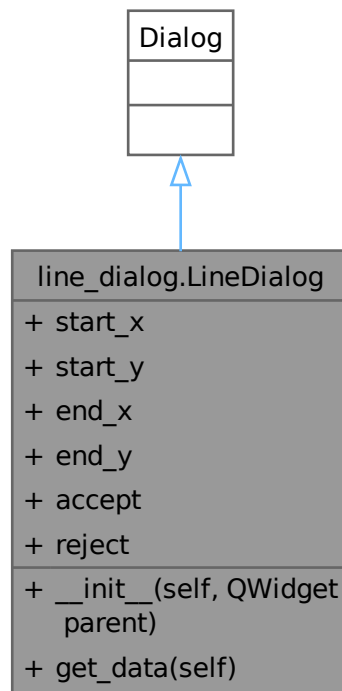
- `modules/data/src/physics/turbulence_models.py`

## 6.28 Класс line\_dialog.LineDialog

Граф наследования:line\_dialog.LineDialog:



Граф связей класса `line_dialog.LineDialog`:



Открытые члены

- `__init__` (self, QWidget parent)
- `get_data` (self)

Открытые атрибуты

- `start_x` = QLineEdit('0')
- `start_y` = QLineEdit('0')
- `end_x` = QLineEdit('100')
- `end_y` = QLineEdit('100')
- `accept`
- `reject`

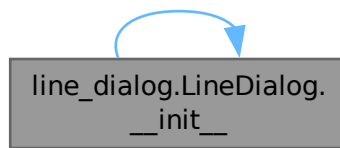
## 6.28.1 Конструктор(ы)

### 6.28.1.1 `__init__`()

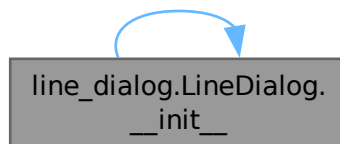
```

line_dialog.LineDialog.__init__ (
    self,
    QWidget parent)
  
```

Граф вызовов:



Граф вызова функции:



## 6.28.2 Методы

### 6.28.2.1 get\_data()

```
line_dialog.LineDialog.get_data (  
    self)
```

## 6.28.3 Данные класса

### 6.28.3.1 accept

```
line_dialog.LineDialog.accept
```

### 6.28.3.2 end\_x

```
line_dialog.LineDialog.end_x = QLineEdit('100')
```

### 6.28.3.3 end\_y

```
line_dialog.LineDialog.end_y = QLineEdit('100')
```

#### 6.28.3.4 reject

```
line_dialog.LineDialog.reject
```

#### 6.28.3.5 start\_x

```
line_dialog.LineDialog.start_x = QLineEdit('0')
```

#### 6.28.3.6 start\_y

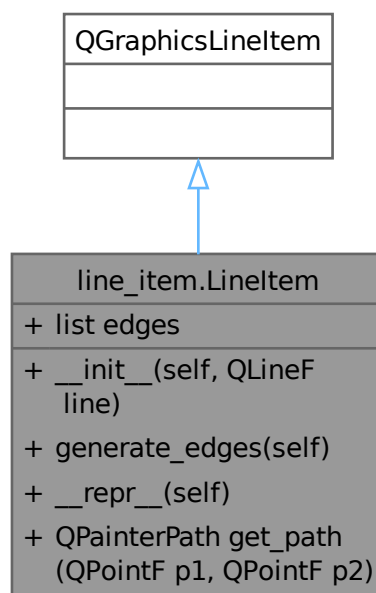
```
line_dialog.LineDialog.start_y = QLineEdit('0')
```

Объявления и описания членов класса находятся в файле:

- `modules/data/src/dialogs/line_dialog.py`

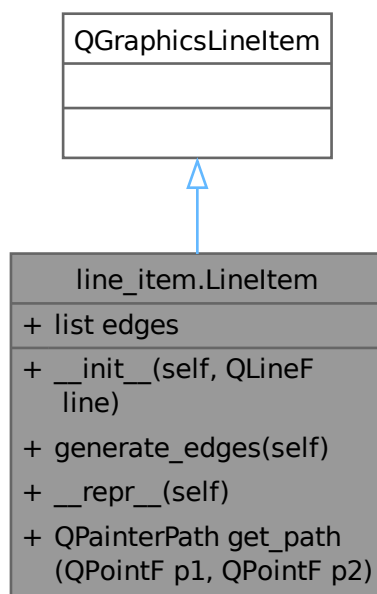
### 6.29 Класс line\_item.LineItem

Граф наследования:line\_item.LineItem:





Граф связей класса line\_item.LineItem:



Открытые члены

- `__init__` (self, QLineF line)
- `generate_edges` (self)
- `__repr__` (self)

Открытые статические члены

- `QPainterPath get_path` (QPointF p1, QPointF p2)

Открытые атрибуты

- `list edges = []`

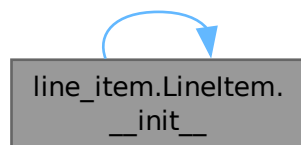
### 6.29.1 Конструктор(ы)

#### 6.29.1.1 `__init__`()

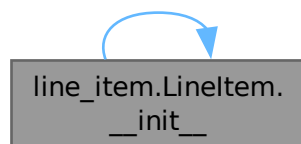
```

line_item.LineItem.__init__ (
    self,
    QLineF line)
  
```

Граф вызовов:



Граф вызова функции:



## 6.29.2 Методы

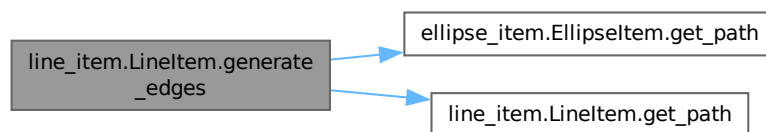
### 6.29.2.1 `__repr__()`

```
line_item.LineItem.__repr__(  
    self)
```

### 6.29.2.2 `generate_edges()`

```
line_item.LineItem.generate_edges(  
    self)
```

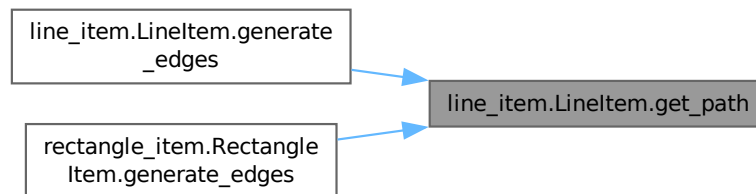
Граф вызовов:



## 6.29.2.3 get\_path()

```
QPainterPath line_item.LineItem.get_path (  
    QPointF p1,  
    QPointF p2) [static]
```

Граф вызова функции:



## 6.29.3 Данные класса

## 6.29.3.1 edges

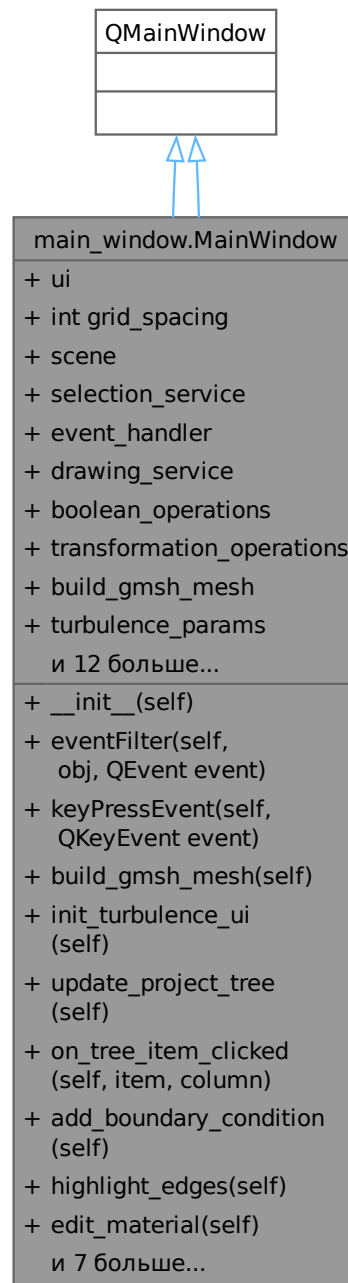
```
list line_item.LineItem.edges = []
```

Объявления и описания членов класса находятся в файле:

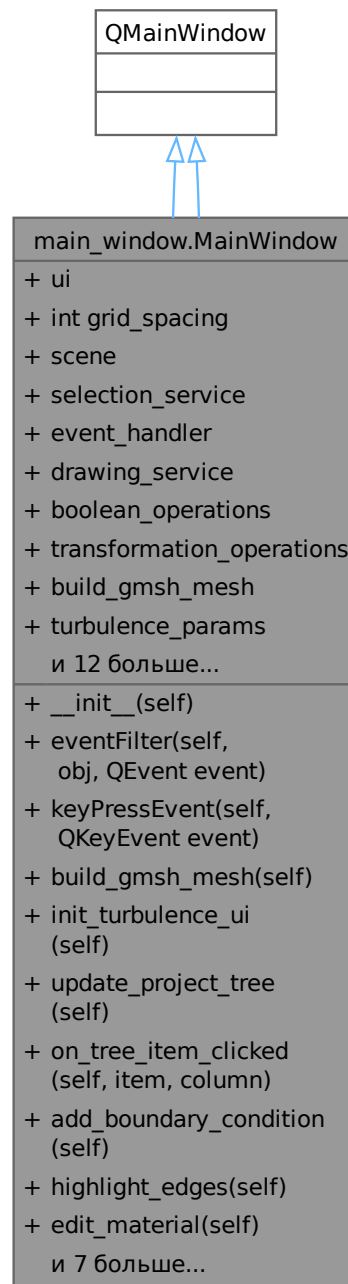
- `modules/data/src/shapes/line\_item.py`

## 6.30 Класс `main_window.MainWindow`

Граф наследования: `main_window.MainWindow`:



Граф связей класса `main_window.MainWindow`:



Открытые члены

- `__init__` (self)
- `eventFilter` (self, obj, QEvent event)
- `keyPressEvent` (self, QKeyEvent event)
- `build_gmsh_mesh` (self)
- `init_turbulence_ui` (self)

- `update_project_tree` (self)
- `on_tree_item_clicked` (self, item, column)
- `add_boundary_condition` (self)
- `highlight_edges` (self)
- `edit_material` (self)
- `edit_turbulence_model` (self)
- `edit_initial_conditions` (self)
- `edit_boundary_condition` (self, item)
- `show_tree_context_menu` (self, position)
- `delete_boundary_condition` (self, item)
- `export_json` (self)
- `__init__` (self)

#### Открытые атрибуты

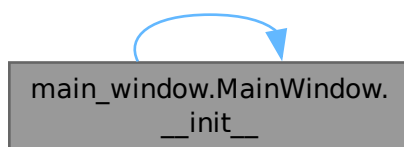
- `ui` = `Ui_MainWindow()`
- `int grid_spacing` = 50
- `scene` = `GridScene(spacing=self.grid_spacing)`
- `selection_service` = `SelectionService(self.scene)`
- `event_handler`
- `drawing_service` = `DrawingService(self, self.scene, command_service, self.selection_service)`
- `boolean_operations`
- `transformation_operations`
- `build_gmsh_mesh`
- `turbulence_params` = `TurbulenceParams()`
- `list boundary_conditions` = []
- `list boundary_edges` = []
- `initial_conditions` = `InitialConditions()`
- `material` = `Material()`
- `on_tree_item_clicked`
- `show_tree_context_menu`
- `add_boundary_condition`
- `logger` = `logging.getLogger(__name__)`
- `canvas_visualization` = `None`
- `toolbar` = `None`
- `layout_visualization` = `QVBoxLayout()`
- `color_bar`

### 6.30.1 Конструктор(ы)

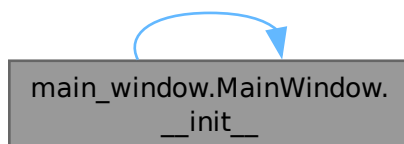
#### 6.30.1.1 `__init__()` [1/2]

```
main_window.MainWindow.__init__(
    self)
```

Граф вызовов:



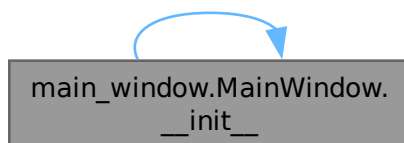
Граф вызова функции:



#### 6.30.1.2 `__init__()` [2/2]

```
main_window.MainWindow.__init__ (  
    self)
```

Граф вызовов:

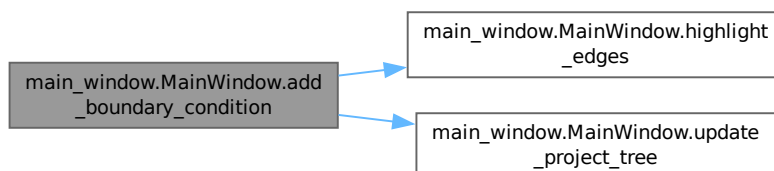


## 6.30.2 Методы

### 6.30.2.1 add\_boundary\_condition()

```
main_window.MainWindow.add_boundary_condition (  
    self)
```

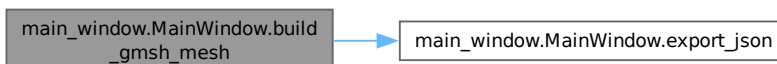
Граф вызовов:



### 6.30.2.2 build\_gmsh\_mesh()

```
main_window.MainWindow.build_gmsh_mesh (  
    self)
```

Граф вызовов:



### 6.30.2.3 delete\_boundary\_condition()

```
main_window.MainWindow.delete_boundary_condition (  
    self,  
    item)
```

Граф вызовов:

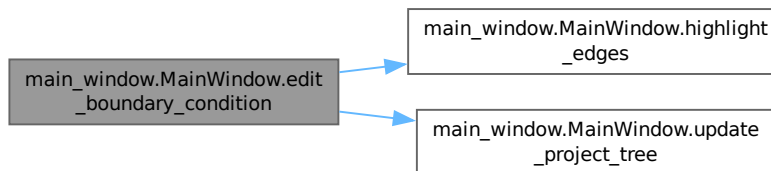




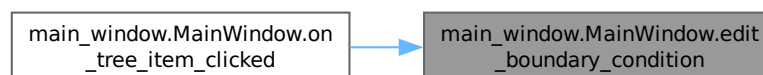
## 6.30.2.4 edit\_boundary\_condition()

```
main_window.MainWindow.edit_boundary_condition (  
    self,  
    item)
```

Граф вызовов:



Граф вызова функции:



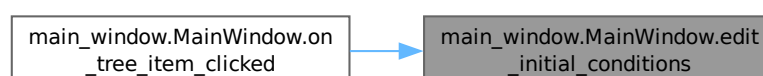
## 6.30.2.5 edit\_initial\_conditions()

```
main_window.MainWindow.edit_initial_conditions (  
    self)
```

Граф вызовов:



Граф вызова функции:



## 6.30.2.6 edit\_material()

```
main_window.MainWindow.edit_material (  
    self)
```

Граф вызовов:



Граф вызова функции:



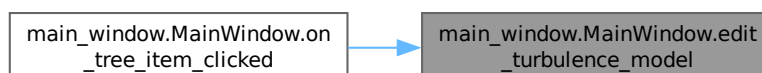
## 6.30.2.7 edit\_turbulence\_model()

```
main_window.MainWindow.edit_turbulence_model (  
    self)
```

Граф вызовов:



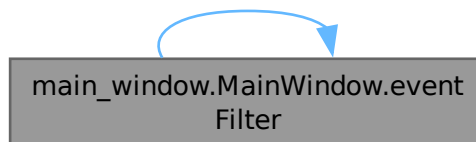
Граф вызова функции:



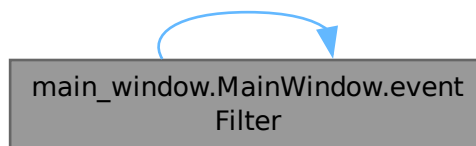
## 6.30.2.8 eventFilter()

```
main_window.MainWindow.eventFilter (  
    self,  
    obj,  
    QEvent event)
```

Граф вызовов:



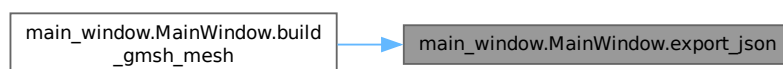
Граф вызова функции:



## 6.30.2.9 export\_\_json()

```
main_window.MainWindow.export__json (  
    self)
```

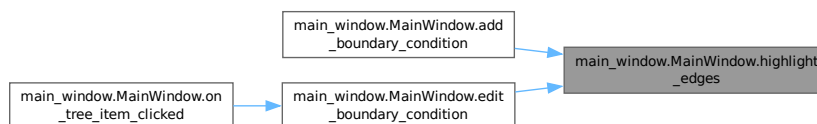
Граф вызова функции:



## 6.30.2.10 highlight\_edges()

```
main_window.MainWindow.highlight_edges (
    self)
```

Граф вызова функции:



## 6.30.2.11 init\_turbulence\_ui()

```
main_window.MainWindow.init_turbulence_ui (
    self)
```

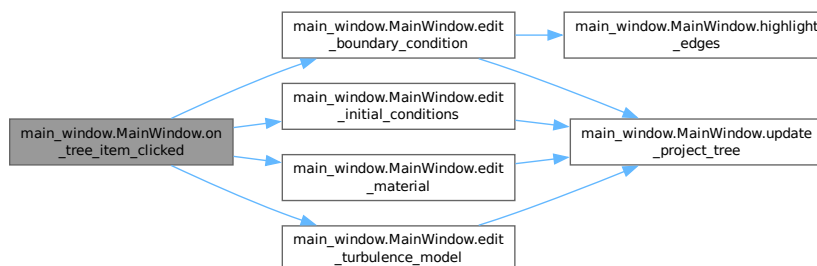
## 6.30.2.12 keyPressEvent()

```
main_window.MainWindow.keyPressEvent (
    self,
    QKeyEvent event)
```

## 6.30.2.13 on\_tree\_item\_clicked()

```
main_window.MainWindow.on_tree_item_clicked (
    self,
    item,
    column)
```

Граф вызовов:



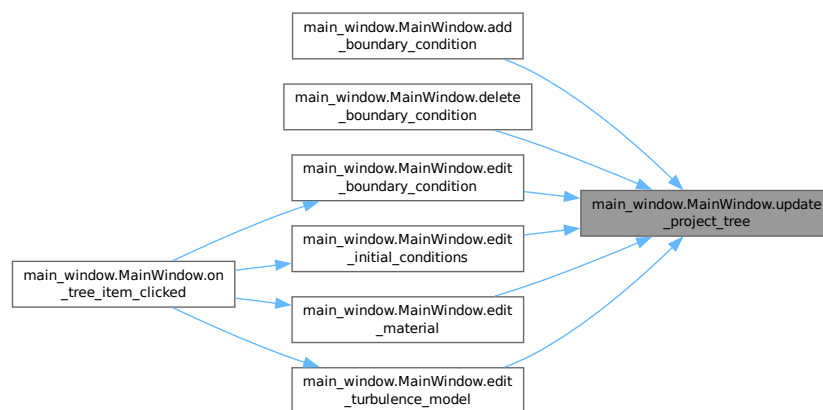
## 6.30.2.14 show\_tree\_context\_menu()

```
main_window.MainWindow.show_tree_context_menu (
    self,
    position)
```

## 6.30.2.15 update\_project\_tree()

```
main_window.MainWindow.update_project_tree (
    self)
```

Граф вызова функции:



## 6.30.3 Данные класса

## 6.30.3.1 add\_boundary\_condition

```
main_window.MainWindow.add_boundary_condition
```

## 6.30.3.2 boolean\_operations

```
main_window.MainWindow.boolean_operations
```

## Инициализатор

```
= BooleanOperations(
    self,
    self.scene,
    command_service,
    self.drawing_service,
    self.selection_service
)
```

## 6.30.3.3 boundary\_conditions

```
list main_window.MainWindow.boundary_conditions = []
```

## 6.30.3.4 boundary\_edges

```
main_window.MainWindow.boundary_edges = []
```

## 6.30.3.5 build\_gmsh\_mesh

```
main_window.MainWindow.build_gmsh_mesh
```

## 6.30.3.6 canvas\_visualization

```
main_window.MainWindow.canvas_visualization = None
```

## 6.30.3.7 color\_bar

```
main_window.MainWindow.color_bar
```

## Инициализатор

```
= Plotter.plot_surface(
    self.canvas_visualization.fig,
    self.canvas_visualization.ax,
    data['x'],
    data['y'],
    data['p' if expression else 'U'],
    levels=levels,
    cmap='rainbow',
)
```

## 6.30.3.8 drawing\_service

```
main_window.MainWindow.drawing_service = DrawingService(self, self.scene, command_service, self.selection_service)
```

## 6.30.3.9 event\_handler

```
main_window.MainWindow.event_handler
```

## Инициализатор

```
= EventHandler(
    self,
    self.scene,
    self.ui.graphicsView,
    self.ui.propertiesLayout,
    self.selection_service,
    command_service
)
```

## 6.30.3.10 grid\_spacing

```
int main_window.MainWindow.grid_spacing = 50
```

## 6.30.3.11 initial\_conditions

```
main_window.MainWindow.initial_conditions = InitialConditions()
```

## 6.30.3.12 layout\_visualization

```
main_window.MainWindow.layout_visualization = QVBoxLayout()
```

## 6.30.3.13 logger

```
main_window.MainWindow.logger = logging.getLogger(__name__)
```

## 6.30.3.14 material

```
main_window.MainWindow.material = Material()
```

## 6.30.3.15 on\_tree\_item\_clicked

```
main_window.MainWindow.on_tree_item_clicked
```

## 6.30.3.16 scene

```
main_window.MainWindow.scene = GridScene(spacing=self.grid_spacing)
```

## 6.30.3.17 selection\_service

```
main_window.MainWindow.selection_service = SelectionService(self.scene)
```

## 6.30.3.18 show\_tree\_context\_menu

```
main_window.MainWindow.show_tree_context_menu
```

## 6.30.3.19 toolbar

```
main_window.MainWindow.toolbar = None
```

## 6.30.3.20 transformation\_operations

```
main_window.MainWindow.transformation_operations
```

**Инициализатор**

```
= TransformationOperations(  
    self,  
    self.scene,  
    command_service,  
    self.selection_service  
)
```

### 6.30.3.21 turbulence\_params

```
main_window.MainWindow.turbulence_params = TurbulenceParams()
```

### 6.30.3.22 ui

```
main_window.MainWindow.ui = Ui_MainWindow()
```

Объявления и описания членов классов находятся в файлах:

- modules/data/src/[main\\_window.py](#)
- modules/visualization/src/[main\\_window.py](#)

## 6.31 Класс turbulence\_models.Material

Граф связей класса turbulence\_models.Material:

| turbulence_models.Material |           |
|----------------------------|-----------|
| +                          | float rho |
| +                          | float mu  |
|                            |           |

Статические открытые данные

- float [rho](#) = 1.204
- float [mu](#) = 1.81397e-5

### 6.31.1 Данные класса

#### 6.31.1.1 mu

```
float turbulence_models.Material.mu = 1.81397e-5 [static]
```

#### 6.31.1.2 rho

```
float turbulence_models.Material.rho = 1.204 [static]
```

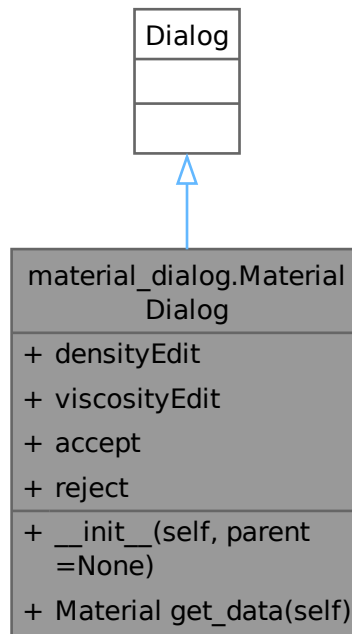
Объявления и описания членов класса находятся в файле:

- modules/data/src/physics/[turbulence\\_models.py](#)

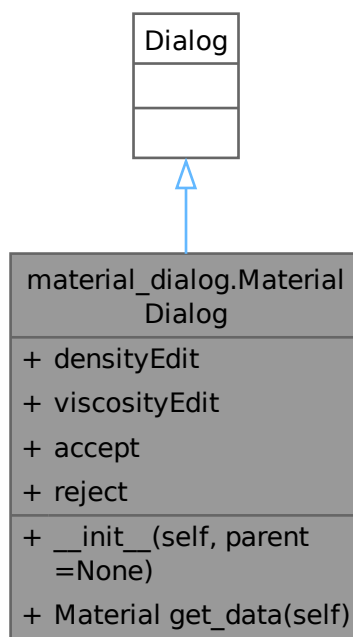


## 6.32 Класс material\_dialog.MaterialDialog

Граф наследования:material\_dialog.MaterialDialog:



Граф связей класса `material_dialog.MaterialDialog`:



Открытые члены

- `__init__` (self, parent=None)
- `Material get_data` (self)

Открытые атрибуты

- `densityEdit` = `QDoubleSpinBox()`
- `viscosityEdit` = `QDoubleSpinBox()`
- `accept`
- `reject`

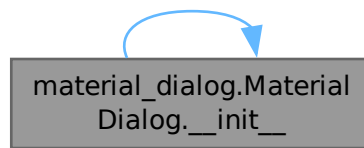
### 6.32.1 Конструктор(ы)

#### 6.32.1.1 `__init__()`

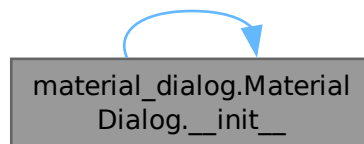
```

material_dialog.MaterialDialog.__init__ (
    self,
    parent = None)
  
```

Граф вызовов:



Граф вызова функции:



## 6.32.2 Методы

### 6.32.2.1 get\_data()

```
Material material_dialog.MaterialDialog.get_data (  
    self)
```

## 6.32.3 Данные класса

### 6.32.3.1 accept

```
material_dialog.MaterialDialog.accept
```

### 6.32.3.2 densityEdit

```
material_dialog.MaterialDialog.densityEdit = QDoubleSpinBox()
```

### 6.32.3.3 reject

```
material_dialog.MaterialDialog.reject
```

#### 6.32.3.4 viscosityEdit

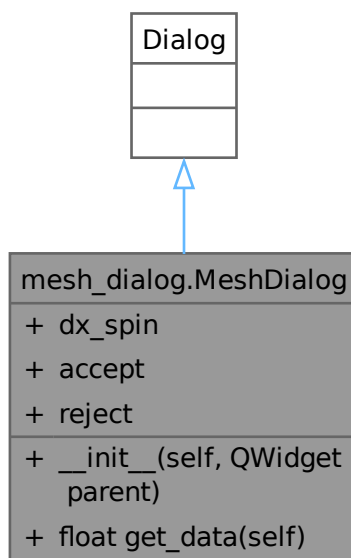
```
material_dialog.MaterialDialog.viscosityEdit = QDoubleSpinBox()
```

Объявления и описания членов класса находятся в файле:

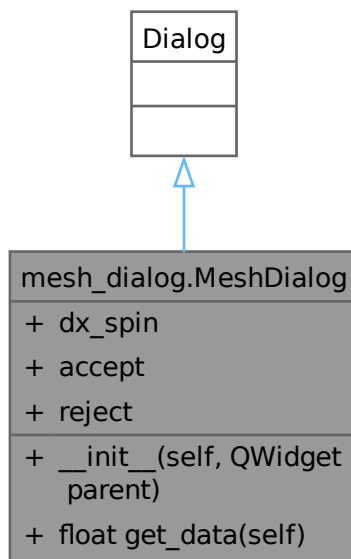
- `modules/data/src/dialogs/material_dialog.py`

### 6.33 Класс mesh\_dialog.MeshDialog

Граф наследования:mesh\_dialog.MeshDialog:



Граф связей класса mesh\_dialog.MeshDialog:



Открытые члены

- `__init__` (self, QWidget parent)
- float `get_data` (self)

Открытые атрибуты

- `dx_spin` = QDoubleSpinBox()
- `accept`
- `reject`

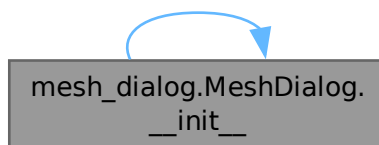
### 6.33.1 Конструктор(ы)

#### 6.33.1.1 \_\_init\_\_()

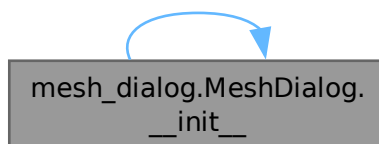
```

mesh_dialog.MeshDialog.__init__ (
    self,
    QWidget parent)
  
```

Граф вызовов:



Граф вызова функции:



## 6.33.2 Методы

### 6.33.2.1 get\_data()

```
float mesh_dialog.MeshDialog.get_data (  
    self)
```

## 6.33.3 Данные класса

### 6.33.3.1 accept

```
mesh_dialog.MeshDialog.accept
```

### 6.33.3.2 dx\_spin

```
mesh_dialog.MeshDialog.dx_spin = QDoubleSpinBox()
```

6.33.3.3 `reject`

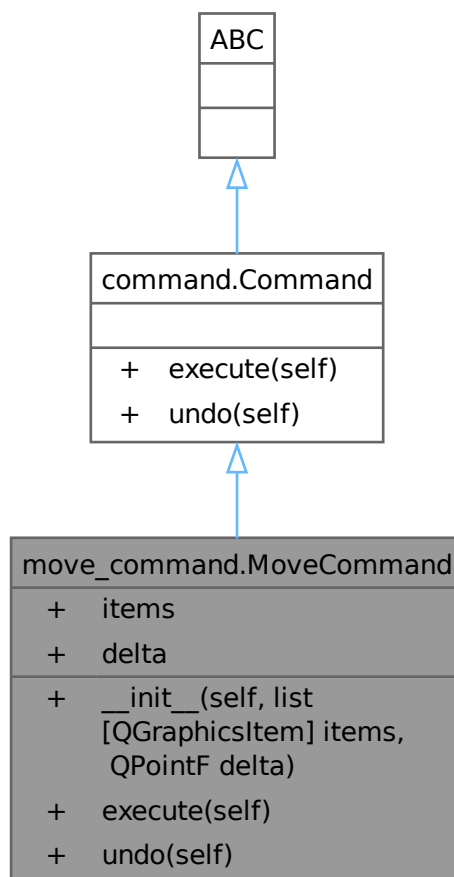
`mesh_dialog.MeshDialog.reject`

Объявления и описания членов класса находятся в файле:

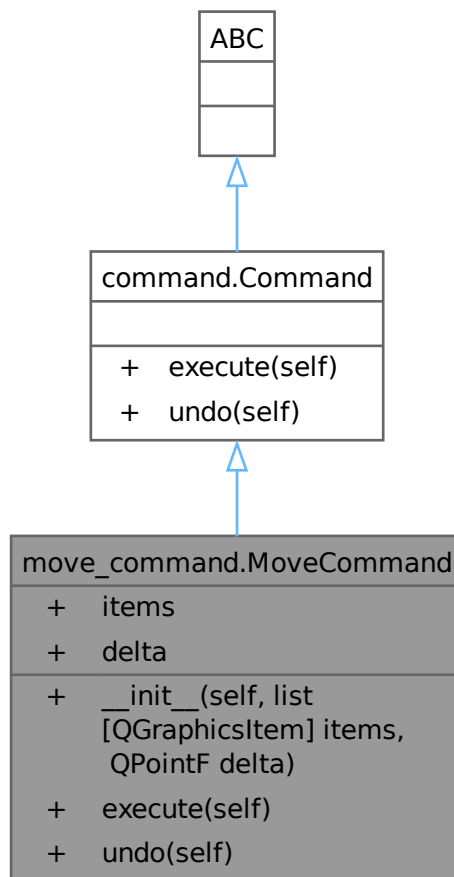
- `modules/data/src/dialogs/mesh_dialog.py`

6.34 Класс `move_command.MoveCommand`

Граф наследования:`move_command.MoveCommand`:



Граф связей класса `move_command.MoveCommand`:



Открытые члены

- `__init__` (self, list[QGraphicsItem] `items`, QPointF `delta`)
- `execute` (self)
- `undo` (self)

Открытые атрибуты

- `items` = items
- `delta` = delta

### 6.34.1 Конструктор(ы)

#### 6.34.1.1 `__init__`()

```

move_command.MoveCommand.__init__ (
    self,
    list[QGraphicsItem] items,
    QPointF delta)
  
```



## 6.34.2 Методы

6.34.2.1 `execute()`

```
move_command.MoveCommand.execute (
    self)
```

Переопределяет метод предка [command.Command](#).

6.34.2.2 `undo()`

```
move_command.MoveCommand.undo (
    self)
```

Переопределяет метод предка [command.Command](#).

## 6.34.3 Данные класса

6.34.3.1 `delta`

```
move_command.MoveCommand.delta = delta
```

6.34.3.2 `items`

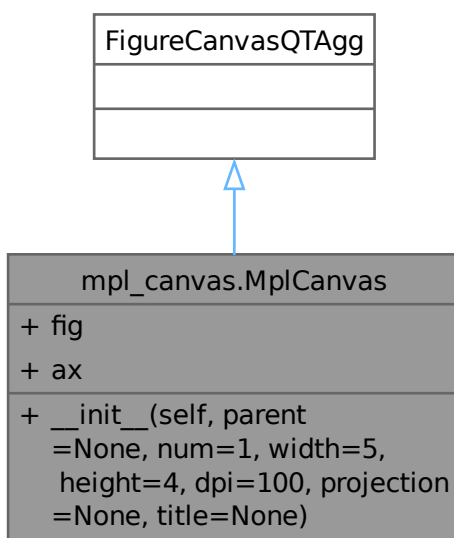
```
move_command.MoveCommand.items = items
```

Объявления и описания членов класса находятся в файле:

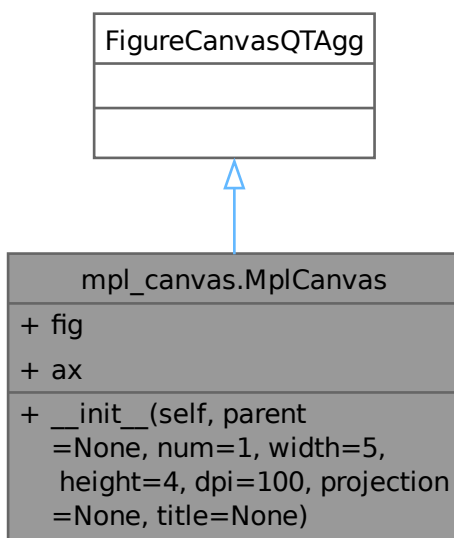
- `modules/data/src/commands/move_command.py`

6.35 Класс `mpl_canvas.MplCanvas`

Граф наследования:`mpl_canvas.MplCanvas`:



Граф связей класса `mpl_canvas.MplCanvas`:



Открытые члены

- `__init__` (self, parent=None, num=1, width=5, height=4, dpi=100, projection=None, title=None)

Открытые атрибуты

- `fig` = `Figure(figsize=(width, height), dpi=dpi, facecolor='#fff')`
- `ax` = `self.fig.add_subplot(111, projection=projection, facecolor='#fff')`

### 6.35.1 Конструктор(ы)

#### 6.35.1.1 `__init__()`

```

mpl_canvas.MplCanvas.__init__ (
    self,
    parent = None,
    num = 1,
    width = 5,
    height = 4,
    dpi = 100,
    projection = None,
    title = None)
  
```

## 6.35.2 Данные класса

6.35.2.1 `ax`

```
mpl_canvas.MplCanvas.ax = self.fig.add_subplot(111, projection=projection, facecolor=' #fff')
```

6.35.2.2 `fig`

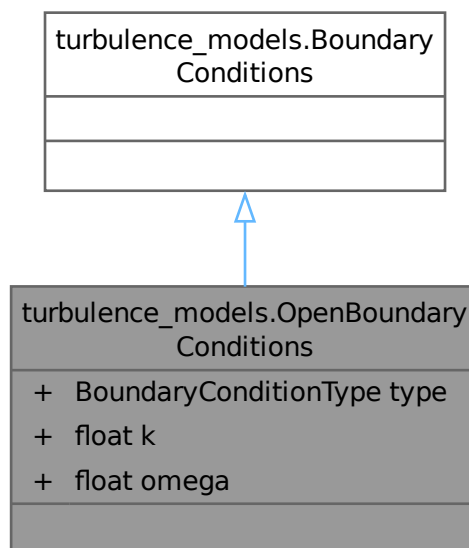
```
mpl_canvas.MplCanvas.fig = Figure(figsize=(width, height), dpi=dpi, facecolor=' #fff')
```

Объявления и описания членов класса находятся в файле:

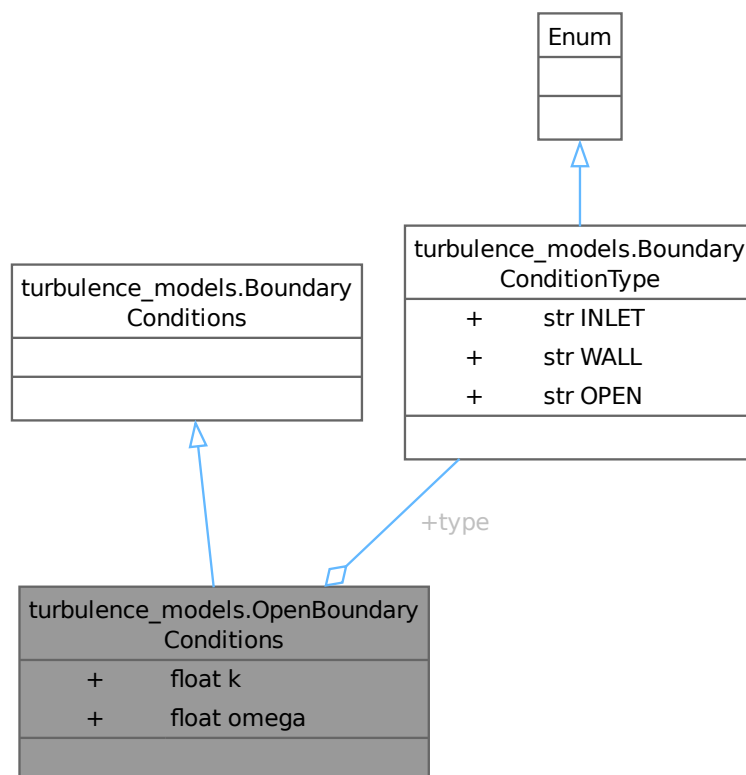
- `modules/visualization/src/mpl_canvas.py`

6.36 Класс `turbulence_models.OpenBoundaryConditions`

Граф наследования: `turbulence_models.OpenBoundaryConditions`:



Граф связей класса `turbulence_models.OpenBoundaryConditions`:



Статические открытые данные

- `BoundaryConditionType type = BoundaryConditionType.OPEN`
- `float k = 4.184e-7`
- `float omega = 2.78`

### 6.36.1 Данные класса

#### 6.36.1.1 k

`float turbulence_models.OpenBoundaryConditions.k = 4.184e-7 [static]`

#### 6.36.1.2 omega

`float turbulence_models.OpenBoundaryConditions.omega = 2.78 [static]`

## 6.36.1.3 type

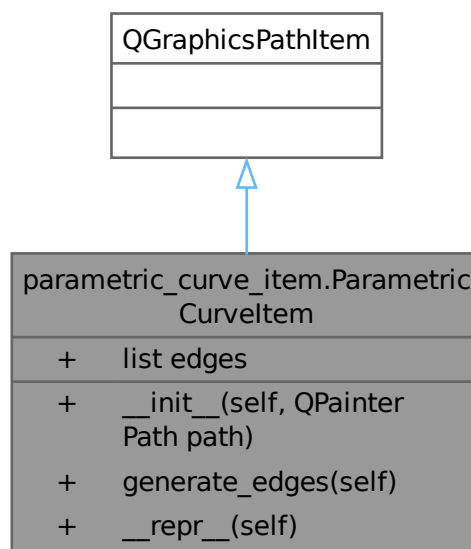
`BoundaryConditionType` turbulence\_models.OpenBoundaryConditions.type = `BoundaryConditionType.OPEN` [static]

Объявления и описания членов класса находятся в файле:

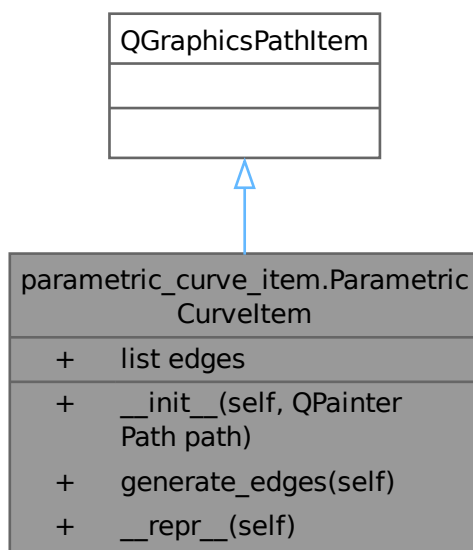
- modules/data/src/physics/turbulence\_models.py

## 6.37 Класс parametric\_curve\_item.ParametricCurveItem

Граф наследования: parametric\_curve\_item.ParametricCurveItem:



Граф связей класса `parametric_curve_item.ParametricCurveItem`:



Открытые члены

- `__init__` (self, QPainterPath path)
- `generate_edges` (self)
- `__repr__` (self)

Открытые атрибуты

- list `edges` = []

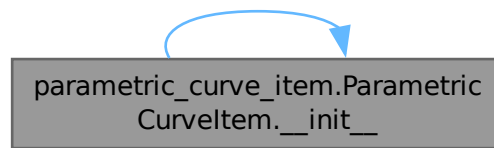
### 6.37.1 Конструктор(ы)

#### 6.37.1.1 `__init__`()

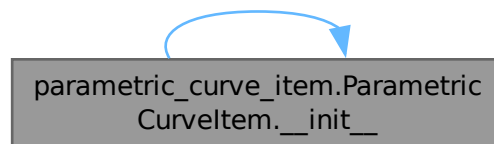
```

parametric_curve_item.ParametricCurveItem.__init__ (
    self,
    QPainterPath path)
  
```

Граф вызовов:



Граф вызова функции:



## 6.37.2 Методы

### 6.37.2.1 `__repr__()`

```
parametric_curve_item.ParametricCurveItem.__repr__ (  
    self)
```

### 6.37.2.2 `generate_edges()`

```
parametric_curve_item.ParametricCurveItem.generate_edges (  
    self)
```

## 6.37.3 Данные класса

### 6.37.3.1 `edges`

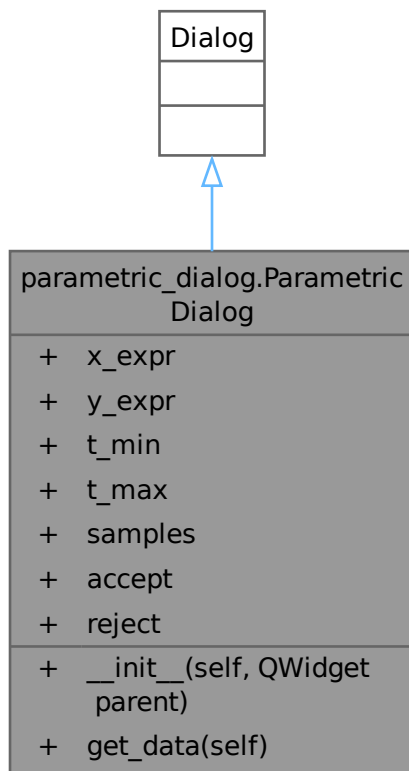
```
list parametric_curve_item.ParametricCurveItem.edges = []
```

Объявления и описания членов класса находятся в файле:

- `modules/data/src/shapes/parametric_curve_item.py`

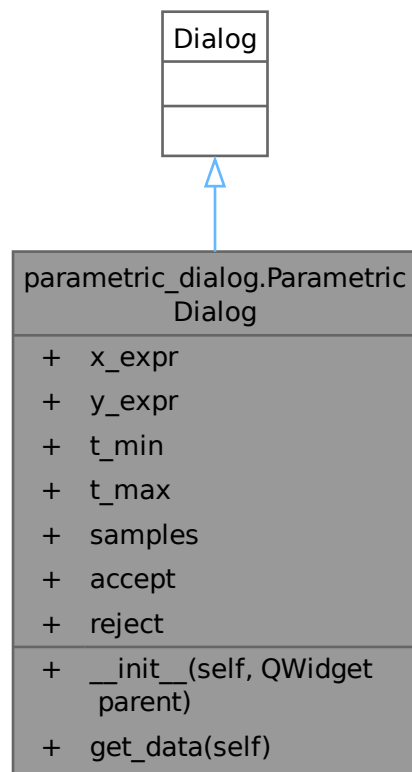
## 6.38 Класс parametric\_dialog.ParametricDialog

Граф наследования: parametric\_dialog.ParametricDialog:





Граф связей класса parametric\_dialog.ParametricDialog:



Открытые члены

- `__init__` (self, QWidget parent)
- `get_data` (self)

Открытые атрибуты

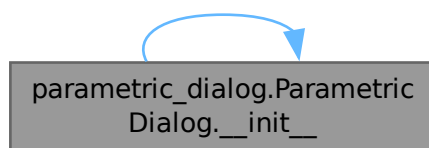
- `x_expr` = `QLineEdit('1.8 * t - 1.8')`
- `y_expr` = `QLineEdit('1.8 * 0.594689181 * (0.298222773 * sqrt(t) - 0.127125232 * t - 0.357907906 * t**2 + 0.291984971 * t**3 - 0.105174696 * t**4)')`
- `t_min` = `QLineEdit('0')`
- `t_max` = `QLineEdit('1')`
- `samples` = `QLineEdit('100')`
- `accept`
- `reject`

### 6.38.1 Конструктор(ы)

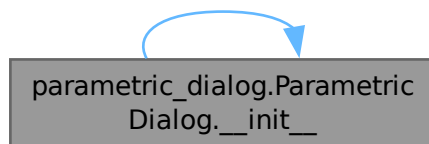
#### 6.38.1.1 `__init__()`

```
parametric_dialog.ParametricDialog.__init__ (  
    self,  
    QWidget parent)
```

Граф вызовов:



Граф вызова функции:



### 6.38.2 Методы

#### 6.38.2.1 `get_data()`

```
parametric_dialog.ParametricDialog.get_data (  
    self)
```

### 6.38.3 Данные класса

#### 6.38.3.1 `accept`

```
parametric_dialog.ParametricDialog.accept
```

6.38.3.2 `reject`

```
parametric_dialog.ParametricDialog.reject
```

6.38.3.3 `samples`

```
parametric_dialog.ParametricDialog.samples = QLineEdit('100')
```

6.38.3.4 `t_max`

```
parametric_dialog.ParametricDialog.t_max = QLineEdit('1')
```

6.38.3.5 `t_min`

```
parametric_dialog.ParametricDialog.t_min = QLineEdit('0')
```

6.38.3.6 `x_expr`

```
parametric_dialog.ParametricDialog.x_expr = QLineEdit('1.8 * t - 1.8')
```

6.38.3.7 `y_expr`

```
parametric_dialog.ParametricDialog.y_expr = QLineEdit('1.8 * 0.594689181 * (0.298222773 * sqrt(t) - 0.127125232 * t - 0.357907906 * t**2 + 0.291984971 * t**3 - 0.105174696 * t**4)')
```

Объявления и описания членов класса находятся в файле:

- `modules/data/src/dialogs/parametric_dialog.py`

6.39 Класс `plotter.Plotter`

Граф связей класса `plotter.Plotter`:

| plotter.Plotter                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>+ <code>plot_geometry(Axes ax, color='black')</code></li> <li>+ <code>Colorbar plot_surface (Figure fig, Axes ax, Series x, Series y, Series z, *args, **kwargs)</code></li> <li>+ <code>plot_streamline(Figure fig, Axes ax, Series x, Series y, Series u, Series v, *args, **kwargs)</code></li> </ul> |

Открытые статические члены

- [plot\\_geometry](#) (Axes ax, color='black')
- Colorbar [plot\\_surface](#) (Figure fig, Axes ax, Series x, Series y, Series z, \*args, \*\*kwargs)
- [plot\\_streamline](#) (Figure fig, Axes ax, Series x, Series y, Series u, Series v, \*args, \*\*kwargs)

### 6.39.1 Методы

#### 6.39.1.1 plot\_geometry()

```
plotter.Plotter.plot_geometry (  
    Axes ax,  
    color = 'black') [static]
```

#### 6.39.1.2 plot\_streamline()

```
plotter.Plotter.plot_streamline (  
    Figure fig,  
    Axes ax,  
    Series x,  
    Series y,  
    Series u,  
    Series v,  
    * args,  
    ** kwargs) [static]
```

#### 6.39.1.3 plot\_surface()

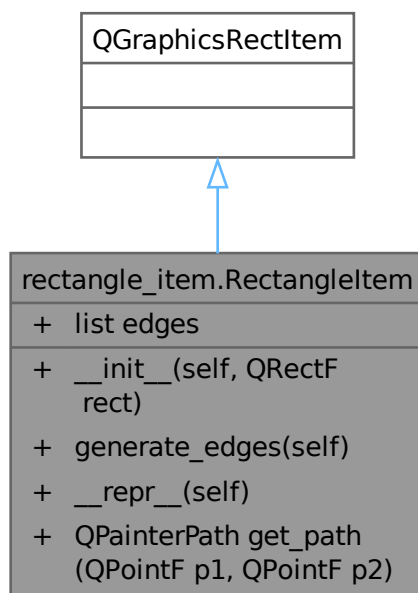
```
Colorbar plotter.Plotter.plot_surface (  
    Figure fig,  
    Axes ax,  
    Series x,  
    Series y,  
    Series z,  
    * args,  
    ** kwargs) [static]
```

Объявления и описания членов класса находятся в файле:

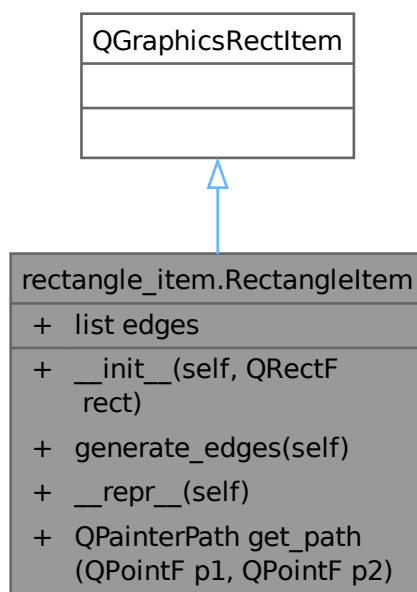
- `modules/visualization/src/services/plotter.py`

6.40 Класс `rectangle_item.RectangleItem`

Граф наследования:`rectangle_item.RectangleItem`:



Граф связей класса `rectangle_item.RectangleItem`:



Открытые члены

- `__init__` (self, `QRectF` rect)
- `generate_edges` (self)
- `__repr__` (self)

Открытые статические члены

- `QPainterPath` `get_path` (`QPointF` p1, `QPointF` p2)

Открытые атрибуты

- list `edges` = []

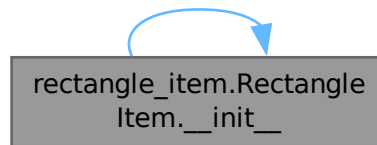
### 6.40.1 Конструктор(ы)

#### 6.40.1.1 `__init__`()

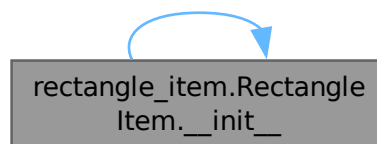
```

rectangle_item.RectangleItem.__init__ (
    self,
    QRectF rect)
  
```

Граф вызовов:



Граф вызова функции:



## 6.40.2 Методы

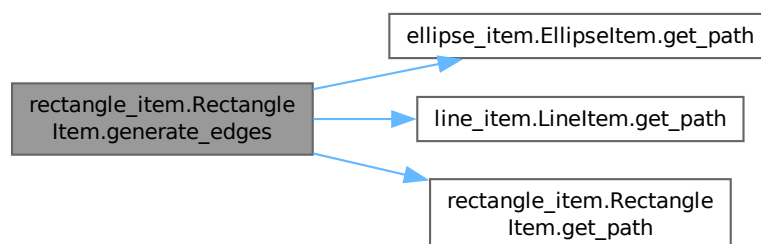
### 6.40.2.1 `__repr__()`

```
rectangle_item.RectangleItem.__repr__ (  
    self)
```

### 6.40.2.2 `generate_edges()`

```
rectangle_item.RectangleItem.generate_edges (  
    self)
```

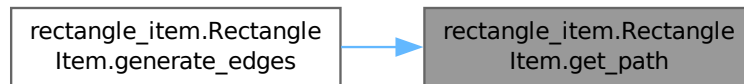
Граф вызовов:



### 6.40.2.3 get\_path()

```
QPainterPath rectangle_item.RectangleItem.get_path (  
    QPointF p1,  
    QPointF p2) [static]
```

Граф вызова функции:



## 6.40.3 Данные класса

### 6.40.3.1 edges

```
list rectangle_item.RectangleItem.edges = []
```

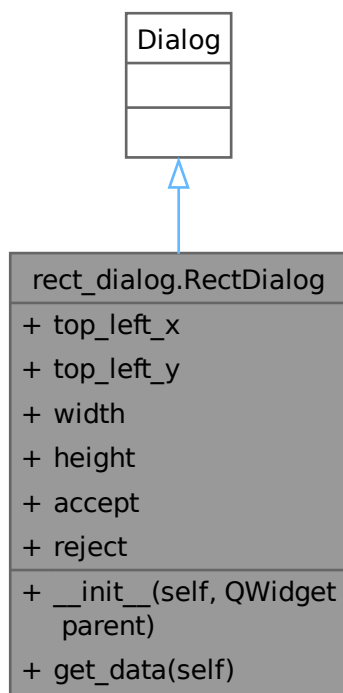
Объявления и описания членов класса находятся в файле:

- `modules/data/src/shapes/rectangle_item.py`

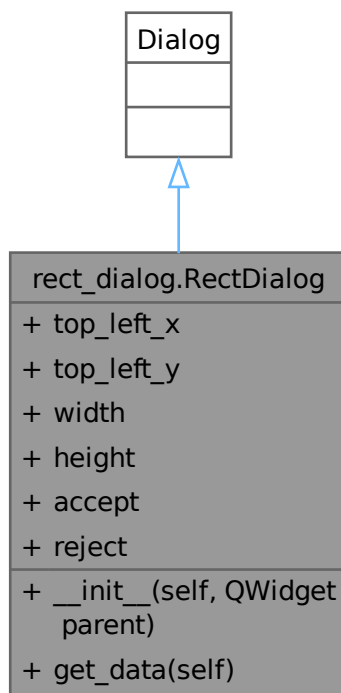


## 6.41 Класс rect\_dialog.RectDialog

Граф наследования:rect\_dialog.RectDialog:



Граф связей класса `rect_dialog.RectDialog`:



Открытые члены

- `__init__` (self, QWidget parent)
- `get_data` (self)

Открытые атрибуты

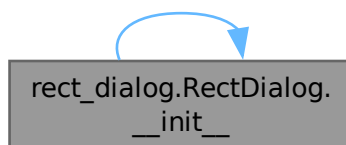
- `top_left_x` = QLineEdit('0')
- `top_left_y` = QLineEdit('0')
- `width` = QLineEdit('180')
- `height` = QLineEdit('180')
- `accept`
- `reject`

### 6.41.1 Конструктор(ы)

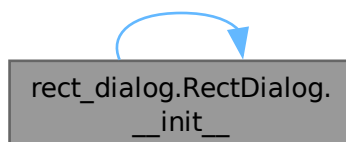
#### 6.41.1.1 `__init__`()

```
rect_dialog.RectDialog.__init__(
    self,
    QWidget parent)
```

Граф вызовов:



Граф вызова функции:



## 6.41.2 Методы

### 6.41.2.1 get\_data()

```
rect_dialog.RectDialog.get_data (  
    self)
```

## 6.41.3 Данные класса

### 6.41.3.1 accept

```
rect_dialog.RectDialog.accept
```

### 6.41.3.2 height

```
rect_dialog.RectDialog.height = QLineEdit('180')
```

### 6.41.3.3 reject

```
rect_dialog.RectDialog.reject
```

#### 6.41.3.4 top\_left\_x

```
rect_dialog.RectDialog.top_left_x = QLineEdit('0')
```

#### 6.41.3.5 top\_left\_y

```
rect_dialog.RectDialog.top_left_y = QLineEdit('0')
```

#### 6.41.3.6 width

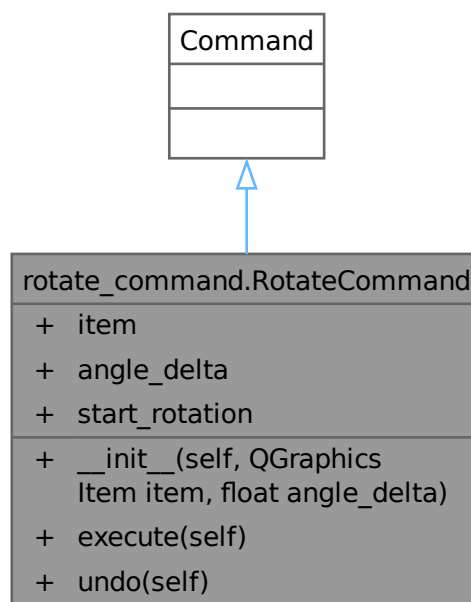
```
rect_dialog.RectDialog.width = QLineEdit('180')
```

Объявления и описания членов класса находятся в файле:

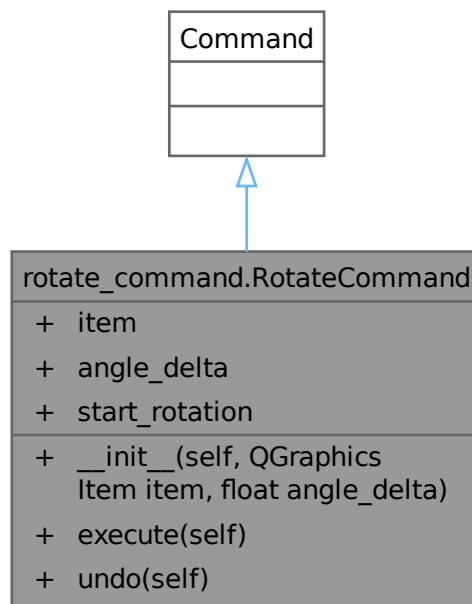
- `modules/data/src/dialogs/rect_dialog.py`

## 6.42 Класс `rotate_command.RotateCommand`

Граф наследования:`rotate_command.RotateCommand`:



Граф связей класса rotate\_command.RotateCommand:



Открытые члены

- `__init__` (self, QGraphicsItem `item`, float `angle_delta`)
- `execute` (self)
- `undo` (self)

Открытые атрибуты

- `item` = item
- `angle_delta` = angle\_delta
- `start_rotation` = item.rotation()

### 6.42.1 Конструктор(ы)

#### 6.42.1.1 `__init__`()

```

rotate_command.RotateCommand.__init__ (
    self,
    QGraphicsItem item,
    float angle_delta)
  
```

## 6.42.2 Методы

### 6.42.2.1 execute()

```
rotate_command.RotateCommand.execute (  
    self)
```

### 6.42.2.2 undo()

```
rotate_command.RotateCommand.undo (  
    self)
```

## 6.42.3 Данные класса

### 6.42.3.1 angle\_delta

```
rotate_command.RotateCommand.angle_delta = angle_delta
```

### 6.42.3.2 item

```
rotate_command.RotateCommand.item = item
```

### 6.42.3.3 start\_rotation

```
rotate_command.RotateCommand.start_rotation = item.rotation()
```

Объявления и описания членов класса находятся в файле:

- `modules/data/src/commands/rotate\_command.py`

## 6.43 Класс selection\_service.SelectionService

Граф связей класса selection\_service.SelectionService:

| selection_service.SelectionService       |
|------------------------------------------|
| + scene                                  |
| + list bool_selection                    |
| + list selected_edges                    |
| + selected_pen                           |
| + selection_pen                          |
| + selection_brush                        |
| + selection_rect                         |
| + last_selection_rect                    |
| + __init__(self, GridScene scene)        |
| + add_edge(self, edge)                   |
| + remove_edge(self, edge)                |
| + clear_edges(self)                      |
| + clear_and_select_item(self, item)      |
| + select_item(self, item)                |
| + clear_selection(self)                  |
| + update_selection(self, selection_rect) |
| + select_all(self)                       |
| + highlight_edge(self, edge)             |

Открытые члены

- `__init__` (self, GridScene [scene](#))
- `add_edge` (self, edge)
- `remove_edge` (self, edge)
- `clear_edges` (self)
- `clear_and_select_item` (self, item)
- `select_item` (self, item)
- `clear_selection` (self)

- `update_selection` (self, `selection_rect`)
- `select_all` (self)
- `highlight_edge` (self, `edge`)

#### Открытые атрибуты

- `scene` = `scene`
- list `bool_selection` = []
- list `selected_edges` = []
- `selected_pen` = `QPen(Qt.red, 0)`
- `selection_pen` = `QPen(Qt.blue, 0, Qt.DashLine)`
- `selection_brush` = `QBrush(QColor(0, 0, 255, 50))`
- `selection_rect` = `None`
- `last_selection_rect` = `QRectF()`

### 6.43.1 Конструктор(ы)

#### 6.43.1.1 `__init__()`

```
selection_service.SelectionService.__init__ (
    self,
    GridScene scene)
```

### 6.43.2 Методы

#### 6.43.2.1 `add_edge()`

```
selection_service.SelectionService.add_edge (
    self,
    edge)
```

Граф вызовов:

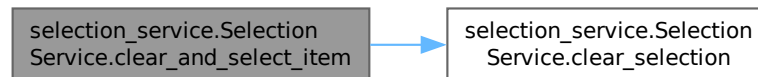




## 6.43.2.2 clear\_and\_select\_item()

```
selection_service.SelectionService.clear_and_select_item (  
    self,  
    item)
```

Граф вызовов:



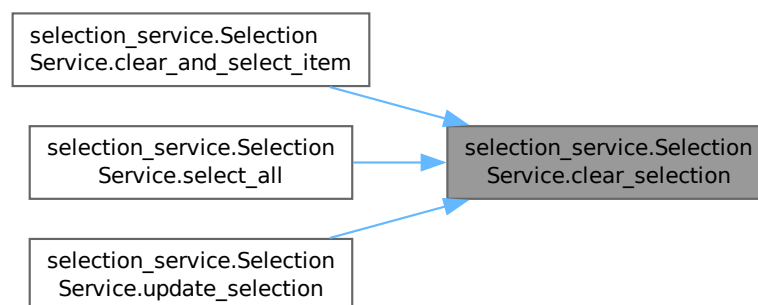
## 6.43.2.3 clear\_edges()

```
selection_service.SelectionService.clear_edges (  
    self)
```

## 6.43.2.4 clear\_selection()

```
selection_service.SelectionService.clear_selection (  
    self)
```

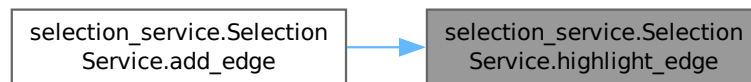
Граф вызова функции:



## 6.43.2.5 highlight\_edge()

```
selection_service.SelectionService.highlight_edge (  
    self,  
    edge)
```

Граф вызова функции:



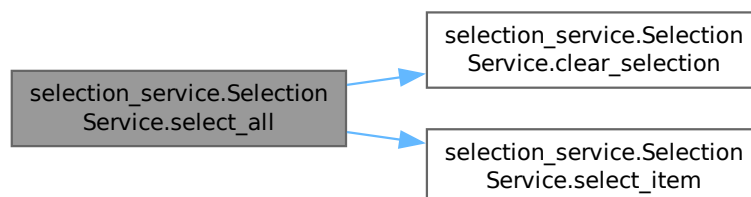
## 6.43.2.6 remove\_edge()

```
selection_service.SelectionService.remove_edge (  
    self,  
    edge)
```

## 6.43.2.7 select\_all()

```
selection_service.SelectionService.select_all (  
    self)
```

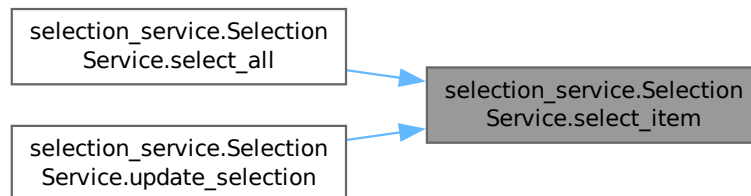
Граф вызовов:



## 6.43.2.8 select\_item()

```
selection_service.SelectionService.select_item (  
    self,  
    item)
```

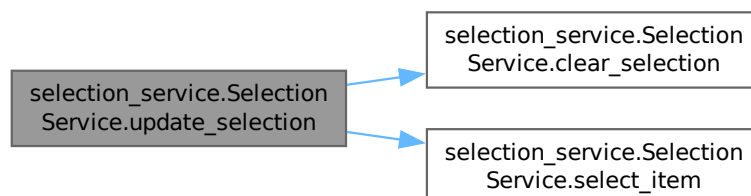
Граф вызова функции:



## 6.43.2.9 update\_selection()

```
selection_service.SelectionService.update_selection (  
    self,  
    selection_rect)
```

Граф вызовов:



## 6.43.3 Данные класса

## 6.43.3.1 bool\_selection

```
list selection_service.SelectionService.bool_selection = []
```

## 6.43.3.2 last\_selection\_rect

```
selection_service.SelectionService.last_selection_rect = QRectF()
```

#### 6.43.3.3 scene

```
selection_service.SelectionService.scene = scene
```

#### 6.43.3.4 selected\_edges

```
list selection_service.SelectionService.selected_edges = []
```

#### 6.43.3.5 selected\_pen

```
selection_service.SelectionService.selected_pen = QPen(Qt.red, 0)
```

#### 6.43.3.6 selection\_brush

```
selection_service.SelectionService.selection_brush = QBrush(QColor(0, 0, 255, 50))
```

#### 6.43.3.7 selection\_pen

```
selection_service.SelectionService.selection_pen = QPen(Qt.blue, 0, Qt.DashLine)
```

#### 6.43.3.8 selection\_rect

```
selection_service.SelectionService.selection_rect = None
```

Объявления и описания членов класса находятся в файле:

- `modules/data/src/services/selection\_service.py`

### 6.44 Класс transformation\_operations.TransformationOperations

Граф связей класса transformation\_operations.TransformationOperations:

| transformation_operations.<br>TransformationOperations                                                                             |
|------------------------------------------------------------------------------------------------------------------------------------|
| + parent                                                                                                                           |
| + scene                                                                                                                            |
| + command_service                                                                                                                  |
| + selection_service                                                                                                                |
| + __init__(self, QWidget<br>parent, GridScene scene,<br>CommandService command<br>_service, SelectionService<br>selection_service) |
| + perform_mirror(self)                                                                                                             |
| + perform_rotate(self)                                                                                                             |

## Открытые члены

- `__init__` (self, QWidget parent, GridScene scene, CommandService command\_service, SelectionService selection\_service)
- `perform_mirror` (self)
- `perform_rotate` (self)

## Открытые атрибуты

- `parent` = parent
- `scene` = scene
- `command_service` = command\_service
- `selection_service` = selection\_service

## 6.44.1 Конструктор(ы)

6.44.1.1 `__init__()`

```
transformation_operations.TransformationOperations.__init__ (
    self,
    QWidget parent,
    GridScene scene,
    CommandService command_service,
    SelectionService selection_service)
```

## 6.44.2 Методы

6.44.2.1 `perform_mirror()`

```
transformation_operations.TransformationOperations.perform_mirror (
    self)
```

6.44.2.2 `perform_rotate()`

```
transformation_operations.TransformationOperations.perform_rotate (
    self)
```

## 6.44.3 Данные класса

6.44.3.1 `command_service`

```
transformation_operations.TransformationOperations.command_service = command_service
```

6.44.3.2 `parent`

```
transformation_operations.TransformationOperations.parent = parent
```

#### 6.44.3.3 scene

```
transformation_operations.TransformationOperations.scene = scene
```

#### 6.44.3.4 selection\_service

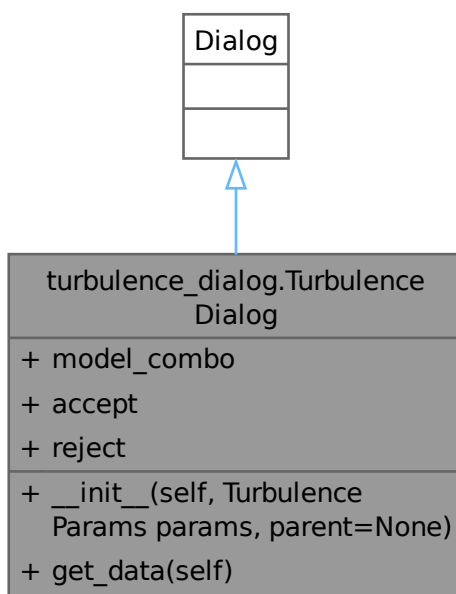
```
transformation_operations.TransformationOperations.selection_service = selection_service
```

Объявления и описания членов класса находятся в файле:

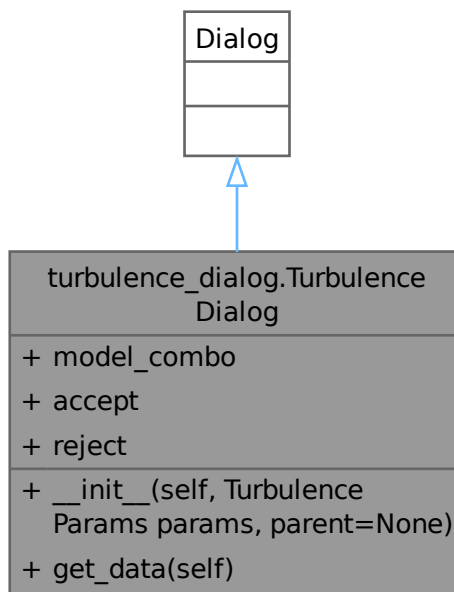
- `modules/data/src/operations/transformation\_operations.py`

### 6.45 Класс `turbulence_dialog.TurbulenceDialog`

Граф наследования: `turbulence_dialog.TurbulenceDialog`:



Граф связей класса `turbulence_dialog.TurbulenceDialog`:



Открытые члены

- `__init__` (self, `TurbulenceParams` params, parent=None)
- `get_data` (self)

Открытые атрибуты

- `model_combo` = `QComboBox()`
- `accept`
- `reject`

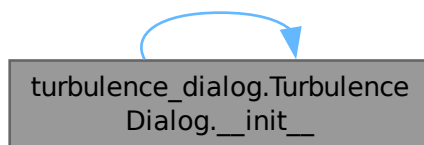
### 6.45.1 Конструктор(ы)

#### 6.45.1.1 `__init__()`

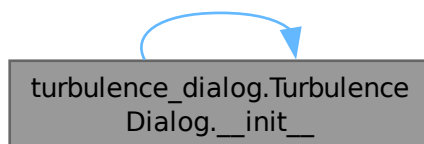
```

turbulence_dialog.TurbulenceDialog.__init__ (
    self,
    TurbulenceParams params,
    parent = None)
  
```

Граф вызовов:



Граф вызова функции:



## 6.45.2 Методы

### 6.45.2.1 get\_data()

```
turbulence_dialog.TurbulenceDialog.get_data (  
    self)
```

## 6.45.3 Данные класса

### 6.45.3.1 accept

```
turbulence_dialog.TurbulenceDialog.accept
```

### 6.45.3.2 model\_combo

```
turbulence_dialog.TurbulenceDialog.model_combo = QComboBox()
```



6.45.3.3 `reject`

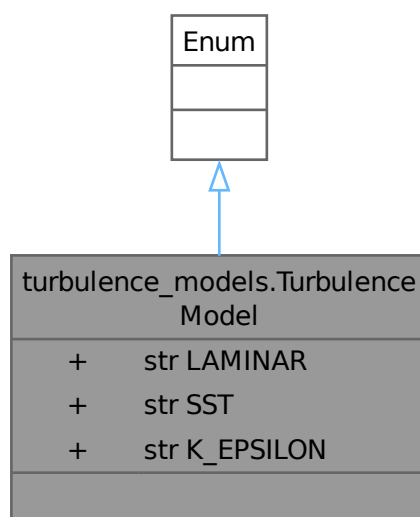
`turbulence_dialog.TurbulenceDialog.reject`

Объявления и описания членов класса находятся в файле:

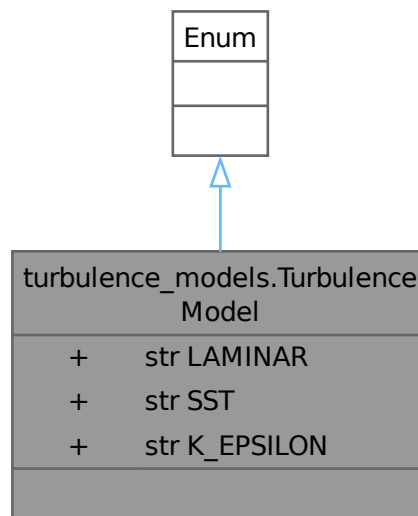
- `modules/data/src/dialogs/turbulence_dialog.py`

6.46 Класс `turbulence_models.TurbulenceModel`

Граф наследования:`turbulence_models.TurbulenceModel`:



Граф связей класса `turbulence_models.TurbulenceModel`:



Статические открытые данные

- `str LAMINAR = 'Laminar'`
- `str SST = 'SST'`
- `str K_EPSILON = 'k-epsilon'`

### 6.46.1 Данные класса

#### 6.46.1.1 K\_EPSILON

```
str turbulence_models.TurbulenceModel.K_EPSILON = 'k-epsilon' [static]
```

#### 6.46.1.2 LAMINAR

```
str turbulence_models.TurbulenceModel.LAMINAR = 'Laminar' [static]
```

#### 6.46.1.3 SST

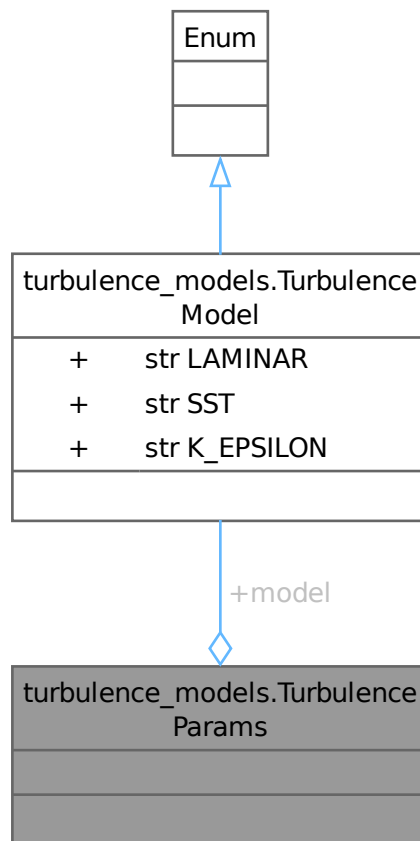
```
str turbulence_models.TurbulenceModel.SST = 'SST' [static]
```

Объявления и описания членов класса находятся в файле:

- `modules/data/src/physics/turbulence_models.py`

## 6.47 Класс `turbulence_models.TurbulenceParams`

Граф связей класса `turbulence_models.TurbulenceParams`:



Статические открытые данные

- `TurbulenceModel model = TurbulenceModel.LAMINAR`

### 6.47.1 Данные класса

#### 6.47.1.1 `model`

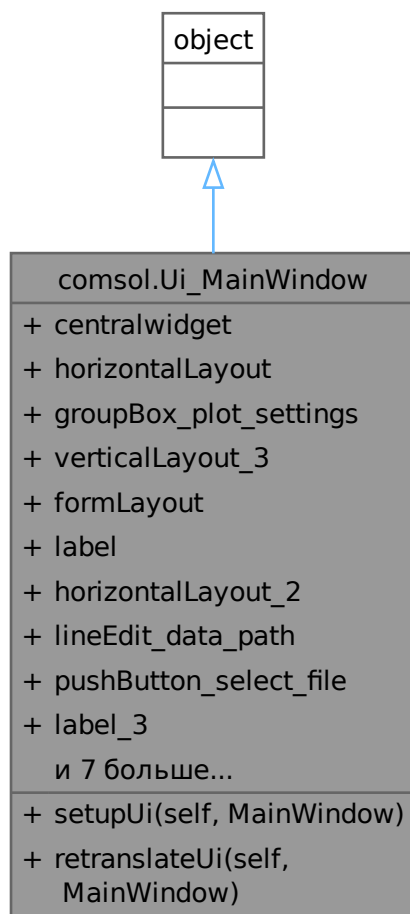
`TurbulenceModel` `turbulence_models.TurbulenceParams.model = TurbulenceModel.LAMINAR` [static]

Объявления и описания членов класса находятся в файле:

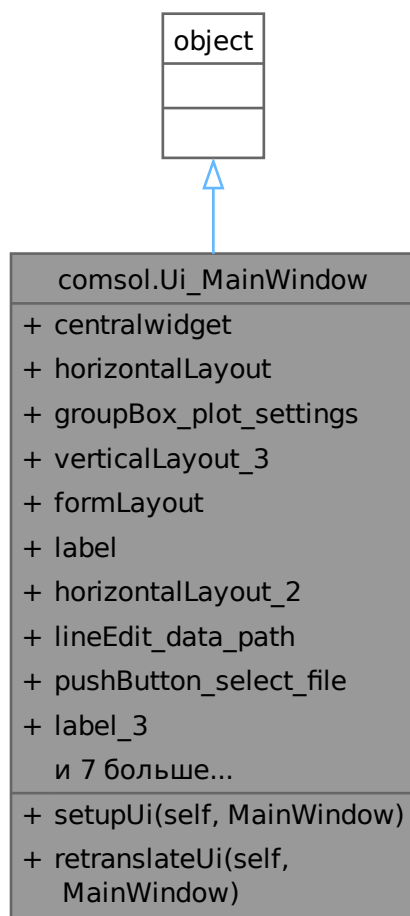
- `modules/data/src/physics/turbulence_models.py`

## 6.48 Класс comsol.Ui\_MainWindow

Граф наследования:comsol.Ui\_MainWindow:



Граф связей класса comsol.Ui\_MainWindow:



Открытые члены

- [setupUi](#) (self, MainWindow)
- [retranslateUi](#) (self, MainWindow)

Открытые атрибуты

- [centralwidget](#) = QWidget(MainWindow)
- [horizontalLayout](#) = QHBoxLayout(self.centralwidget)
- [groupBox\\_plot\\_settings](#) = QGroupBox(self.centralwidget)
- [verticalLayout\\_3](#) = QVBoxLayout(self.groupBox\_plot\_settings)
- [formLayout](#) = QFormLayout()
- [label](#) = QLabel(self.groupBox\_plot\_settings)
- [horizontalLayout\\_2](#) = QHBoxLayout()
- [lineEdit\\_data\\_path](#) = QLineEdit(self.groupBox\_plot\_settings)
- [pushButton\\_select\\_file](#) = QPushButton(self.groupBox\_plot\_settings)

- `label_3 = QLabel(self.groupBox_plot_settings)`
- `comboBox_expression = QComboBox(self.groupBox_plot_settings)`
- `label_2 = QLabel(self.groupBox_plot_settings)`
- `checkBox_stream_lines = QCheckBox(self.groupBox_plot_settings)`
- `label_4 = QLabel(self.groupBox_plot_settings)`
- `lineEdit_levels = QLineEdit(self.groupBox_plot_settings)`
- `pushButton_plot = QPushButton(self.groupBox_plot_settings)`
- `widget_visualisation = QWidget(self.centralwidget)`

## 6.48.1 Методы

### 6.48.1.1 `retranslateUi()`

```
comsol.Ui_MainWindow.retranslateUi (  
    self,  
    MainWindow)
```

### 6.48.1.2 `setupUi()`

```
comsol.Ui_MainWindow.setupUi (  
    self,  
    MainWindow)
```

## 6.48.2 Данные класса

### 6.48.2.1 `centralwidget`

```
comsol.Ui_MainWindow.centralwidget = QWidget(MainWindow)
```

### 6.48.2.2 `checkBox_stream_lines`

```
comsol.Ui_MainWindow.checkBox_stream_lines = QCheckBox(self.groupBox_plot_settings)
```

### 6.48.2.3 `comboBox_expression`

```
comsol.Ui_MainWindow.comboBox_expression = QComboBox(self.groupBox_plot_settings)
```

### 6.48.2.4 `formLayout`

```
comsol.Ui_MainWindow.formLayout = QFormLayout()
```

### 6.48.2.5 `groupBox_plot_settings`

```
comsol.Ui_MainWindow.groupBox_plot_settings = QGroupBox(self.centralwidget)
```

## 6.48.2.6 horizontalLayout

```
comsol.Ui_MainWindow.horizontalLayout = QHBoxLayout(self.centralwidget)
```

## 6.48.2.7 horizontalLayout\_2

```
comsol.Ui_MainWindow.horizontalLayout_2 = QHBoxLayout()
```

## 6.48.2.8 label

```
comsol.Ui_MainWindow.label = QLabel(self.groupBox_plot_settings)
```

## 6.48.2.9 label\_2

```
comsol.Ui_MainWindow.label_2 = QLabel(self.groupBox_plot_settings)
```

## 6.48.2.10 label\_3

```
comsol.Ui_MainWindow.label_3 = QLabel(self.groupBox_plot_settings)
```

## 6.48.2.11 label\_4

```
comsol.Ui_MainWindow.label_4 = QLabel(self.groupBox_plot_settings)
```

## 6.48.2.12 lineEdit\_data\_path

```
comsol.Ui_MainWindow.lineEdit_data_path = QLineEdit(self.groupBox_plot_settings)
```

## 6.48.2.13 lineEdit\_levels

```
comsol.Ui_MainWindow.lineEdit_levels = QLineEdit(self.groupBox_plot_settings)
```

## 6.48.2.14 pushButton\_plot

```
comsol.Ui_MainWindow.pushButton_plot = QPushButton(self.groupBox_plot_settings)
```

## 6.48.2.15 pushButton\_select\_file

```
comsol.Ui_MainWindow.pushButton_select_file = QPushButton(self.groupBox_plot_settings)
```

## 6.48.2.16 verticalLayout\_3

```
comsol.Ui_MainWindow.verticalLayout_3 = QVBoxLayout(self.groupBox_plot_settings)
```

## 6.48.2.17 widget\_visualisation

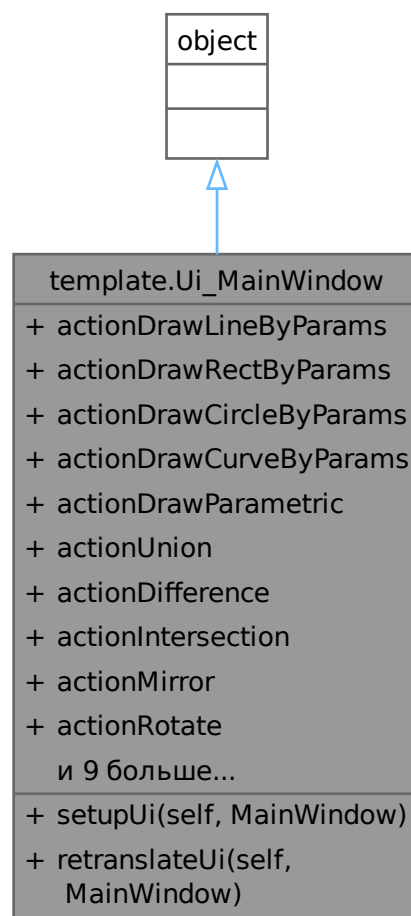
```
comsol.Ui_MainWindow.widget_visualisation = QWidget(self.centralwidget)
```

Объявления и описания членов класса находятся в файле:

- `modules/visualization/src/ui/comsol.py`

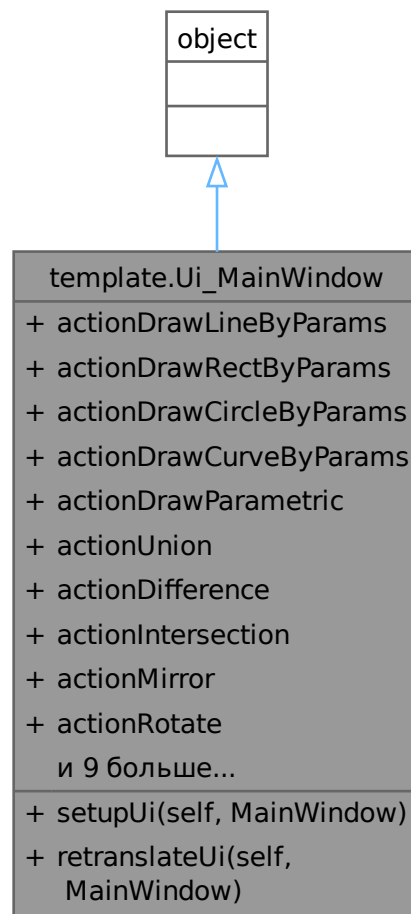
6.49 Класс `template.Ui_MainWindow`

Граф наследования: `template.Ui_MainWindow`:





Граф связей класса `template.Ui_MainWindow`:



Открытые члены

- [setupUi](#) (self, MainWindow)
- [retranslateUi](#) (self, MainWindow)

Открытые атрибуты

- [actionDrawLineByParams](#) = QAction(MainWindow)
- [actionDrawRectByParams](#) = QAction(MainWindow)
- [actionDrawCircleByParams](#) = QAction(MainWindow)
- [actionDrawCurveByParams](#) = QAction(MainWindow)
- [actionDrawParametric](#) = QAction(MainWindow)
- [actionUnion](#) = QAction(MainWindow)
- [actionDifference](#) = QAction(MainWindow)
- [actionIntersection](#) = QAction(MainWindow)
- [actionMirror](#) = QAction(MainWindow)

- `actionRotate` = `QAction(MainWindow)`
- `actionBuildMesh` = `QAction(MainWindow)`
- `centralwidget` = `QWidget(MainWindow)`
- `horizontalLayout` = `QHBoxLayout(self.centralwidget)`
- `projectTree` = `QTreeWidget(self.centralwidget)`
- `verticalLayout` = `QVBoxLayout()`
- `graphicsView` = `GraphicsView(self.centralwidget)`
- `propertiesLayout` = `QHBoxLayout()`
- `statusbar` = `QStatusBar(MainWindow)`
- `mainToolBar` = `QToolBar(MainWindow)`

## 6.49.1 Методы

### 6.49.1.1 `retranslateUi()`

```
template.Ui_MainWindow.retranslateUi (  
    self,  
    MainWindow)
```

### 6.49.1.2 `setupUi()`

```
template.Ui_MainWindow.setupUi (  
    self,  
    MainWindow)
```

## 6.49.2 Данные класса

### 6.49.2.1 `actionBuildMesh`

```
template.Ui_MainWindow.actionBuildMesh = QAction(MainWindow)
```

### 6.49.2.2 `actionDifference`

```
template.Ui_MainWindow.actionDifference = QAction(MainWindow)
```

### 6.49.2.3 `actionDrawCircleByParams`

```
template.Ui_MainWindow.actionDrawCircleByParams = QAction(MainWindow)
```

### 6.49.2.4 `actionDrawCurveByParams`

```
template.Ui_MainWindow.actionDrawCurveByParams = QAction(MainWindow)
```

6.49.2.5 `actionDrawLineByParams`

```
template.Ui_MainWindow.actionDrawLineByParams = QAction(MainWindow)
```

6.49.2.6 `actionDrawParametric`

```
template.Ui_MainWindow.actionDrawParametric = QAction(MainWindow)
```

6.49.2.7 `actionDrawRectByParams`

```
template.Ui_MainWindow.actionDrawRectByParams = QAction(MainWindow)
```

6.49.2.8 `actionIntersection`

```
template.Ui_MainWindow.actionIntersection = QAction(MainWindow)
```

6.49.2.9 `actionMirror`

```
template.Ui_MainWindow.actionMirror = QAction(MainWindow)
```

6.49.2.10 `actionRotate`

```
template.Ui_MainWindow.actionRotate = QAction(MainWindow)
```

6.49.2.11 `actionUnion`

```
template.Ui_MainWindow.actionUnion = QAction(MainWindow)
```

6.49.2.12 `centralwidget`

```
template.Ui_MainWindow.centralwidget = QWidget(MainWindow)
```

6.49.2.13 `graphicsView`

```
template.Ui_MainWindow.graphicsView = GraphicsView(self.centralwidget)
```

6.49.2.14 `horizontalLayout`

```
template.Ui_MainWindow.horizontalLayout = QHBoxLayout(self.centralwidget)
```

## 6.49.2.15 mainToolBar

```
template.Ui_MainWindow.mainToolBar = QToolBar(MainWindow)
```

## 6.49.2.16 projectTree

```
template.Ui_MainWindow.projectTree = QTreeWidget(self.centralwidget)
```

## 6.49.2.17 propertiesLayout

```
template.Ui_MainWindow.propertiesLayout = QHBoxLayout()
```

## 6.49.2.18 statusbar

```
template.Ui_MainWindow.statusbar = QStatusBar(MainWindow)
```

## 6.49.2.19 verticalLayout

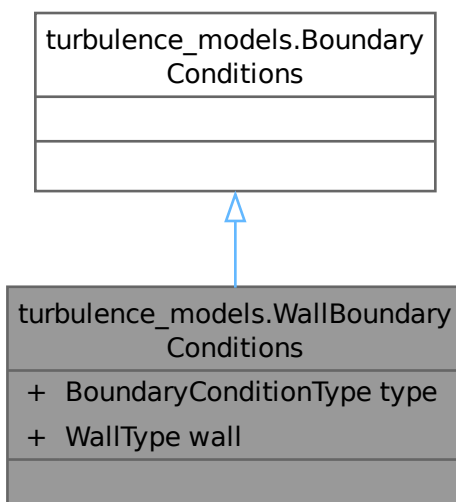
```
template.Ui_MainWindow.verticalLayout = QVBoxLayout()
```

Объявления и описания членов класса находятся в файле:

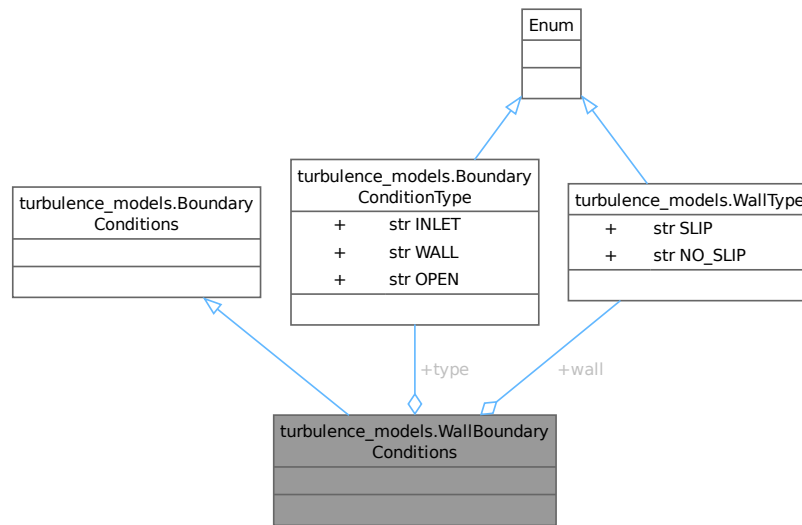
- `modules/data/src/ui/template.py`

6.50 Класс `turbulence_models.WallBoundaryConditions`

Граф наследования: `turbulence_models.WallBoundaryConditions`:



Граф связей класса `turbulence_models.WallBoundaryConditions`:



Статические открытые данные

- `BoundaryConditionType type = BoundaryConditionType.WALL`
- `WallType wall = WallType.NO_SLIP`

### 6.50.1 Данные класса

#### 6.50.1.1 type

`BoundaryConditionType` `turbulence_models.WallBoundaryConditions.type = BoundaryConditionType.WALL` [static]

#### 6.50.1.2 wall

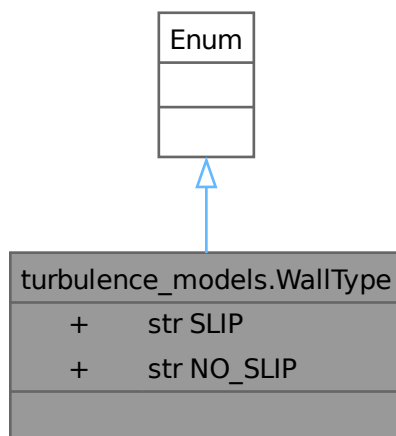
`WallType` `turbulence_models.WallBoundaryConditions.wall = WallType.NO_SLIP` [static]

Объявления и описания членов класса находятся в файле:

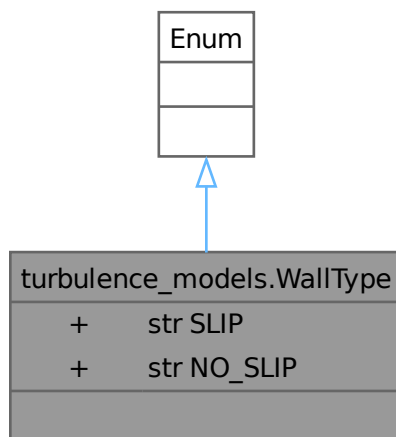
- `modules/data/src/physics/turbulence_models.py`

## 6.51 Класс `turbulence_models.WallType`

Граф наследования: `turbulence_models.WallType`:



Граф связей класса `turbulence_models.WallType`:



Статические открытые данные

- `str SLIP = 'slip'`
- `str NO_SLIP = 'no slip'`

### 6.51.1 Данные класса

#### 6.51.1.1 NO\_SLIP

```
str turbulence_models.WallType.NO_SLIP = 'no slip' [static]
```

#### 6.51.1.2 SLIP

```
str turbulence_models.WallType.SLIP = 'slip' [static]
```

Объявления и описания членов класса находятся в файле:

- `modules/data/src/physics/turbulence_models.py`





## Глава 7

# Файлы

### 7.1 Файл `modules/data/main.py`

Пространства имен

- namespace `main`

Переменные

- `main.level`
- `main.DEBUG`
- `main.format`
- `main.stream`
- `main.app = QApplication(sys.argv)`
- `main.window = MainWindow()`

### 7.2 Файл `modules/visualization/main.py`

Пространства имен

- namespace `main`

Переменные

- dict `main.font`

### 7.3 Файл `modules/data/src/commands/add_command.py`

Классы

- class `add_command.AddCommand`

Пространства имен

- namespace [add\\_command](#)

## 7.4 Файл `modules/data/src/commands/command.py`

Классы

- class [command.Command](#)

Пространства имен

- namespace [command](#)

## 7.5 Файл `modules/data/src/commands/delete_command.py`

Классы

- class [delete\\_command.DeleteCommand](#)

Пространства имен

- namespace [delete\\_command](#)

## 7.6 Файл `modules/data/src/commands/move_command.py`

Классы

- class [move\\_command.MoveCommand](#)

Пространства имен

- namespace [move\\_command](#)

## 7.7 Файл `modules/data/src/commands/rotate_command.py`

Классы

- class [rotate\\_command.RotateCommand](#)

Пространства имен

- namespace [rotate\\_command](#)

## 7.8 Файл modules/data/src/dialogs/bezier\_dialog.py

### Классы

- class [bezier\\_dialog.BezierDialog](#)

### Пространства имен

- namespace [bezier\\_dialog](#)

## 7.9 Файл modules/data/src/dialogs/boundary\_conditions\_dialog.py

### Классы

- class [boundary\\_conditions\\_dialog.BoundaryConditionsDialog](#)

### Пространства имен

- namespace [boundary\\_conditions\\_dialog](#)

## 7.10 Файл modules/data/src/dialogs/dialog.py

### Классы

- class [dialog.Dialog](#)

### Пространства имен

- namespace [dialog](#)

## 7.11 Файл modules/data/src/dialogs/dialog\_factory.py

### Классы

- class [dialog\\_factory.DialogFactory](#)

### Пространства имен

- namespace [dialog\\_factory](#)

## 7.12 Файл modules/data/src/dialogs/ellipse\_dialog.py

### Классы

- class [ellipse\\_dialog.EllipseDialog](#)

Пространства имен

- namespace [ellipse\\_dialog](#)

## 7.13 Файл `modules/data/src/dialogs/initial_conditions_dialog.py`

Классы

- class [initial\\_conditions\\_dialog.InitialConditionsDialog](#)

Пространства имен

- namespace [initial\\_conditions\\_dialog](#)

## 7.14 Файл `modules/data/src/dialogs/line_dialog.py`

Классы

- class [line\\_dialog.LineDialog](#)

Пространства имен

- namespace [line\\_dialog](#)

## 7.15 Файл `modules/data/src/dialogs/material_dialog.py`

Классы

- class [material\\_dialog.MaterialDialog](#)

Пространства имен

- namespace [material\\_dialog](#)

## 7.16 Файл `modules/data/src/dialogs/mesh_dialog.py`

Классы

- class [mesh\\_dialog.MeshDialog](#)

Пространства имен

- namespace [mesh\\_dialog](#)

## 7.17 Файл modules/data/src/dialogs/parametric\_dialog.py

### Классы

- class [parametric\\_dialog.ParametricDialog](#)

### Пространства имен

- namespace [parametric\\_dialog](#)

## 7.18 Файл modules/data/src/dialogs/rect\_dialog.py

### Классы

- class [rect\\_dialog.RectDialog](#)

### Пространства имен

- namespace [rect\\_dialog](#)

## 7.19 Файл modules/data/src/dialogs/turbulence\_dialog.py

### Классы

- class [turbulence\\_dialog.TurbulenceDialog](#)

### Пространства имен

- namespace [turbulence\\_dialog](#)

## 7.20 Файл modules/data/src/event\_handler.py

### Классы

- class [event\\_handler.EventHandler](#)

### Пространства имен

- namespace [event\\_handler](#)

## 7.21 Файл modules/data/src/main\_window.py

### Классы

- class [main\\_window.MainWindow](#)

Пространства имен

- namespace `main_window`

Переменные

- `main_window.app = QApplication([])`
- `main_window.window = MainWindow()`

## 7.22 Файл `modules/visualization/src/main_window.py`

Классы

- class `main_window.MainWindow`

Пространства имен

- namespace `main_window`

## 7.23 Файл `modules/data/src/operations/boolean_operations.py`

Классы

- class `boolean_operations.BooleanOperations`

Пространства имен

- namespace `boolean_operations`

## 7.24 Файл `modules/data/src/operations/transformation_operations.py`

Классы

- class `transformation_operations.TransformationOperations`

Пространства имен

- namespace `transformation_operations`

## 7.25 Файл modules/data/src/physics/turbulence\_models.py

### Классы

- class [turbulence\\_models.TurbulenceModel](#)
- class [turbulence\\_models.BoundaryConditionType](#)
- class [turbulence\\_models.WallType](#)
- class [turbulence\\_models.TurbulenceParams](#)
- class [turbulence\\_models.BoundaryConditions](#)
- class [turbulence\\_models.InletBoundaryConditions](#)
- class [turbulence\\_models.OpenBoundaryConditions](#)
- class [turbulence\\_models.WallBoundaryConditions](#)
- class [turbulence\\_models.InitialConditions](#)
- class [turbulence\\_models.Material](#)

### Пространства имен

- namespace [turbulence\\_models](#)

## 7.26 Файл modules/data/src/services/command\_service.py

### Классы

- class [command\\_service.CommandService](#)

### Пространства имен

- namespace [command\\_service](#)

## 7.27 Файл modules/data/src/services/drawing\_service.py

### Классы

- class [drawing\\_service.DrawingService](#)

### Пространства имен

- namespace [drawing\\_service](#)

## 7.28 Файл modules/data/src/services/gmsh\_mesh\_builder.py

### Классы

- class [gmsh\\_mesh\\_builder.GmshMeshBuilder](#)

Пространства имен

- namespace [gmsh\\_mesh\\_builder](#)

## 7.29 Файл `modules/data/src/services/selection_service.py`

Классы

- class [selection\\_service.SelectionService](#)

Пространства имен

- namespace [selection\\_service](#)

## 7.30 Файл `modules/data/src/shapes/boolean_item.py`

Классы

- class [boolean\\_item.BooleanShapeItem](#)

Пространства имен

- namespace [boolean\\_item](#)

## 7.31 Файл `modules/data/src/shapes/ellipse_item.py`

Классы

- class [ellipse\\_item.EllipseItem](#)

Пространства имен

- namespace [ellipse\\_item](#)

Переменные

- [ellipse\\_item.el](#) = [EllipseItem](#)(QRectF(0, 0, 100, 100))

## 7.32 Файл `modules/data/src/shapes/line_item.py`

Классы

- class [line\\_item.LineItem](#)



Пространства имен

- namespace [line\\_item](#)

## 7.33 Файл modules/data/src/shapes/parametric\_curve\_item.py

Классы

- class [parametric\\_curve\\_item.ParametricCurveItem](#)

Пространства имен

- namespace [parametric\\_curve\\_item](#)

## 7.34 Файл modules/data/src/shapes/rectangle\_item.py

Классы

- class [rectangle\\_item.RectangleItem](#)

Пространства имен

- namespace [rectangle\\_item](#)

## 7.35 Файл modules/data/src/ui/template.py

Классы

- class [template.Ui\\_MainWindow](#)

Пространства имен

- namespace [template](#)

## 7.36 Файл modules/data/src/widgets/edge\_item.py

Классы

- class [edge\\_item.EdgeItem](#)

Пространства имен

- namespace [edge\\_item](#)

## Функции

- `int edge_item._next_edge_id()`

## Переменные

- `edge_item._edge_id_counter = itertools.count(1)`
- `edge_item.p = QPainterPath()`
- `edge_item.e = EdgeItem(p)`
- `edge_item.p1 = QPointF(0.0, 0.0)`
- `edge_item.p2 = QPointF(0.0, 0.0)`

7.37 Файл `modules/data/src/widgets/editable_bezier.py`

## Классы

- `class editable_bezier.DraggablePoint`
- `class editable_bezier.EditableBezierCurveItem`

## Пространства имен

- `namespace editable_bezier`

7.38 Файл `modules/data/src/widgets/graphics_view.py`

## Классы

- `class graphics_view.GraphicsView`

## Пространства имен

- `namespace graphics_view`

7.39 Файл `modules/data/src/widgets/grid_scene.py`

## Классы

- `class grid_scene.GridScene`

## Пространства имен

- `namespace grid_scene`

## 7.40 Файл modules/doxytest/doxytest.py

### Классы

- class `doxytest.DoxyTest`

### Пространства имен

- namespace `doxytest`

## 7.41 Файл modules/visualization/src/mpl\_canvas.py

### Классы

- class `mpl_canvas.MplCanvas`

### Пространства имен

- namespace `mpl_canvas`

## 7.42 Файл modules/visualization/src/services/data\_parser.py

### Классы

- class `data_parser.DataParser`

### Пространства имен

- namespace `data_parser`

### Переменные

- `data_parser.parser = DataParser('../data/semicolon.txt')`
- `data_parser.data = parser.parse()`
- `int data_parser.c = 1`

## 7.43 Файл modules/visualization/src/services/plotter.py

### Классы

- class `plotter.Plotter`

### Пространства имен

- namespace `plotter`

## 7.44 Файл modules/visualization/src/ui/comsol.py

### Классы

- class `comsol.Ui_MainWindow`

### Пространства имен

- namespace `comsol`



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